



Dragonforge

Game Design Document

Mobile online, social & skill-based gothic fantasy mobile game



Dragonforge | Changelog

<truncated from v25>

8/31 – v18 updated pages with new HUD designs, added crafting, trading pages. Pages 48 through 53, Added pages 76 & 83 for crafting and NPC UIs

9/21 – v19 Pages 67, 68, 69 regarding item generation in regards to crafting.

9/24 – v19 Pages 70 thru 75 – Item generation

10/12 – v21 Revised page 29 for armor locations (combined shoulders into torso and legs+feet into a single location).

10/16 – v21 Updated armor locations some more, added page 71 regarding materials in crafting.

11/5 – v22 Added pages 33 and 34 detailing damage calculations for weapons

11/8 – v22 – Revised pages 30 and 31 in regards to armor protection calculation and durability loss.

11/9 – v22 Added page 72 regarding affix tiers, created pages 35 for weapon damage calcs and pages 32 for elemental/affinity resist calcs and 33+34 for shield blocking and damage mitigation design

11/19-11/29 – v24 Added pages 101 through 111 in regards to enemies, AI and pathfinding.

12/4-12/6 –v25 Added pages 112 through 116 regarding enemy template data, loot and abilities.

2/11-2/13 –v26 Added pages 69 and 70 for Skill level XP gain and levelling up information.

3/4/2019 – v27 Added pages 38 through 41 detailing combat ability variables and calculations.

3/5 – v27 Completed pages 40 – 42 for spells combat.

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Dragonforge | Game Overview

The World of the Dragonforge

At it's heart and soul, *Dragonforge* is an open-world sandbox MMORPG. It's appearance however is that of an dark fantasy Action-RPG with impressive visual elements and effects.

Our World

The core principle behind *Dragonforge* is based on the premise that a player can freely play anyway they wish. Casual players will enjoy the ease of simple, short and bite-sized game sessions. Where as advanced players will have full customization over their characters to tackle the exceptional challenges that lie ahead. Players of all sorts will enjoy that they can contribute regardless of time spent, equipment or abilities.

All players will have equal access to the rich story-driven quests, deep dank dungeons and all of the wonders of the world at their fingertips. Furthermore, players will be able to band together to form guilds, buy & sell their goods and services at the player-driven marketplace.

Dragonforge developers will employ cutting edge methods to prevent overcrowding, trolling, exploits and unfair play. While we will also offer players the capability to host and join private instances as well as publically available ones which will be policed and overseen by experienced Game Masters.





The Heroic Players of Dragonforge

Each and every player is the hero, savior of mankind and destroyer of monsters!

Narrative

To this end, the story, quests and dialogue shall be written in a way to always place emphasis on the player as important to the goals and objectives in each story arc. Keeping in mind the player is so important and valuable that mundane and common tasks are beneath them. Need someone to slay 12 rats in a cellar? Nope, find some commoner or peasant, the player is busy with *hero* work.

Players can be good or evil

It's just a matter of perspective. A player can choose to be friendly, righteous, villainous, or even down right pure evil if they so choose. The story and interactions with the quests should reflect the player's good reputation or infamy all the same. In some instances, we may also wish to reward, punish for their deeds or even tempt to sway the player one way or another.

The ends justify the means

While players can openly attack and kill other players, sometimes this may necessitate the end goal of fulfill a quest for the greater good. In other cases, players that choose to attack and murder others will likely have a more challenging experience.

Player Hero limits

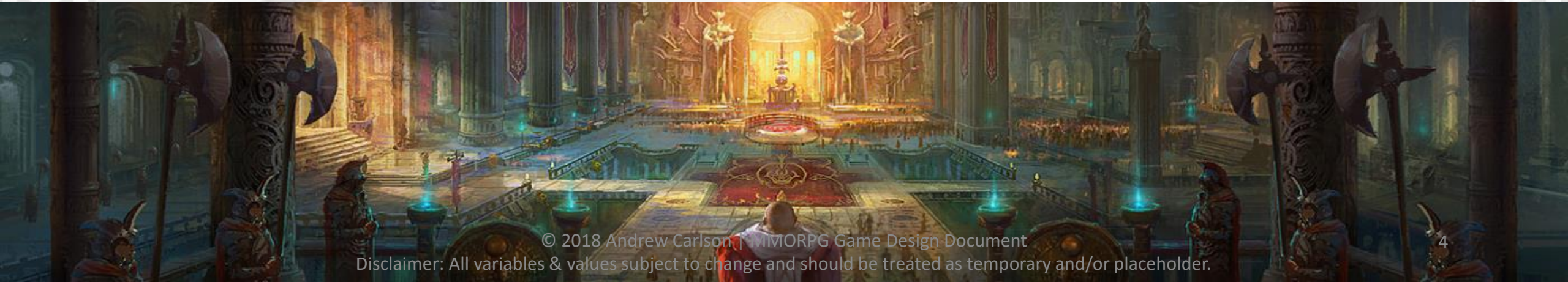
Ultimately, player characters are mortal and as such can still be slain. *see Combat section regarding Life and Death. The player will be able to continue as the same character even after death occurs as Dragonforge shall not have perma-death enabled.



Dragonforge, a city of commerce

At the center of this massive city is the legendary Dragonforge, a forge heated by the fiery fury of the council of dragons. The council consists of seven Elder Dragons, each a Lord and ruler of the lesser, outlying cities of men. The Dragonforge is the only forge capable of smelting some of the most powerful, rare and dangerous metals on the known to mortals.

All players are welcome to visit and make use of the Dragonforge, granted they meet the various requirements and complete a series of Story quests to unlock additional features of the forge. In the case of characters that are “flagged” and cannot enter the city due to bounties or being marked as a murder shall need to wait until their status changes to make use of the forge. Only neutral and good-aligned characters will be able to approach the forge.



Rule of thirds



Mobile Phones

The rule of thirds states that the bottom third is the *control region* (things that can be reached with single hand, usually with the player's thumb).

The middle region typically is where the player's eye focuses, so it is reserved for main visuals. Interactive elements here can still be reached with the player's thumb, but often the player will opt to use their index finger (aka forefinger) on their opposite hand while holding the device with their primary hand.

The top and usually side region(s) typically consist of the the "Head Up Display (HUD)" layer. Visual elements that rarely need to be seen, focused on or interacted with are commonly placed here.

Mobile Tablets

While the rule of thirds still applies to larger surfaces such as tablets, typically players still favor the bottom, middle and right-side of the screen for interaction.

Commonly for English-speaking players that read left-to-right, the left side (and top edge) is typically saved for informational displays and HUD elements.

Orientation

Portrait (vertical)



Why it matters?

Mobile users primarily utilize their device in one hand with portrait orientation. Supporting this orientation should be our focus for phone users as most users will interact with the game in this way due to sheer convenience and familiarity. In this orientation, users are typically restricted to using just their thumb and as such have limited reach, typically a single input gesture (tap or swipe) and slower reaction times.

Benefits of Landscape orientation

Due to limited screen space and display size, landscape view offers the best format for wide-screen content such as well as better supporting two-handed play. Furthermore, landscape view is the most commonly utilized and preferred orientation on mobile tablets.

Dragonforge will primarily only support landscape orientation, although a portrait orientation version or mode may become available at some later point in development.

Landscape (horizontal)



Login methods

New accounts can be registered in the following ways:



- **Email** (account activated upon verifying the activation link)
- **Facebook** (account activated upon authorizing)
- **Twitter** (account activated upon authorizing)
- **Twitch** (account activated upon authorizing)
- **Google (G+ or Play)** (account activated upon authorization)
- **Apple (iTunes)** (account activated upon authorization)

Services not listed above are not valid for signing, registration or account activation. Users will be required to validate & authorize their 3rd party service, app or account prior to completing registration and activation of their game account.

Edit Sign-in method, Auto-login and account info

Players can change their login settings, profile and manage their account from either in-game after signing in, or from the Dragonforge website (TBD).

Auto-login will be enabled by default on all new accounts. Users that disable auto-login will need to sign in manually with username/password. If signed in with a 3rd party application, the user may need to re-generate and verify an email activation link and sign in with a temporary password if they have not yet set one.

Login considerations

Multi-device users

In instances where a user wishes to play on different devices, they can simply login to any supported mobile device and play. However, doing so will automatically log them out of any other in-progress game sessions on other devices.

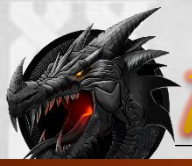
Players will have account settings available to lock or “bind” one device to their account at a time. As a security feature, this allows players to prevent logging in from any unauthorized device or prevent being auto-logged out if one of their other devices is being used and/or accidentally signs into Dragonforge. If a player “binds” the account, they can always unbind it from the website portal and/or in-game by logging into that bound device prior to switching devices and removing the device lock.

Multiple accounts

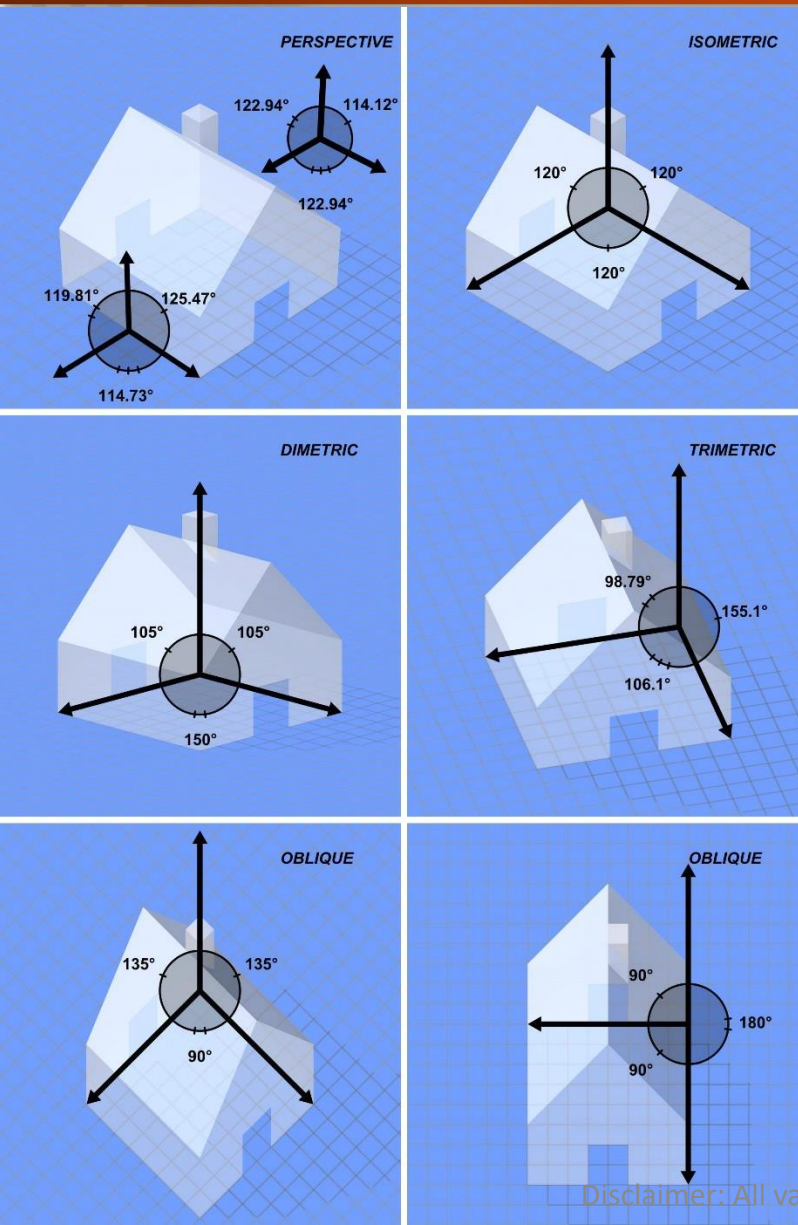
We will take no action against multi-device users as long as they are using unique accounts and one account per device.

Multiple characters

Players are welcome to have more than one character or “alts”. However only one character may be signed in on any single account at a time and Dragonforge servers will enforce this.



Camera & View



Dragonforge's ARPG gameplay style calls for *Perspective* projected camera view. While not true 45-degree isometric, this approximate camera angle allows for isometric projection in addition to use of skewed FOV and perspective which looks more natural to the human eye.

While unlike most 2D games that rely on strictly isometric 2D art or "tiles", Dragonforge is a fully 3D game with all 3D game objects with the exception of the GUI layer which is 2D. In practice, these structures and objects will be created out of Unity prefabs.



Example of perspective angle



Example of isometric angle

Camera & View



Diablo III; camera angle



Path of Exile; camera angle

Camera behavior is as follows:

The player is able to adjust the following view settings:

- Camera distance from the game's terrain (aka "Zoom")
- Camera facing (aka "Rotation") on the Z-axis only (cannot tilt (X-axis) or roll (Y-axis) the camera view), up to 45 degrees in either direction (no 360 degree rotation, to simply camera control and reduce level design work).
- Camera's Field of View (aka "FOV") between 60 and 90 degrees. Greater than 90 may have an unfair advantage in regards to the view offered to players.

Camera restrictions:

- Camera always points-at (looks-at) the head of the player's character, never pointed at the character's feet or base.
- The camera is a follow-camera and cannot be "unlinked" from following the character. It maintains its defined distanced always.
- The camera never shall switch targets or focus. It will always remain fixed on the player

Camera & View

Additional Camera behavior

- Camera shall always retain it's angle, FOV and position if adjusted by the player and “snap” back to this position/orientation after any cinematic, cut-scene or other instance where a camera view override is required. Upon logging out/in the camera will reset to default position, zoom, and orientation. The FOV setting shall be stored and restored upon re-entering the game world.
- The environments will be designed and built to support the camera's expected pathing and to avoid clipping or passing through meshes, objects and buildings. For this purpose we restrict the camera from being able to rotate a full 360 degrees.
- The camera will not have a collision and can float through solid objects as needed. This prevents the camera from being “stuck” or flipping out as the character (target) moves away.



Obstructed view

In any instanced when the camera is obstructed by a game object, that object shall be rendered semi-transparent and/or removed from being rendered as needed.

In the case of being obstructed by other players, monsters or smaller game objects no higher than the player's chest, these game objects/characters will not be made transparent to indicate that they are solid and cannot be walked through.

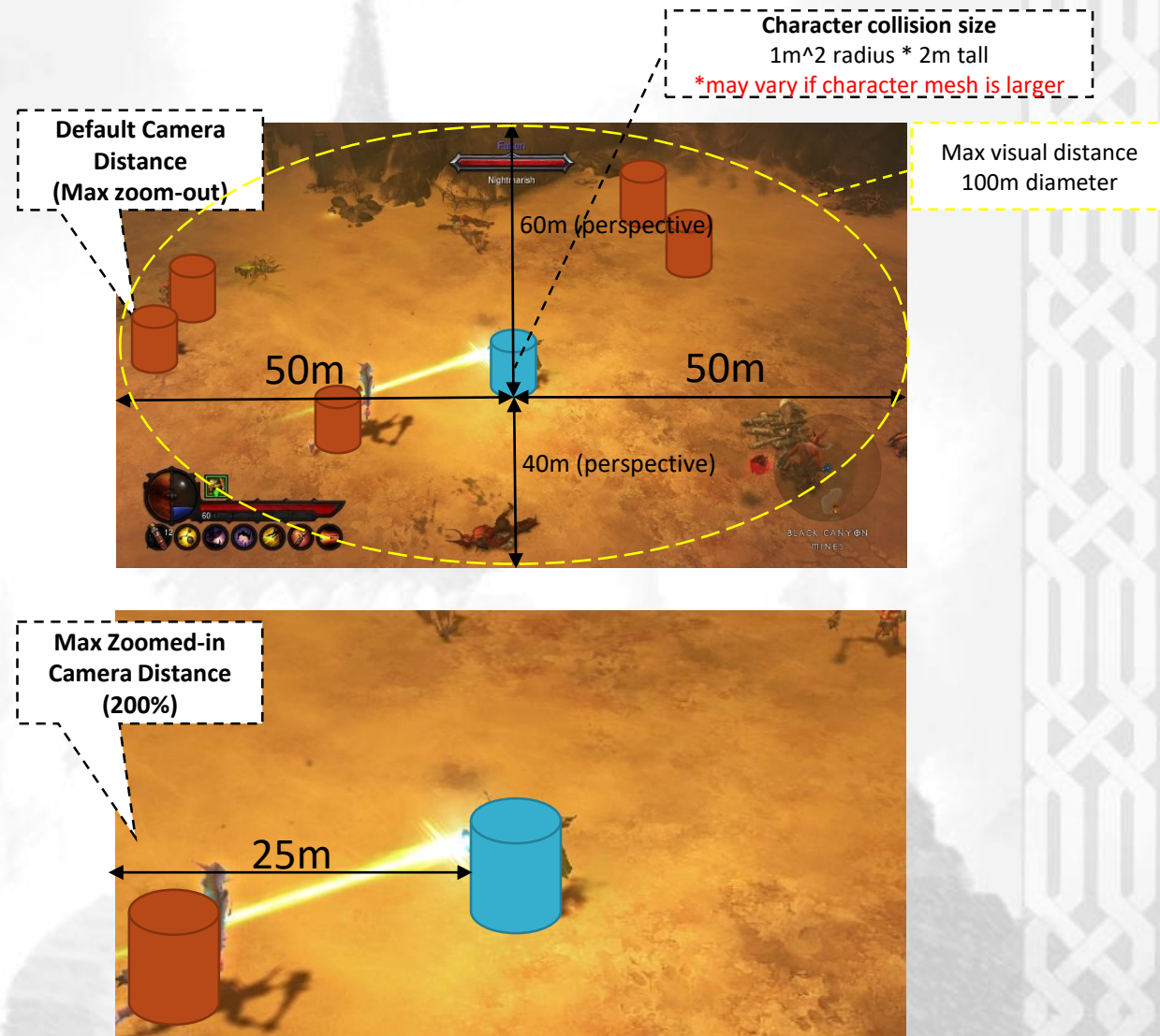
Only walls and other environmental objects will be made transparent within a small radius around the player's character, not the entire viewable screen-space.

View Distance & Scale



Distance is measured in meters (assuming Unity is using meters also)

- Character: 1m (3ft) radius footprint, about 2m to 2.5m (6ft-7.5ft) tall
- Max visual radius, approx. 100m (300ft) diameter
- **Sizes:** *These may vary:
 - **Smallest:** 0.5m footprint * 1m tall (goblin, dwarf, etc)
 - **Humanoid:** 1m footprint * 2m tall (skeleton, human, etc)
 - **Large:** 1.5m footprint * 3m tall (troll, ogre, etc)
 - **Very large:** 2m footprint * 4m tall (giant, boss, etc)
 - **Massive:** 4m+ footprint * 4m+ tall (dragon, titan, etc).
 - Vessel: 10m+ footprint * 10m+ tall (sailing ship, 2-masted)

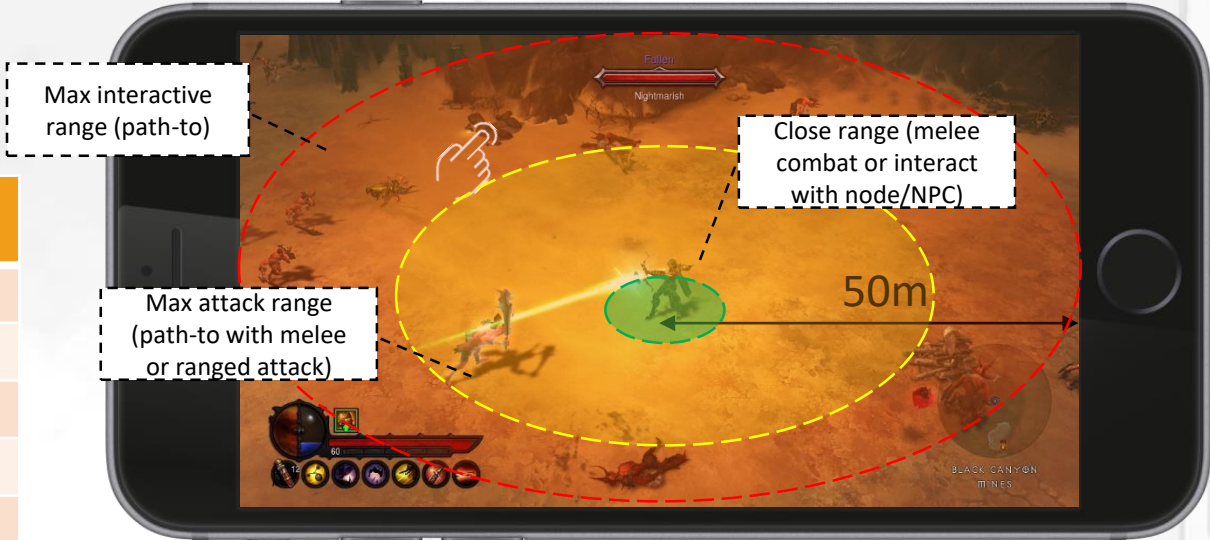


Movements & Traversal Speed

These are defined in meters per second, subject to change!

Type of Movement	Movement Amount *subject to change and skill proficiency
Idle	0 m/s
Walk	3 m/s
Run (default movement)	8 m/s
Sprint	12 m/s, limited duration (based on skill)
Horseback (gallop), Sailing	15 m/s
Flight & Falling	20 m/s
Teleportation / Portal	Load screen required if changing zones, otherwise instant within the current screen and/or zone with a 200ms delay before movement occurs (to buffer).
Pull/Gravity	-2 m/s upwards of -10 m/s in opposite direction. Pull strength TBD depending on circumstance and skills.
Projectile (arrow, fireball, etc).	10 m/s (variable), cannot outrun on foot.

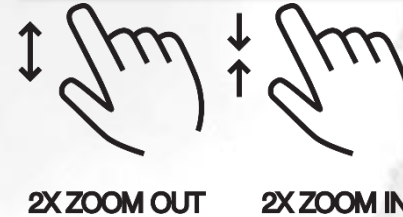
Development note:
 We may need to restrict the maximum movement speed allowed to reduce latency and to maintain a consistent framerate for combat/collision calculations. This is especially true with vessels, vehicles and mounts that are larger in size and will traverse more terrain requiring faster load times and draw distances.



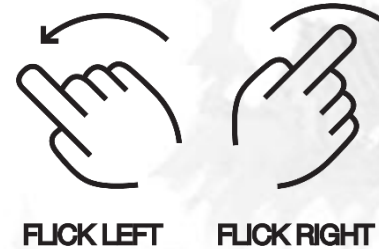
Distance Desc	Diameter around (meters) *subject to change and skill proficiency
Close Range (and melee), player	0.1m to 4m, variable. 0.1m minimum distance.
Minimum attack distance, player	1m, ranged attacks under this minimum do not hit or cause harm (projectile collision safety)
Maximum attack distance, player	25-30m, variable
Interact trigger distance, node/NPC	Player's close range and node trigger envelopes need to overlap by >50% to allow activation.
Maximum draw/visual distance	100m-150m, up to and including corners of view and perspective at max zoom-out.
Low-visibility draw/visual distance	Variable, such as lighting, fog, haze...etc. Minimum draw is 5m lit-radius around player.



View/Camera Controls



Camera Zoom: Using two fingers (thumb and forefinger), touch the screen then perform a pinch or spread motion to adjust the camera's distance from the character between the minimum and maximum zoom limits. (Snaps to the nearest 10%)



Camera Rotate: With a single finger, touch the screen and slide (swipe) quickly left or right to adjust the rotation angle of the camera. The camera will "snap" to 15 degree increments left or right up to 30 degrees in either direction from the default angle).

Example; Rapidly swipe to the left twice to rotate the camera 30 degrees to the left.



Reset Camera Angle & Zoom: To return the camera to default rotation & distance, tap the screen with three (3) fingers.

Touch-based Controls



TAP



DOUBLE TAP



2X DOUBLE TAP



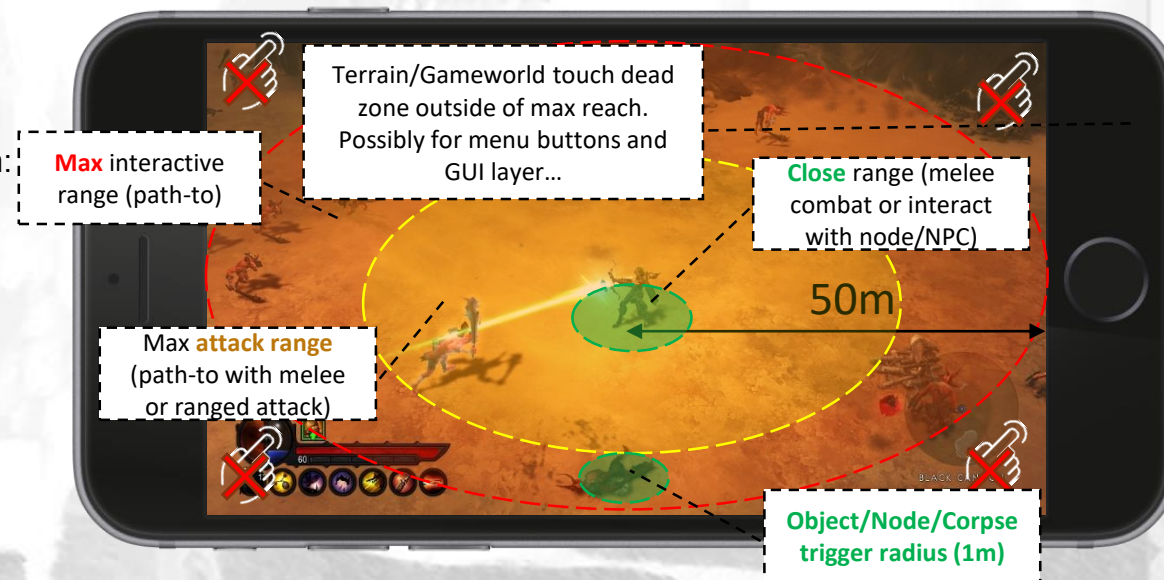
DRAG & FOLLOW

Movement (contextual, depending on what is tapped upon)

- Tap on empty ground/terrain: **Path** and **Run** to tapped point
 - D-tap on empty ground/terrain: **Path** and **Sprint** to point
- Tap & Hold on ground/terrain: **Path** and **Run** towards held point.
 - May slide/move held point for character to path and follow.
- D-Tap & Hold on ground/terrain: **Path** and **Sprint** towards held point.
 - May slide/move held point for character to path and follow.
- **Tap on interactive object:** **Path** and **Run** to **close range** (if able)
 - D-Tap on interactive object: **Path** and **Sprint** to tapped point
- (With Melee equipped) Tap on enemy to **Path** and **Run** to it.
- (With Melee equipped) D-Tap on enemy to **Path** and **Sprint** to it.

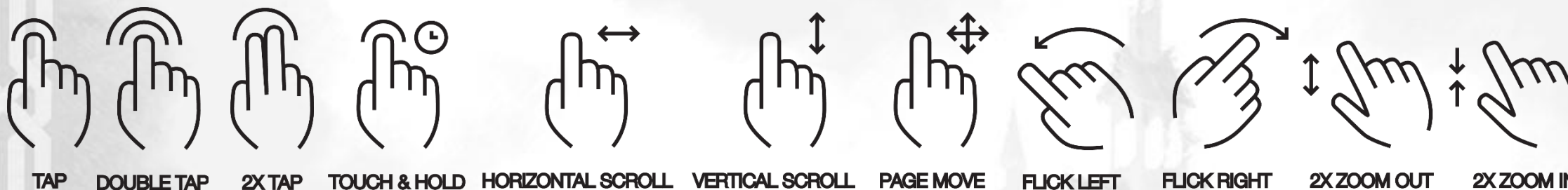
Interaction (within interaction range)

- Tap on Object, NPC or Node: **Path** and **Run** to object (if able), then:
 - **Object:** Pickup/collect
 - **Corpse:** Open loot UI (Tap and hold on corpse for quick-loot-all, when in range)
 - **NPC:** Speak and open dialogue UI
 - **Friendly Player:** Target/Select
 - (Tap and hold on friendly for more options).
 - **Node:** Approach and begin harvest
 - **Enemy:** Attack with ranged weapon/spell or approach for melee. *see combat section for details.
- D-tap will approach these elements by **Sprinting**.



Menu Interactions

For the GUI layer, we will utilize the following gestures:



Notes about gestures;

- 2x tap is a single tap using two fingers (not a double single-finger tap). 3x means three fingers, 4x means 4 fingers...etc.
- Touch & Hold will also be used for “Tap & Drag” as well.
- Horz/Vert scroll is different than a short swipe, it is used as a long, held swipe gesture for scrolling page content. These will be defined if it is a “swipe” or a “scroll” action on the GUI Layer.
- Horz/Vert swipe is a short and quick “flick” movement.
- Page Move is the same as “Free Scroll” or “Move”, if used. Can also be used in conjunction with “Tap, Hold & Free Move”.
- 2X “Zoom” simply means to pinch or spread with the thumb and forefinger (2 finger gesture).

Menu Actions



TAP

Tap

Makes a single selection. The primary navigational action and gesture used to make choices in the GUI. Buttons and other inputs should only accept and register a single Tap input and not mistake a Tap & Hold action as Single-tap.



DOUBLE TAP

Double-Tap

Toggle selection gesture, in the GUI a single tap may select something, where as a second tap will unselect/disable it. In the case of inventory,

Example; A player may D-tap any piece of equipment from their inventory to immediately auto-equip it to their character and replace the existing (if any) piece of worn equipment.

Example; The player D-taps on a stack of logs to select it. They may then D-tap to unselect that item (or single tap elsewhere).

Developer Note:

It is important that a Double Tap is not registered as two unique Single-taps. That could break GUI functionality.



TOUCH & HOLD

Touch & Hold

Used to drag and drop various elements (such as inventory items into various windows). But also used to conjure any related sub-menu to show More Options GUI.

* The selections within this sub-menu may vary item-to-item.

Example; A player may Touch & Hold on an stack of logs to view the options to split the stack, Craft something, view the logs properties/info or drop the stack entirely from their inventory.

Touch, Hold & Flick

Used to “grab” onto an item in the Inventory GUI, then upon releasing the finger after the Flick gesture, the item is dropped from the player’s inventory. A quick way to throw away or drop items.

*See inventory section regarding “dropping” items.

Menu Actions



VERTICAL SCROLL

Scrolling (Tap + Hold and Slide)

In GUIs that feature scroll bars (such as dialogue, descriptions or informational panels), a player may scroll the content up and down by tapping, holding and sliding up or down with a single finger.



HORIZONTAL SCROLL

The same gesture can be utilized in cases where content is wider than the display, however it is very likely that only the Vertical Scroll gesture will be utilized.



2X ZOOM OUT

Zoom (in & out), within a GUI widget

Spread (outward pinch), Change amount downwards in open GUI widget, if applicable.



2X ZOOM IN

Pinch (Inward pinch), Change amount downwards in open GUI widget, if applicable.

*Example; A player is placing coins in their bank account, spreading outwards **increases** the amount of coin being selected in increments of 1000. Pinching **decreases** the amount selected to be deposited.*



FLICK LEFT

Flick Left / Flick Right (aka Swipe Left / Right)

Using this Gesture will change the selected GUI panel and tab to the left.



FLICK RIGHT

Example; A player is viewing their "Player Inventory", to the left of that is the "Player Stats", simply swiping left changes the view to the "Player Stats" screen as if they tapped the tab/button to switch the GUI.



2X TAP

Single, Two-finger Tap

Short cut for "Cancel", "Close" or "Back". If applicable, may also activate "Accept and Close" actions. Useful for closing GUIs or going Backwards in menus quickly.



3X TAP

Single, Three-finger Tap

Short cut for "Close and exit" all GUIs, dialogs...etc.

Gamepad Controls

Dragonforge shall support paired Bluetooth 4.2 (or greater) gamepads.

The popular (unofficially supported, but sold via Apple.com) is the SteelSeries game controllers as they are Apple MFi supported Bluetooth game pads. (pictured here).

On Android, any compatible controller should work, however we shall only support up to 10-buttons inputs, with 2 directional sticks and a single d-pad.

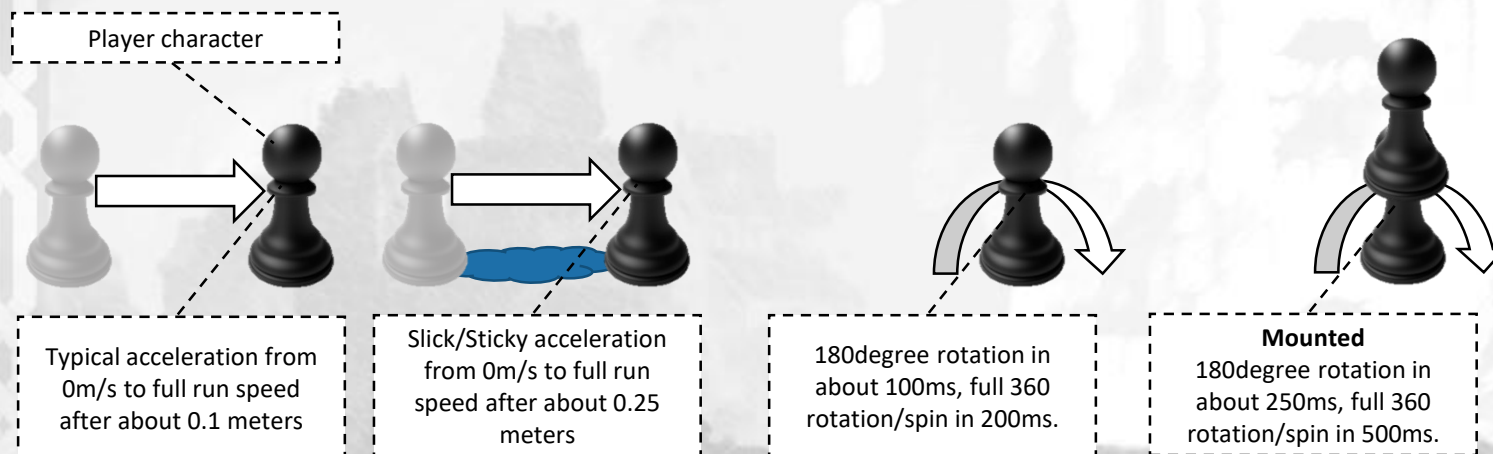
Gamepad Inputs

- **Left Analog stick** (8-directional); Move Character, navigate menus (L3 action not supported)
- **Right Analog stick** (8-directional); X-axis: Adjust Camera Rotation, Y-axis, Adjust Camera zoom distance.(R3 action not supported)
- **D-pad**; Cycles the “hotbar” or “tool belt” shortcuts related to selecting an assigned skill, spell, item or other hot keyed action.
- (A) button: Accept/Use or Interact (menus), Attack/Target closest (or weakest) enemy.
- (B) button: Cancel/Close or Quit
- (X) button: Attack/Target furthest (or strongest) enemy.
- (Y) button: Open Player Menu system (equipment, inventory, map, stats...etc)
- R1 button: Cycle to next target / Tab pages to the right (menus)
- L1 button: Cycle to prev target / Tab pages to the left (menus)
- R2 trigger: Cast/Use selected Skill, Spell or Item (after selecting it from Dpad)
- L2 trigger: Sprint (in direction held)
- Menu/Home (Start button); Open Player Menu inventory, directly.



Character Movement

- **Walking** – Player characters by default do not walk. NPCs and monsters will only walk when patrolling or if a pathing script requires walking.
- **Running** – All players by default traverse at run-speed.
- **Sprinting** – Players may optionally sprint for a short time, the sprint bar on the HUD will display the remaining sprint stamina.
- **Mounted vs unmounted** – Players may approach and “mount” rideable NPCs (horses for example), if owned by the player. The benefit to riding a mount is added maximum movement speed without the need to sprint while on foot. Acceleration is unaffected.
- **Turning** – All characters pivot (turn) *nearly instantly* (rotational speed TBD), this means that a character can rotate to face opponents without the need for a turn/transitional pivot animation, thus increasing reaction time. When mounted, this turning speed is reduced by half.
- **Acceleration** – All characters should have a subtle acceleration/deacceleration when moving from point A to B. When on some surfaces (such as ice, mud..etc) this value can be affected to simulate terrain traversal difficulty.



Mounts



A mount is any creature, monster or other single-rider vehicle that a player can interact with to ride upon to gain additional movement & travel speed. Mounts do not have offensive abilities (cannot attack) and players must dismount to perform combat.

Mounts are restricted to being mainly a means of travel in Dragonforge (1.0.0).

Players are allowed to approach and interact while mounted, such as speaking with NPCs, shopping and trading. However specific actions such as crafting, harvesting, combat and such require a player to dismount. Engaging in one of these “dismount” actions will automatically dismount the player. *Some mounts may be faster/slower than others. *see data tables for details.*

Mounts can be killed/destroyed as with any NPC game object and are treated as if you attacked a player themselves.

Vessels

Vessels are classified as massive, movable objects capable of carrying more than one player and have lots of storage space. While similar to Mounts, these vessels will be capable of traversing at greater speeds, over water, through the air and are immune to attack unlike individually owned Mounts.

Vessels may also act as mobile player homes, however that feature is yet to be determined.

Vessels will make use of pre-defined paths or “lanes” in which they can travel between docks, ports and other specific locations. Players do not steer vessels as they would a Mount and they travel automatically once a path is selected.



Player actions

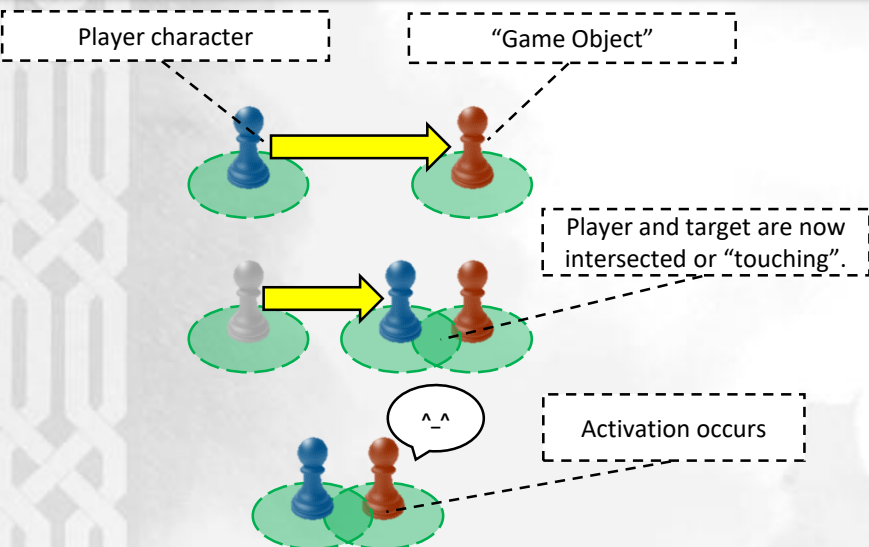
Each game object will have a specific “action” tag associated with it that affects what a player may do when they “interact” with it. Interacting with a game object involves approaching the game object and tapping once. Once a game object has been activated, specific skills then govern the result of the interaction. Some game objects may require specific skills or skill level to activate.

For example: A player taps on an Iron Ore node (game object). The character will approach the node and perform a skill check if the player’s Mining Skill meets the required skill level. If yes, the character then begins mining and harvesting the ore from the node. If no, the character does not begin harvesting, be warned they do not meet the skill level needed and will stand idle until commanded to take a different action.

Typical Common actions (list subject to change):

- Talk to NPC (for info/quests)
- Shop with NPC
- Theft from NPC (or player)
- Target (select) game object
- Inspect game object’s properties/info
- Attack monster, NPC or player
- Perform skill, spell and/or ability on target
- Begin collection/harvesting from node
- Activate crafting station or appliance (such as campfire, oven, forge...etc)
- Activate switch, button or lever
- Use Furniture (such as open storage chest or close door)
- Mount/Dismount
- Perform emote
- Trade with player(s)
- Loot corpse (or container)

Performing actions



Common activation:

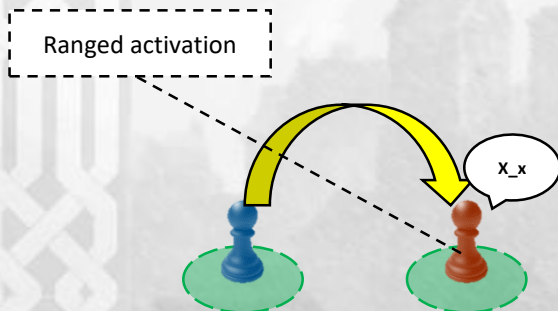
1. Player taps (or d-taps) to approach object.
2. When both player and object intersect with their “close range” collisions, they are considered “touching”, in which case pathing & movement ceases.
3. If both intersection and pathing is stopped, the “activation” commences after a short 100ms delay (latency/server safeguard to ensure pathing is completed)

Mounted activation:

1. Player (mounted) taps or (d-taps) to approach object *while still mounted*.
2. If “*dismount to activate*” = true on the game object properties, this decides that when “intersection” occurs between the player’s mount and game object, player ceases movement and dismounts, then finishes pathing 0.5m to the object on foot to ensure dismount has valid spacing. Otherwise the player will remain mounted and activate the game object upon intersection as usual while still mounted.
3. Activation occurs as usual.

Ranged activation:

1. Player (mounted) taps or (d-taps) object as usual, however if a ranged weapon, ability, skill or spell is first selected that allows ranged-activation, the player will move to the maximum activation range distance (variable). (*example; casting a healing spell at a player 50m away.*)
2. Once within maximum range distance, the activation occurs. If mounted, the player may be required to dismount first prior to skill/spell or ranged activation occurs.



Targeted Actions

What is an targeted action?

This is any action that requires the player to activate a skill, hotkey, ability or spell then select a valid target for that action.

For example #1; Ken taps his “healing spell” icon from his hot bar and is promoted to select a valid target to cast the spell upon if no valid target is already selected. Ken then taps Robert, a friendly-allied player, the spell is then cast and Robert is healed.

Another example; Ken has an enemy player selected that he is currently dueling. Ken presses his “healing spell” icon, however the game will not heal a hostile target that is selected, instead the spell will queue and wait for Ken to select a target to cast the spell upon.

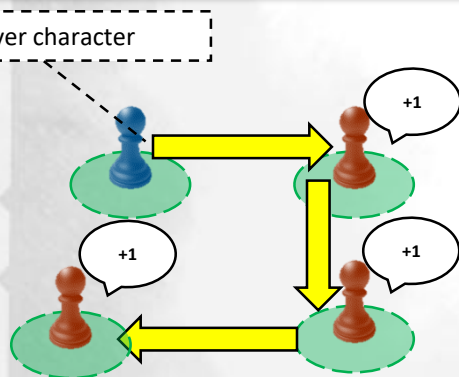
In the case of spells no magic power will be consumed or items that consume charges or ammo, and no action will be taken until the valid target is selected. Meaning to say that no spell is allowed to “pre-cast” or be prepared in anticipation to be applied in advance. The action, animation and resource(s) are not deducted from a player until the target is chosen and action is performed and/or completed.

Selecting targets and cancelling a targeted action

When an action requires a target, the player simply needs to single-tap on the target to select it (if valid) as they normally would for any other activity. The system message will read “Select Target”. If an invalid target is selected for the chosen action, then the player will receive an “error” sound effect and system message that reads “Invalid Target” and the player can tap again to choose another target selection.

However, if the player decides to cancel the spell, action or ability before a valid target is chosen, they simply need to double-tap on the screen to cancel the targeted action. The system message will then read “Action cancelled”.

Auto actions



What is an auto action?

When enabled, the player will select the type of auto action for the character to perform. Such as, path, harvest and drop off resources or repeat some action several times to increase skill levels or produce goods or materials. An auto action is a pre-defined, supported and built-in macro of sorts.

On the primary HUD we will have a pulldown/button that allows the player to set the type of auto action they wish to enable. Tapping this button will toggle the auto action off while it is in progress.

Duration and timers

Auto actions will have default time limits and durations in which an action can be unattended for before requiring to be “restarted”.

In most cases this time allotment will be plenty for most casual players to set their tasks to auto perform and check in from time to time (between 1 to 3 hours). However, we do not want to completely automate everything, so some actions will automate up to a point where the action then may require manual player confirmation to proceed to performing a batch-craft or consumption of rare resources as a safeguard against undesired results performed by the auto action system.

Leave some for everyone else

Because many players will be utilizing the auto-action feature, auto-harvesting will never gather 100% of resources from a single node, but rather harvest up to half of the available resources from a single node and then move onto the next node (allowing others to also harvest from these also). Likewise if a node is exhausted, the auto-harvest action will know also to skip a harvest attempt and seek another node. Nodes that are depleted from an auto harvesting player will regenerate/regrow much faster than those hand-harvested manually.

Example; Robert sets his character to auto-harvest logs from trees for 1 hour. His character will then remain active and path automatically from tree node to tree node, stopping at each to extract 50% of the available resources of that node before moving onto the next node. Upon completing this 1 hour of work, the character will stop and log-off automatically unless the player resets the auto-action.

Efficiency of auto-actions

Naturally as auto-actions are performed, the skills that govern the actions need to calculate how well the action is carried out and return the results when a skill is performed. However, any and all skills operate at a cap of 75% efficiency when used with the auto-action system. This means that inherently, any player that is manually playing gains a fair advantage over the “bots” and macroing players.

While the auto-action system allows for casual play and quality of life access, it does not completely negate the player’s ability to compete if they decide to pick up and play the game directly. Players that manually harvest resources will not only gain skill XP faster, but they will also accumulate more resources in a shorter amount of time as well as better quality resources.

- *Example A; Jim’s character is auto-harvesting iron ore for 1 hour, his base Mining skill is 100. Because of the auto-action system, when he auto-mines iron ore, the skill is calculating the action as if it were a Mining skill of 75, which slows down how fast the skill works and how much ore per harvest he earns.*
- *Example B; Steve’s mining iron ore nearby Jim, but because he is mining manually, his Mining skill of 100 is unaffected, thus resulting in collecting more ore per harvest and slightly faster than Jim due to the calculations. After one hour, Steve would have more ore than Jim and will have gained more skill XP than Jim also.*

Limitations

The auto-action system will require the player to have the tools, materials and appropriate skill level for the auto-action desired. If, for example, a player breaks their only pickaxe while auto-mining, the auto-action will cease and the player alerted their character requires attention. Likewise if the player is auto-crafting arrows and runs out of feathers in their held inventory, the action will cease. The auto-action system will not attempt to seek out to replenish supplies, tools or materials.

Furthermore, the auto-action system will not allow players to buy/sell, trade or interact with NPCs in a complex manner (such as auto-complete quests). While we will want the auto-action system to sort or “drop-off” inventory, this will need to be limited to a player’s bank, inventory or secured container.



Combat auto-actions

A player will be able to enable auto-cast, auto-attack and auto-renewal actions during combat as well as configure these auto-combat features with simple filters. To toggle on an auto-combat action, the player will simply tap and hold on the action's icon from the hotbar and set it to "auto" from within the sub-menu, which will then enable this action for all combat

Simple Filters (not a final list)

Filter	Description of what this combat filter does
TOGGLE Auto-mode	Simply enables/disables this specific action to automatically target and fire on valid targets depending on the set combat filters below.
Target Filter	Sets the behavior for selecting a valid target for this specific action within the player's line of sight and appropriate range. Detection Settings: Target closest, furthest, weakest, strongest Type Settings: Target monsters first, last, only; Target players first, last, only. Target Mode: Manual, Auto-monsters, Auto-players, Auto-All (<i>controls the target system overall</i>) Target Search: Held, Scan, Players-only, Monsters-only, Smart (<i>Held means the target selected and held until death or lost LOS (which is good for offensive-focused actions). Scan means once the action is taken, scan for a new target immediately (useful for support actions). Smart (default) will determine if the target lock should be held or released to rescan, useful for switching between healing and fighting as some targets may need multiple attacks or heals before it is no longer a valid target for the action.</i>)
Approach Filter	Determines auto-pathing action of this action. Useful to prevent a character from rushing into combat foolishly. Distance-to-target Setting: Maintain maximum range, minimum range, force melee range always, auto range. (<i>auto = allow all ranges</i>) Chase Target Setting: Pursue target to maintain distance? True/False (<i>if false and the target leaves LOS, character will re-target</i>).
Behavior Filter	Allows the player to set the attack parameters for the action. Priority: Support, Standard, Warfare (<i>Support gives auto-action favor to healing and buff abilities, where as Warfare will favor offensive abilities</i>) Aggression: Low, Normal, High (<i>Low aggression begins combat with lower-damage and faster attacks first, saving larger attacks for last. High means the first attack will be the player's strongest, followed by their weaker attacks</i>).
Self-preservation	Self-preservation: Off, Low, Normal, High (<i>Off means that a character will fight to the death, setting this to High means the character will more likely attempt to flee and disengage from combat if mildly wounded</i>).

Combat (Players)

Combat Attributes

this section is full of temporary and incomplete variables! Be warned, here be dragons, literally

In Dragonforge there are no derived attributes, for example, *Constitution* that determines how much health a player has, or strength that determines damage output. Instead, all variables are simplified and straight forward. Each of these can be increased through various skills, equipment, enchantments and temporarily altered with potions and magical buffs/debuffs.

Player Attributes

- **Health Pool (HP)** – A character's life force, when this reaches 0, the character is defeated and dies (see section regarding death and resurrection). May regenerate over time. Players begin with 1000 HP. Players cannot increase health through “leveling up”, but can affect it with various armor prefixes and enchantments.
- **Stamina Pool (STAM)** – A linear measure of a character's physical ability to perform actions. Sprinting drains stamina as does using skills and performing attacks. If this reaches 0, a character cannot move nor attack and may be more vulnerable to incoming damage. May regenerate over time. Players begin with 100 STAM
- **Mental Power (MP)** – A pool of energy in which a spell caster draws upon, which also represents a character's mental ability to perform actions. Some skills may drain MP as well as STAM. When this reaches 0, a character will be unable to perform most skills and only very simple actions. May regenerate over time. Players begin with 100 MP.
- **Timing (aka Attack Speed) (TIME)** – Controls the frequency and animation speed of how often attacks occur. This is a variable governed by skill, weapon's speed & size, buffs, stamina and more. The default is 1.0.
- **Movement Speed (MOVE)** – Governs the speed of which a character travels on foot. This is adjustable in some circumstances (through skills, buffs or equipment), otherwise player's movement speeds are constant variables.
- **ATTACK Rating* (ATK)** – This variable represents the *estimated* potential of outgoing damage on a neutral-affinity target (melee vs melee) with the currently equipped weapon (and any alterations). Magic Damage displayed will be similar. Damage is reduced by target's defensive rating, skills, buffs or other aspects.
- **DEFENSE Rating* (DEF)** – This variable represents the *estimated* potential of mitigating incoming damage. The higher the value the more likely the character will dodge, block or absorb the damage received. This is determined from all worn pieces of equipment, buffs, skills and such. However when determining damage calculations, we defer to the hit location table. This is a derived average from all armor worn armor slots (all protection values).



Combat Affinity

All combat in Dragonforge is comprised of a series of checks and balances, all based on “paper-rock-scissors” style of gameplay. In this system, all combat has a strength, neutral affinity and a weakness. Skills influence and govern these calculations and determine things like hit chance, damage dealt & damage received as well as other aspects of combat.

Primary Combat Affinities:

- **Melee** – This includes all weapons or means of dealing/taking damage within the “close range” radius. Melee combatants are tougher than other primary methods of combat. Melee players can further specialize by gearing and skilling their character towards being proficient at dealing damage (DPS), heavy defense and durability (TANK) or taking a more balanced mix of the two.
- **Ranged** – While ranged combat can also take place within melee range, the effectiveness of the attack launched within the melee range will be reduced. Typically a ranged character will want to avoid engaging at close-range to minimize risks during combat. While spell casters also have ranged capabilities, purely ranged characters are superior in regards to mobility and more effective at maintaining that range. Ranged characters can specialize between DPS, Range or Mobility.
- **Magic** – Spell casting is a mix of both melee (close range) spells and spells that can be cast from a distance. Magic users also abide by similar limitations as melee or ranged users in that their magic power (MP) restricts the spells that can be cast. While magic users can cast distance spells within melee range, they will be vulnerable to attacks of opportunity by doing so. Spell casters can specialize between Power, Duration or Efficiency.



Inherent Weaknesses

- Ranged attacks are best against Magic users, but weak against Melee. Ranged attacks are neutral against other Ranged characters.
- Magic attacks are best against Melee users, but weak against Ranged. Magic attacks are neutral against other Magic characters.
- Melee attacks are best against Ranged users, but weak against Magic. Melee attacks are neutral against other Melee characters.

Development note – Weapons, armor and other attributes will incorporate these weaknesses inherently. The amount of the weakness or strength will vary.

Combat (Players)

Combat Attributes, continued

this section is full of temporary and incomplete variables! Be warned, here be dragons, literally

More Player Attributes

- **Hit Chance %** – The chance that that an attack will connect with the target and wound them. This will greatly depend on the player's skill, their equipment, enemy's attributes as well as any applied buffs. This calculation is done automatically for each attack attempt between players and enemies. This value can exceed 99% when modified!
- **Player Hit Locations** - Whenever an enemy attacks you with any attack or skill, an area on your body will be randomly selected to take the hit. The chance for each area to be hit is: (NPCs do not have hit locations, all hits on an NPC land in the center torso/chest).

Body Location:	Base chance to hit specific location:
Head	15%
Torso/Back	35%
Arms/Hands	28%
Legs/Feet	22%
Rings/Amulet/Belt/Other	0% (cannot be hit)



Upgrading your equipment is equally important across all armor slots. If you have low protection in a slot, you are vulnerable to that body location.

Critical Chance % – The chance that after a hit occurs, if it will increase the damage dealt to that location. This cannot exceed 100%.

Critical Modifier – The amount the critical damage is multiplied by. The critical modifier is increased from equipment, skills and buffs.



Combat (Players)

Armor

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Armor Protection & Efficiency

Armor has 2 primary attributes a player sees:

- Protection Value (PV) – The amount of damage the armor segment mitigates. (this value is dynamic based on base armor, skill, durability..etc)
- Durability (DUR) – The amount of health and overall effectiveness the armor has left.

Armor skills affect how well the equipment mitigates damage (per location). While any player can wear any armor, it's base effectiveness is governed by the related skill. The effective protection value (PV) can exceed 100% of the base protection value the armor offers (which then require de-buffs, weaknesses and other ailments to bypass the armor's increased defense).

Protection calc: $(\text{PrimarySkill} * (\text{basePV} * \text{armorDur}\%)) + ((\text{SecondarySkill} * (\text{basePV} * \text{armorDur}\%)) * 0.5) * 0.1$ = mitigated damage.

With a minimum of 5% of total basePV at 0 skill level.

Example #1:

Jeff is wearing a chest plate that has a base protection value (PV) of 100.

The armor is at 80% out of 100% durability (20% damaged). Let's assume this is 800/1000 total durability.

Jeff's Heavy Armor skill is 50 and his Defending skill is 100.

Because the armor is only at 80% repair, it suffers a -20% base PV reduction on both skill calculations, meaning it's effective base PV is only 80 (instead of the full 100).

*The calc would look like: $((50 * 80) + ((100 * 80) * 0.5)) = 8000 * 0.1 = 800$ total damage mitigated if he were hit in the chest.*

Getting Hit Example A:

Jeff takes a hit to the chest from a Fire Elemental for 936 damage. His armor calc above would mitigate 800 of that $(936 - 800 = 136)$ damage to Jeff's health). 5% of that mitigated damage is removed from the armor's durability $(800 - 40 = 760)$ out of 1000). Jeff's chest armor is now at 76%, further lessening his protection on the next blow.

Getting Hit Example B:

Jeff takes a hit to the chest from a Small Goblin for 90 damage. His armor calc above would mitigate up to 800 of that $(800 - 90 = 710)$ damage still needed to harm Jeff's health). In any case where the armor protects fully, the armor takes durability damage equal to 20% of the incoming blow. In this instance, that would be 18 damage to the durability of the armor and no health lost to Jeff). Repeated blows seek to wear down the armor and eventually allow damage to Jeff's health as the armor begins to fail and become heavily damaged.





Combat (Players)

Durability

this section is full of temporary and incomplete variables! Be warned, here be dragons, literally

Weakening the armor

As armor locations are hit (regardless if damage was dealt to the player's health pool or not), the armor in that location loses durability. As the armor's durability is lost, the total protection value (PV) is reduced. When an armor location reaches zero (0) durability, the equipped piece of armor is rendered "damaged" and is automatically falls off the player (ejected) from the equipped slot and returned to the player's inventory, leaving an empty and unarmored section on the player's body. By default, we shall assume that all armors degrade at a value of 1.0 (affected by skills, etc). When armor received incoming damage, that damage deducts from the armor's durability value instead of the player's health pool.

Increasing maximum durability

Maximum durability increases as the player gains additional skill as well as reinforces their armor with various affixes, bonuses, materials and enhancements. The basic calculation for durability is: **((DefendingSkill * 0.5) * baseDur)** when the armor is worn. Any further durability enhancements stack after this calculation. When the armor is not worn, it merely shows the % status of the armor, such as 85% out of 100% durability. When this is then worn, the calculation is used to determine the precise number of what the armor's hit points are at and the % of damage is deducted from that total.

Jeff picks up (but does not wear) a helmet with 50% durability remaining on it, it has a basePV value of 5 and a baseDur of 10. When Jeff wears this helmet, the armor calculation then takes Jeff's Heavy Armor skill of 50 and his Defending skill of 100 into account to recalculate the helmet's current hit points and adjusted protection value (PV).

*Example: $((100 \text{ defending skill} * 0.5) * 10 \text{ baseDur}) = 500 \text{ maximum durability}$. Since the helmet is at 50% repair, it shows as 250/500 (50%).*

Naturally this helmet's PV value is also calculated from Jeff's armor skills when worn also.

*$((50 * 5) + ((100 * 5) * 0.5)) = 750 * 0.1 = 75 \text{ damage mitigated at } 100\%$.*

*Since its at 50% repair, then we deduct that from the calc, so $75 * 0.50 = 37.5 \text{ damage that can be mitigated if hit in the head with this specific helmet when worn}$.*

A troll then hits Jeff in the head for 250 damage, this helmet mitigates 38 (rounded up), and Jeff's health pool loses 212 hp. The helmet takes 5% (12.5, 13 rounded up) of this incoming damage to it's durability, and reducing it from 250/500 (50%) down to 237/500 (47.4%)

Repairing armor

The more damaged armor becomes, the harder it is to repair. However armor with 0 durability is never destroyed and removed from the game entirely, however it also cannot be equipped and thus offers zero protection. When armor is repaired, maximum durability is lost (this will vary on material, tools, repair skill..etc). If a piece of armor reaches zero (0) maximum durability, the armor is then completely destroyed and deleted from the game. Repairing armor may require materials and NPC vendors can only repair some types of armor up to a limited value. No common set of armor is meant to last forever, however in some special cases, armors designated with "Indestructible" suffix/prefix shall never loose duration when damaged and cannot be repaired (will always register as having max durability).



Combat (Players)

Resistances

this section is full of temporary and incomplete variables! Be warned, here be dragons, literally

Affinity Resists

- Ranged attacks are best against Magic users, but weak against Melee. Ranged attacks are neutral against other Ranged characters.
- Magic attacks are best against Melee users, but weak against Ranged. Magic attacks are neutral against other Magic characters.
- Melee attacks are best against Ranged users, but weak against Magic. Melee attacks are neutral against other Melee characters.

By default and upon character creation, a player receives a native flat 20% reduction from all damage (incoming) from other players (and NPCs) of the opposing affinity type you are strong against, regardless of attack, spell or other means of inflicting harm. This damage reduction is applied to raw damage before hit location, block chance or any mitigation calculations occur.

Resists and “vs.” Modifiers

All resist and “vs.” modifiers are applied to the final damage calculation & prior to damage mitigation being assigned and resolved.

Example: A fire elemental physically hits Kevin for 100 heat damage to his Torso, Kevin's torso armor has 15% Resist Heat affix. Before damage is mitigated, the heat damage is reduced by 15% down to 85 damage, then mitigation occurs (if no other resistances are applicable).

Let's assume Kevin also has 10% physical resist on the same armor as well, because the fire elemental attack was a physical attack as well as heat, the fire elemental's damage is reduced in total by -25%, so instead of having to mitigate the full 100 damage, Kevin's armor only has to mitigate 75 total damage.

Likewise, when an attack occurs (outgoing damage being dealt), any damage modifications are applied to the raw outgoing damage value, prior to mitigation.

PvP example:

Jim with a magic affinity is attacking Kevin with a melee affinity. Because Magic is strong against Melee, Kevin is going to suffer an additional +20% to all damage taken by Jim. This occurs prior to the mitigation phase. Jim hits Kevin with a fireball, dealing 100 damage to Kevin's leg. Automatically, that fireball is given +20% damage due to the affinity bonus, raising it to 120 damage.

Kevin however is wearing leggings that grant +10% Heat resist and +10% Resist Spell Damage which reduce that 120 total damage down by 20% (-24 dmg), which actually lowers the damage down to 96, which his armor then mitigates 80 and he takes 16 total damage to his health.





Shields

How shields work

Unlike armor, shields do not have a “hit location”. Instead, when equipped, shields have an equal chance to mitigate damage from any hit location when an incoming attack connects. This is referred to as “Block Chance”. Which is a flat chance that the shield will bear the force of the attack rather than the armor location.

Order of operation:

1. Determine if incoming attack “connects” and a hit is registered
2. Check what location the attack would land upon
3. Determine if a block intercepts the incoming attack.
4. If yes - the shield intercepts the attack, mitigates damage to the shield first rather than to the armor location as normal.
5. If the incoming attack damage is not fully mitigated by the shield, the armor location then is checked against the remaining “damage overflow” from the attack and treated as normal before dealing damage to the player’s health.

Block Chance %

The chance to block increases with better shields as well as scales with the Shield and Defending skills. A shield’s durability does not affect it’s chance to block an incoming blow. However if a shield is heavily damaged, it will mitigate far less damage.

1. **BlockChance calc:** $((\text{ShieldsSkill} * 0.01) * \text{baseBlockChance}) + (\text{DefendingSkill} * 0.01) = \text{maxBlockChance}$, **capped at 95%.**
2. **Then we re-calc based on the shield’s item Tier value:** $(\text{maxBlockChance} + \text{itemTierLVL}) = \text{finalBlockChance}$, **capped at 95%.**

Damage Mitigation

If the shield does not block the incoming blow, damage is mitigated to the armor location as usual. When a shield does block a blow, the shield’s PV value will determine the damage mitigated as if it were a “first layer” of armor.

Protection calc: $(\text{ShieldSkill} * (\text{basePV} * \text{armorDur\%}) + ((\text{DefendingSkill} * (\text{basePV} * \text{armorDur\%})) * 0.5) * 0.1) = \text{mitigated damage}$.

With a minimum of 5% of total basePV at 0 skill level.

Example #1:

Stan is using a Kite shield T4 that has a base protection value (PV) of 14. Base Block Chance of 4.5.

The shield is at 80% out of 100% durability (20% damaged). Let’s assume this is 800/1000 total durability.

Stan’s Shields skill is 500 and his Defending skill is 400.

Mitigation Calc: $((500 * (14 * 0.8) + ((400 * (14 * 0.8)) * 0.5) * 0.1) == 784.0$ damage mitigation on next hit, if shield blocks.

Block Chance Calc: $((500 * 0.01) * 4.5) + (400 * 0.01) + (4 \text{ item tier level}) == 30.5\%$ chance to block with the shield per hit, this is unaffected by durability % remaining.

If a Ogre hits Stan for 850 damage, Stan successfully blocks with his shield, the shield mitigates 784 damage, 66 damage spills over and hits Stan’s torso, in which the remaining 66 damage is mitigated by his chest armor. Stan’s shield durability takes 5% of the original 850 damage from the Ogre (for 43 damage), reducing his shield from 800/1000 down to 757/1000 (75.7%).





Example A: $((200 \text{ shields skill} * 0.01) * 2 \text{ baseBlockChance}) = 4$. Then add $(150 \text{ DefendingSkill} * 0.01) = 1.5$, total maxBlockChance is 6.5%. The shield's item tier level is 2, which we recalc as $(6.5 + 2 \text{ item tier}) = 8.5\%$ chance to block with such a low level skill and poor shield.

Example B: $((400 \text{ shields skill} * 0.01) * 6 \text{ baseBlockChance}) = 24$. Then add $(350 \text{ DefendingSkill} * 0.01) = 3.5$, total maxBlockChance is 27.5%. The shield's item tier level is 4, which we recalc as $(27.5 + 4 \text{ item tier}) = 31.5\%$ chance to block with average level skill and decent shield.

Example C: $((1000 \text{ shields skill} * 0.01) * 8 \text{ baseBlockChance}) = 80$. Then add $(1000 \text{ DefendingSkill} * 0.01) = 90$, total maxBlockChance is 90.0%. The shield's item tier level is 10, which we recalc as $(90 + 10 \text{ item tier}) = 100\%$ chance, however we cap this at 95% to allow a 5% chance for the shield miss a block at maximum skill and tier.



Combat (Players)

Weapons & Attacking

**this section is full of temporary and incomplete variables!*

Weapon attributes

- Base Damage is the default attribute every offensive weapon, spell or ability has. This value is then modified by the player's skills and buffs (excluding durability) to calculate the Attack Value (AV) when damage is dealt.
- Accuracy (ACC) is another base weapon attribute that is affected by skills and other attributes (including the target's skills). This determines how easy the weapon is to wield in combat. Higher skill with this weapon type will increase this value. Weapons with higher ACC values are easier to wield and require less skill to land a blow.
- Crit Bonus (CRIT) determines the percentage the weapon will use to determine if a hit is upgraded to a critical hit.
- Speed/Size – This represents how fast a player can swing, use, shot or wield the weapon (and is affected by skills/buffs). The greater the size, typically the slower the speed and vice-versa. Speed is displayed as Very Slow, Slow, Average, Fast, Very Fast. Size is displayed as Tiny, Small, Medium, Large, Extra Large.

Example; Kim wielding a Long Sword (ACC 40), with Tactics (20), Focus (30), Blades (40) and Anatomy (10) against a Skeleton with a DV of 160.

160 DV – 40 ACC = 120 CV (combat value). This is the expected value to meet or exceed to be successful. Kim's combat skills add up to 20+30+40+10 for a total of 100 CV. 100 CV / 120 CV = 83.33% Chance to Hit the skeleton successfully and deal damage.

The sword's CRIT is 20(%) of the chance to hit of 83.33 = An additional 16.66% chance that this HIT will become a CRITICAL HIT and deal additional damage. (once a successful hit has been determined, damage calculation then is figured out).

For example: Jack's Steel Sword has a base of 50 damage and ACC 50. Jack's a swordsman however, so his skills and buffs increase these values during combat calculations.

Durability, Ammo/Shots & Charges (aka USES)

Weapons lose durability differently than armor, a weapon's durability is effectively how many times the weapon can be used rather than a "pool of health" like armor. Weapon durability loses 1 use/charge per hit (durability is not deducted if weapon fails to hit target). However for wands, bows and other weapons with charges or ammo, a charge/shot is deducted regardless if the weapon hits or not.

- Also unlike armor, a weapon's available uses/charges/shots do not affect its combat effectiveness. Each hit will always deal the appropriate unmodified damage value.

Example; Jack's Steel Sword has 50/50 uses remaining. When he enters combat against a goblin and slays it with 4 blows. His sword's uses would be at 46/50. His sword would still deal full damage on each hit down until the sword becomes unusable (broken) at 0/50 uses.

*Uses is displayed as both a current/max value as well as a %. Like this: **Steel Broadsword Uses: (85/100) (85%)***

Combat (Players)

Weapons & Attacking

**this section is full of temporary and incomplete variables!*

Damage Calc

Base Damage is the default attribute every offensive weapon, spell or ability has. This value is then modified by the player's skills and buffs (excluding durability) to calculate the Attack Value (AV) when damage is dealt.

- Base damage; The hidden damage value on the player's weapon or monster attack. Players do not see this base value.
- Damage-per-second; The final calculated damage a player potentially can inflict. Players will only see this end value on weapons and damage dealt. This value SCALES along with player's relevant skills and allows us to modify the base damage values and calculations without players noticing too much (if we need to adjust these later on.)

Base DPS calculation

$(\text{PrimarySkill} * \text{BaseDamage}) + (\text{SecondarySkill} * \text{BaseDamage}) * 0.5 + (\text{ControllerSkill} * \text{BaseDamage}) * \text{AttackSpeedMod} * 0.1 = \text{DPS}$

*In the case of melee, the <ControllerSkill> is the skill "Force". For ranged, it is the skill "Archery" and for magic, it is the skill "Potency".

Example; Longsword base damage 22, Blade skill 30, Pierce skill 30 (secondary), Force skill 30, Speed is Avg (mod 2.0) then divide by 10.

*$30 * 22 = 660 + (30 * 22) * 0.5 = 330 + 30 * 22 = 660 * 2.0 \text{ (speed)} * 0.1 \text{ (value reduction)} = 330.0 \text{ DPS (or 165.0 Damage per Hit, considering two hits per second)}$*

***While the player will see a DPS value of 330.0 in this example on their sword (which also takes in scaling of their skills), in reality they are dealing a total potential of 165.0 damage twice every second, so their visible damage output will actually be less.*

RECOMMEND we change "DPS" to "Damage per Hit" or "DPH" so we can reflect the accurate damage values instead of per-second. When dealing with raw DPS, slower weapons will appear to be far stronger versus a weapon that hits more often. This is actually a common issue in many games that calculate raw DPS...

Attack Speeds

- Very Slow = attack calculation every 800ms. (speed mod 1.2)
- Slow = attack every 650ms. (speed mod 1.35)
- Average = attack every 500ms. (2 potential hits per second) (speed mod 2.0)
- Fast = attack every 333ms. (speed mod 3.0)
- Very Fast = attack every 250ms. (4 potential hits per second) (speed mod 4.0)



Combat (Players)

Weapons & Attacking

**this section is full of temporary and incomplete variables!*

Bonus Damage Calc with Affixes and Buffs

- Bonus damage (or Additional Damage) is taken from the end DPS calculation, such as any bonus that grants +X% and then added on top of the total damage.
- “Flat damage” or anything that adds “Base damage” is added onto the weapon’s base damage value at the start of the calculation. This means that it also gets bonuses from any +X% effect at the end of the calculation as well.

Standard damage (no bonuses)

$30*22=660 + (30*22)*0.5 = 330 + 30*22=660 * 2.0 \text{ (speed)} * 0.1 \text{ (value reduction)} = \mathbf{330.0 \text{ DPS (165.0 damage per hit)}}$

Example A – Flat Damage boost only (base is 22, base boost is +2, TOXX elemental damage)

$30*22+2= 660(60) + (30*22+2)*0.5 = 360(30) + 30*22+2= 660(60) * 2.0 \text{ (speed)} * 0.1 \text{ (value reduction)} = \mathbf{330.0 \text{ DPS \& } 30.0 \text{ Toxx DPS}}$

Example B – No flat damage boost, only additional +10% to total damage affix

$30*22=660 + (30*22)*0.5 = 330 + 30*22=660 * 2.0 \text{ (speed)} * 0.1 \text{ (value reduction)} * 1.1 \text{ (boost)} = \mathbf{363.0 \text{ DPS}}$

Example C – Flat damage boost AND, additional +10% to total damage affix

$30*22+2= 660(60) + (30*22+2)*0.5 = 360(30) + 30*22+2= 660(60) * 2.0 \text{ (speed)} * 0.1 \text{ (value reduction)} * 1.1 \text{ (boost)} = \mathbf{363.0 \text{ DPS \& } 33.0 \text{ Toxx DPS}}$

Example D - MAX Standard damage (no bonuses, maxed skill)

$1000*22=22000 + (1000*22)*0.5 = 11000 + 1000*22=22000 * 2.0 \text{ (speed)} * 0.1 \text{ (value reduction)} = \mathbf{11,000.0 \text{ DPS (or 5,500 damage per hit)}}$

Example E- Absolutely maxed out skill, max (capped) base damage and full affixes (assuming 8 affixes with +100% damage each), at fastest attack speed

$1000*100=100000 + (1000*100)*0.5 = 50000 + 1000*100=100000 * 4.0 \text{ (speed)} * 0.1 \text{ (value reduction)} = \mathbf{100,000.0 \text{ DPS} * 8.0 \text{ (affixes)} = 800,000 \text{ DPS (or 200,000 DPH)}}$

Why choose flat damage over +X% damage affixes?

Elemental damages are added as flat damage, this means that in the example above, such as adding +Toxx damage to your weapon would also benefit from any +X% bonuses applied at the end of the calculation, especially with “vs.” affixes, which means adding more damage overall in the form of other elements.



Combat (Players)

Abilities

Variable definitions

- These fields are located within DragonForge Data (and subject to change).

Variable	Description
Name	Visible name of the ability or spell
Affinity	Combat affinity this is related to (Melee, Ranged or Magic)
Type	How this ability is used, such as for Attack or Support..etc.
PrimarySkill	Skill that governs the use of this ability or spell.
SecondarySkill	Minor skill that governs the use of this ability or spell.
DV_learn	Difficulty value to learn this ability or spell (TBD).
DV_primaryskillcheck	Difficulty value to use this ability or spell (TBD).
WeaponRequired	Classification of weapon required to be equipped.
AmmoRequired	Ammo required to use this ability (unused, reserved for future expansion)
AccessoryRequired	Trinket required (unused, reserved for future expansion)
ItemRequired	For example, consumable items such as Health Potions...etc.
Target	Allowed targets; Other means other than yourself, such as an enemy.
TargetType	Single target, multiple targets or a target area.
TargetCountMax	Maximum allowed targets if Type is set to Multi or Area.
TargetAOEMeters	Range of Area effect in meters if Type is set to Area.
HPCost	How much Health to deduct when this is activated, per tier
MPCost	How much Mental Power to deduct when this is activated, per tier
STCost	How much Stamina to deduct when this is activated, per tier
AmmoCost	Ammo to consume (unused, reserved for future expansion)
DurCost	How much durability to remove from the equipped weapon when used

Variable	Description
TriggerCount	How many times the ability is triggered (each attack/use is calculated individually)
ActivationDelayMS	Delay to allow for animation and activation before targeting is allowed. Also used to prevent ability/spell spamming. In milliseconds.
DelayBetweenAttacksMS	If the ability has more than one attack count, this is the delay before the next attack is triggered and calculated. In milliseconds.
ATKSpeed	How fast the attack completes, based on Very Slow to Very Fast (or -1 for Instant).
CastSpeed	Same as ATKSpeed, but for spells. -1 = Instant cast.
ProjSpeedMod	How fast any launched projectile will travel (for ranged and magic). Based on Very Slow to Very Fast (or -1 for Instant). Comparable to movement speeds for game objects.
BlockChance	Chance for this ability to block any incoming attack while in use.
CoolDownMS	How long before this ability or spell can be used again (in milliseconds).
MaxRangeMeters	Maximum distance this ability can be used upon a target. If out of range, ability will not activate or have an effect. Includes projectile travel distance.
ModBaseDMG	How much damage is calculated from the equipped weapon. Does not count as elemental damage and is considered damage from all combat affinities.
ModHeal	How much healing is calculated from the equipped weapon's ATK value.
ModVs(Affinity)	Damage augment dealt to the target with this affinity. 0 = no damage dealt.
ModACC	Accuracy modification to the weapon's ACC value when this ability is used.
ModCRIT	Modification to the weapon's CRIT value when this ability is used.
Mod(Element)	Adds or removes elemental properties to the ability. 0 = no elemental effect.
ModRange	Increases/decreases the range of the ability. 1 = no change.
ApplyEffect#	Effect or spell effect to apply upon successful hit. Up to 5 unique.
EffectTarget#	Determines if the effect is applied is to OTHER (target) or SELF or both...etc.
Effect#Val	Magnitude of the effect applied based from the ability's initial damage after modifiers.
Effect#Duration	How long the effect lasts (DoT/HoT). -1 is used for static effects that linger until removed.
InterruptChance	Chance that this ability/spell can be disrupted while being activated from an attack.
IgnoreArmorChance	Chance that this ability/spell will ignore the target's PV (armor rating).
UnblockableChance	Chance that this ability/spell cannot be blocked or deflected.
Desc	Visible description of the ability or spell.

Abilities (Steps)

How abilities work

The Base ability damage is calculated from the equipped weapon (also the weapon type is required for specific abilities) as well as the primary and secondary skills. All abilities consume stamina (in the same way spells consume MP).

Order of operations

1. Player presses hotkey to activate the ability/spell.
2. Targeting
 - a. If no target is selected, targeting reticle will appear and prompt the player to choose a target.
 - b. If no target is selected and no target has been selected, the player can cancel the ability/spell activation without triggering usage or a cooldown.
 - c. If a target is pre-selected when the ability/spell is activated, then the COST is deducted and the ActivationDelay begins (along with animation and vfx).
 - d. If no target is selected and player selects a target with the reticle, then the COST is deducted and the ActivationDelay begins.
3. Deduct the activation cost (see above if target is valid). If player cannot pay the activation COST (low MP/ST etc) then the ability/spell simply does not activate (and player is warned they lack the resources to use/cast), but the target will remain selected regardless.
4. Interruption – *During the activation delay sequence (and animation)*, if a successful attack lands and is calculated to have interrupted the ability/spell, the COST is still deducted and ActivationDelay and animation is immediately cancelled (in the case of spells, it “fizzles” and casting ends). The character receives incoming damage as usual and has a hit reaction as normal. In the case being interrupted, the cooldown timer also triggered for the interrupted ability/spell. Once the activation delay and animation are over, interrupts are no longer possible.
5. Once the ActivationDelay ends and animation/vfx completes, then calculations begin. *NOTE – Attack/Cast speeds dictate how long the animation length is to play. The ActivationDelay just delays the start of the animation. The ability Attack/Cast speeds override the weapon speed or vise-versa, whichever is slower. This means if you use a “Fast” ability with a “Slow” weapon, the ability will use the “Slow” speed.
6. First, we calculate ACCURACY and ModACC from the weapon and skills to determine if the ability/spell hits.
 - a. If a MISS occurs, then the ability cooldown starts and no damage or effects occurs.
 - b. If successful, then we begin damage calculations.
7. Each “attack” will then calculate DAMAGE using all the Damage mods, starting with ModBaseDMG, followed by ModVs(Affinity), Mod(Element).
8. Check how much armor is ignored. IgnoreArmor is a % of target's PV (protection value) to be ignored in the damage calc. This is always calculated even if the ability is deemed unblockable. This variable is always used, even if defined as 0% (no PV ignored). Most spells will ignore armor (because Magic affinity is strongest against Melee). Which is why Magical Resistances are important.
9. Determine if the Ability/Spell is deemed Unblockable. This variable is merely a random chance from 0% to 100%. For example a value of 0.1 = 10% chance this ability is unblockable, which is different than IgnoreArmor. Unblockable ability/spell damage can still be mitigated by PV and resistances, such as the Magical Resistance skill.
10. Lastly, we apply (or remove) various effects, starting with any CRITICAL that may have occurred. These may require timers for the effect to continue even after the ability/spell activation is completed.



Abilities (Calculations)

We already know what a default (basic) attack looks like:

Base DPS calculation

$(\text{PrimarySkill} * \text{BaseDamage}) + (\text{SecondarySkill} * \text{BaseDamage}) * 0.5 + (\text{ControllerSkill} * \text{BaseDamage}) * \text{AttackSpeedMod} * 0.1 = \text{DPS/DPH}$

Example; Longsword base damage 22, Blade skill 30, Pierce skill 30 (secondary), Force skill 30, Speed is Avg (mod 2.0) then divide by 10.

*$30 * 22 = 660 + (30 * 22) * 0.5 = 330 + 30 * 22 = 660 * 2.0 \text{ (speed)} * 0.1 \text{ (value reduction)} = 330.0 \text{ DPS (or 165.0 Damage per Hit, considering two hits per second)}$*

Using the same Longsword, here's what a Power Slash ability looks like

- When the ability is activated, use **ModACC** and **ModCRIT** of the Power Slash's values to augment the weapon's ACC and CRIT to determine if a hit occurs on the specified target(s) for each AttackCount.

*Example; Kim wielding a Long Sword (ACC 40), with Tactics (20), Focus (30), Blades (40) and Anatomy (10) against a Skeleton with a DV of 160. Power Slash **ModACC is 0.85** and **ModCRIT is 0.8***

- 160 DV (skeleton) – (40 ACC * **0.85**) = 126 CV (modified combat value, which is -15% lower than a normal attack to use Power Slash).
- Kim's combat skills add up to 20+30+40+10 for a total of 100 CV.
- 100 CV / 126 CV * **ModACC** = 79.37% Chance to Hit the skeleton successfully and deal damage.
- The ability's CRIT uses the weapon's CRIT value of 20(%) * **ModCRIT** = 16% out of 79.37 = 12.7% chance that this HIT will become a CRITICAL HIT and deal additional damage.

- Damage from the Power Slash has **ModBaseDMG of 1.5** (no other modifiers for Power Slash).

Ability DPS calculation

$(\text{PrimarySkill} * \text{BaseDamage}) + (\text{SecondarySkill} * \text{BaseDamage}) * 0.5 + (\text{ControllerSkill} * \text{BaseDamage}) * \text{AttackSpeedMod} * 0.1 = \text{DPS} * (\text{each modifier and added together}) \text{ (see below)}$

Longsword base damage 22, Blade skill 40, Pierce skill 30 (secondary), Force (controller skill) 30, Speed is Avg (because the weapon is slower than the ability) (spd mod 2.0, see Attack Speeds) then divide by 10.

*$40 * 22 = 880 + (30 * 22) * 0.5 = 330 + 30 * 22 = 660 * 2.0 \text{ (speed)} * 0.1 \text{ (value reduction)} = 330.0 \text{ DPS (or 165 Damage per Hit, considering two hits per second)}$*

ADD IT UP

We take the output of 165 and multiply by **ModBaseDMG** + any/all other damage modifiers + any added effects (in the case of Power Slash, only ModBaseDMG is used which is 1.5), which comes out to a total raw damage output of $165 * 1.5 = 248$ damage per hit (DPH, roundup)... IgnoreArmor for PowerSlash = 0 (no PV bypassed), which means we can then take that 248 damage dealt by Power Slash and determine the armor hit location and mitigations on the victim.

- If the ability had a ModHeat value of 0.1 for example, we'd add this after ModBaseDMG, like this: $165 * \text{ModBaseDMG} + 165 * \text{ModHeat} = 248$ damage with an additional 25 elemental heat damage.
- If the spell had ModBaseDMG of 0, but had ModHeat of 2, then: $165 * \text{ModBaseDMG} + 165 * \text{ModHeat} =$ Only 330 elemental heat damage. (common of spells that are elemental anyway).
- If the ability had a ModVsMelee value of 0.2 and the target was of the Melee Affinity, then we include this as: $165 * \text{ModBaseDMG} + 165 * \text{ModVsMelee} = 248$ damage with an additional 50 non-elemental damage against the Melee affinity target. If target was Ranged or Magic affinity, then we fully ignore ModVsMelee. This similar to ignoring vs. Monster types, like vs. Beasts when fighting Serpents...etc.
- Other effects and damages are applied on top of these calcs, such as if the attack adds the vs. Beasts (tier 1) effect or heat/fire attacks adding a "Burning" effect.



Combat (Players)

Spell Casting

**this section is full of temporary and incomplete variables!*

Spell variables

- Base spell damage is calculated from the caster's equipped weapon (also it is required to be equipped for casting a spell) just like any other attack ability. This includes wands, staves and spellbooks...etc.
- Healing spells make use of ModHeal instead of ModBaseDMG, you will notice all healing spells deal 0 damage. While we could make ModBaseDMG a negative value to "give health back", I think its good to separate this into it's own value so we can control the healing factor more closely.
- ACC and CRIT are still calculated from the weapon stats and ModACC and ModCRIT as with any ability. It is possible for spells to fail, be interrupted or "miss" their target entirely. For friendly spells, we offset this with extra ModACC to make those less likely to miss, even with low skill.
- Spells cannot be "pre-cast" and can only be cast at the time of activation upon a valid target.
- Some spells deal no damage at all and simply apply (or remove) effects, for example Revive an Cleanse.

Spell damage

This works the same as with any other attack or ability, except we're using different skills:

$(\text{PrimarySkill} * \text{BaseDamage}) + (\text{SecondarySkill} * \text{BaseDamage}) * 0.5 + (\text{ControllerSkill} * \text{BaseDamage}) * \text{AttackSpeedMod} * 0.1 = \text{DPS} * (\text{each modifier and added together})$
(see below)

*Wand base damage 10, Spellcraft skill 40, Potency skill 30 (secondary), Essence (controller skill) 30, Speed is Fast (spd mod 3.0, see Attack Speeds) then divide by 10.
 $40 * 10 = 400 + (30 * 10) * 0.5 = 150 + 30 * 10 = 300 * 3.0 (\text{speed}) * 0.1 (\text{value reduction}) = 255.0 \text{ DPS (or 85 Damage per Hit (DPH), considering three hits per second)}$*

- The spell being cast is Flamebolt, which only has ModHeat 1.5... so this comes out to $85\text{dph} * 1.5\text{heat} = 128 \text{ heat damage only, per hit. (x3 because of the fast speed mod for DPS, which is 384)}$

Combat (Players)



Spell Casting

**this section is full of temporary and incomplete variables!*

Taking and Resisting spell damage

** See armor RESISTANCES for more details!*

First – lets assume the spell being cast is Flamebolt, which comes out to $85dph * 1.5heat = 128$ heat damage only, per hit. Flamebolt has IgnoreArmor of 0.5 (50% ignored).

- Spell hits the target in the chest, no critical occurs.
- The target has Resist Heat 10% prefix on their chest armor and 200 Magical Resistance and is of the Melee affinity (-20% vs Magic)

Calc for Spell vs Player

1. Adjust for Affinity damage, in this case add +20% damage because magic vs melee: $128 * 1.2 = 154$
2. Deduct all resists from single armor hit location, in this case the chest: $154 * 0.9$ (heat resist of 10%) = 139
3. Apply Magical Resistance skill which is $(200 * 0.5) * 0.1 = 10\%$ reduction. (at skill level 1000, player is given a maximum of -50% reduction). $139 * 0.9 = 125$
4. Deal damage with IgnoreArmor %. In which case $125 * 0.5$ (ignorearmor 50%) = 63 direct health damage.
5. Take the remaining damage and mitigate with the armor's PV (see below or Armor section).

Protection calc: $(PrimarySkill * (basePV * armorDur\%) + ((SecondarySkill * (basePV * armorDur\%)) * 0.5) * 0.01) =$ mitigated damage.

With a minimum of 5% of total basePV at 0 skill level.

Example #1:

*Jeff is wearing a chest plate that has a base protection value (PV) of 50.
The armor is at 80% out of 100% durability (20% damaged). Let's assume this is 25.6/32 total durability.
Jeff's Heavy Armor skill is 50 and his Defending skill is 100.*

*Because the armor is only at 80% repair, it suffers a -20% base PV reduction on both skill calculations, meaning it's effective base PV is only 40 (instead of the full 50).
The calc would look like: $((50 * 40) + ((100 * 40) * 0.5)) = 6500 * 0.01 = 65$ total damage mitigated if he were hit in the chest.*

65-63 = 2 dmg more damage required before the armor is penetrated with the remaining damage from the spell.

Combat (PvM)



Player vs Monsters (or creatures)

Combat against non-player entities is fairly straight forward. Once the player has selected their target (by tapping or d-tapping the NPC), combat will initiate once the player character is within the appropriate range for the weapons, spells or abilities equipped. In these instances, the player typically strikes first, catching the NPC unaware and thus alerting the NPC to the player's presence.

Once combat begins (even before either entity lands a single hit), attacking and defending is automatic and will run through a defined "queue" of automatic combat actions to perform (*see combat queue). When the end of the queue is reached, the combat cycle repeats until the NPC is defeated (and looted). Players will be able to toggle on/off auto-combat features such as "auto-approach", "strike first", and "auto-combat-queue".

Affinity – Monsters have the same combat affinity attribute as players (Melee, Ranged or Magic). As such they will also suffer and benefit from the same strengths and weaknesses as players.

Aggression and target priority

By default, NPC enemies are "blind". Depending on the enemy (and it's script) it may patrol, seek or actively pursue a player within a range. Nearly all enemies will ignore a player that is targeting them without having been attacked. Unless the script says to detect and attack someone targeting them explicitly.

Enemies that have been attacked will immediately target, pursue and attempt to attack that player dealing the first blow (being attacked and missing does not count for triggering this initial aggression).

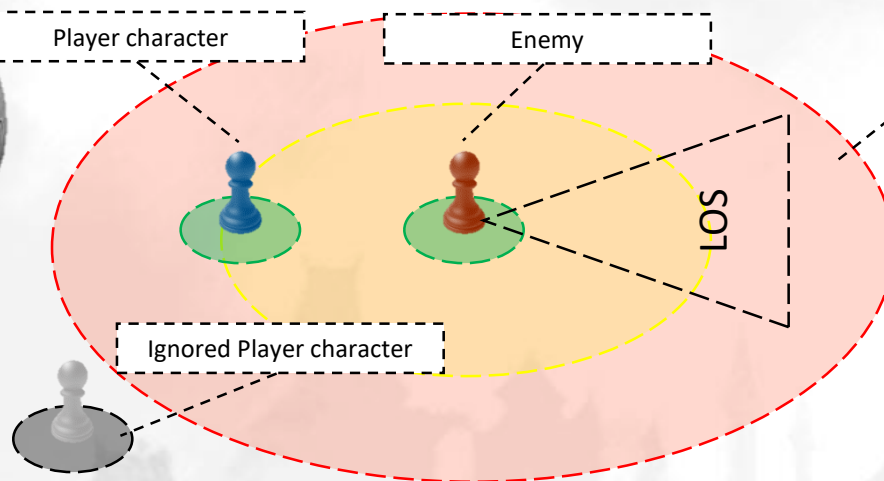
While in pursuit of the initial target, any other player that deals damage to the enemy will draw that enemy's focus (be re-targeted) if the damage received is greater than the previous target of pursuit. Meaning to say, an enemy will target, chase and attack the greater threat. An enemy that is repeatedly attacked by multiple players will only change targets up to a maximum of 3 times to prevent poor pathfinding and help resolve "kiting" or leading an enemy around by drawing it's focus too often. Likewise, a monster will only chase a player for so long (time/distance) before giving up or choosing a new target.



Combat (PvM)

Enemy's area of influence

**this section is full of temporary and incomplete variables! Be warned, here be dragons, literally **



Radial Legend (invisible to the player):

Green = Melee distance. Any player within this area is automatically detected and targeted (even if hidden). Enemy will turn to attack if no other target/threat is available.

Yellow = Ranged distance and caution area where the enemy may detect an unhidden player. If targeting a player and able to perform ranged attacks, the enemy will remain stationary and launch projectiles.

Red = Maximum line of sight and chase distance. Any player targeted and being actively chased can escape the enemy by leaving their maximum LOS radius. The enemy may also have additional AI behaviors in addition to the maximum LOS chase area. It is also the area of which a wounded/fleeing enemy will look to try and escape from players.

LOS = Line of sight. A cone-shaped detection area that is projected in front of the enemy depending on which direction the enemy is facing. Unhidden players within this LOS cone are targeted/detected. Patrolling and pathing enemies will rely primarily on their LOS cone to find and attack players. Some LOS cone sizes will vary (some can be longer/shorter, wider/narrow...etc).

Dev Note: Generally the enemy's radius values will mirror those of a player. However we may desire to decrease them in size to reduce pathing/AI needs during gameplay for performance concerns if we have many enemies on screen. In these cases we may wish to reduce the NPC radius values by 25-33%.

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Damage Display

The amount of health lost per hit (after combat calculations) shall appear overhead on the damaged enemy and/or player. Numbers in RED are lost health, numbers in GREEN represent health restored.

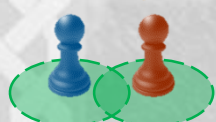
In the case of a critical hit, the number will appear larger and in GOLD for damage and TEAL for critical healing.

As the numbers appear, they float upwards for 1 second then vanish. At the most, up to 5 values can be seen (and overlap oldest-to-newest).

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Combat (General AI)



Fight or Flee

Most every enemy, creature or NPC will fight to the death, especially those with the Melee affinity. Some magic or ranged NPCs may have scripts and behaviors that trigger them to flee or path away to gain distance from their targets or to avoid melee-aligned players. The individual scripts of each creature should be tuned so that they have a min-max preferred combat range as well as a min-max health range before their behavior is changed.

For example, if an goblin were to be at less than 10% health, it may have a higher chance of breaking combat and fleeing to his outer LOS range to try and avoid the player so it can regenerate. Once it's above 10% health again, it may then turn and return to combat. Other creatures such as skeletons or mindless foes may just fight until the death without any such retreat behavior.

Death Blow

When an NPC reaches zero (0) health, they are immediately killed and their corpse falls to the ground.

- Within 3 seconds of the deathblow, only the player that landed the killing blow (or did the most damage) can loot the corpse. (locked to other players)
- Within 5 seconds of the deathblow, only the player and the player's group/party or allies may loot the corpse.
- After 10 seconds, the enemy corpse becomes open to all players and can be looted freely.
- After 15 seconds, the enemy's corpse decays and is removed from the game world, leaving behind a "loot bag" container.
- After 30 seconds, the enemy's loot bag decays and all items within are removed from the game world.

NPC Corpses

NPCs upon death will do a skill check on the player character landing the deathblow (or dealt the most damage) to determine additional actions and loot to add to the corpse automatically. A player will never need to manually "skin" or chop up a corpse for additional materials. **Not all NPCs drop loot or all the time.**

Example: Ken slays a skeleton, Ken's Necromancy and Anatomy skills dictate that there will be an additional chance and amount of "Bone Powder" dropped by the skeleton within it's corpse container for Ken to loot.

Player Progress



Attunement (progress checkpoint linking)

Scattered throughout the world in cities, dungeons and in the wilderness are magical crystal shards. These shards have the ability to contain the essence of a player. Meaning to say that if anything were to happen to a player's physical body, their spirit is stored within the last crystal used and thus allows a player to be restored.

Using an attunement crystal

- A player simply needs to approach a crystal and TAP, upon confirming they wish to link to this specific crystal, the process will be completed.
- Upon linking, a player's skills, attributes and progress (including worn equipment) is recorded at the location of the crystal that is utilized.
- The cost to attune to a crystal will vary depending on location. This cost can be paid with Magic Power upon use or paid with consumable "crystal shards" items carried by the player.
- When a player dies and returns to their attuned crystal, the crystal will no longer be attuned to the player (crystal will be dimmed), requiring the player to re-attune if they desire and pay another attunement fee. An attunement is a single-use "respawn". If the player does not re-attune and dies again, they may end up back in town and lose progress and/or have to walk all the way back to where they were!

Benefits of attunement

- Able to respawn physically at a closer location in case of death. Usually near the entrance to a dungeon or area.
- Offers a means for the player to manually save their character's skill progress in the case of death and loss of skill progress.
- Attunement reduces the length of temporary death's ill de-buff effects.
- The crystal offers slightly increased, yet passive health, magic and stamina regeneration within a limited area of effect while nearby.
- *(TBD) Travel to/from the crystal from a travel portal within town/city locations.*

Player Death



Death vs Unconscious (KO)

When a player reaches zero (0) health, they are rendered unconscious (KO) and fall to the ground. Once a player enters the KO state, a visible timer begins for **10 seconds** (base value, skills may affect duration). During this window of opportunity, the KO'd player can be either revived or executed by other players (NPCs will never execute a KO'd player, instead they will ignore the downed player and look for other targets).

When this KO timer expires;

- The player dies (being executed is different, but similar),
- Their corpse appears as an open and loot-able container containing all of their current inventory (no worn equipment), loot-able by any player.
- The slain player is returned and respawned alive at 100% health (*if attuned to a crystal, at 10% health if not attuned or respawn back in town*), but without any inventory (including all gold held) (they will keep their worn equipment) at their attuned crystal, where ever that may be.
- The player receives a death penalty as well as a temporary death debuff.

If a player is Revived during the KO timer;

- The player rises from the ground with some health (skill/spell power determines restored amount)
- Risen player has all items and equipment carried upon time of KO.
- Risen player is de-targeted by nearby enemies and cannot be re-targeted for 2 seconds.
- Risen player receives a temporary "Resurrection" effect (see *effects* for details).
- Resurrected players cannot be targeted, be attacked or perform attacks/actions until this temp debuff is in effect. They can path and get out of harms way however. A player that immediately dies again can be resurrected again in the same way.
- The dead player has the option of refusing to be resurrected (no one can force-rez you, this is to prevent griefing)

If a player Executed during the KO timer;

- The player dies as expected.
- The player that performs the execution is marked as a murderer (see murderer status/penalties)
- The slain player leaves a corpse containing all inventory AND all worn equipment, including Coins.
- The player respawns at their attuned crystal (or in town if not attuned) and receives death penalties/effects as usual.

Player Death



Resurrection and revival

Upon being resurrected, a player will be given a temporary death status effect:

If Resurrected (by another player) “Raised”

- No death debuff is applied (because they did not actually die).
- Player will receive a “Raised” effect for 2 seconds which grants invulnerability and the player cannot be targeted (invisible).
- Following the expiration of “Raised”, the player then receives a “Resurrected” effect. All skill gain for 10mins is reduced by 75%.
- Cannot be resurrected again until the “Resurrected” effect expires. Becoming KO’d and dying will result in a respawn.
- Player respawns with their worn equipment.
- Player respawns with their entire inventory, including all Coins previously held.

If a player dies (KO timer expiration) “Expired”

- Player upon respawn (if unattuned) receives “**Embraced by Death**”, a debuff that lasts 30 minutes and reduces all skill gain by 50%. All skills also have a -50% reduced effectiveness (for calculation purposes, they are calculated at half).
 - If player is attuned, the player receives “**Marked by Death**” upon respawning at the crystal, which these effects are 15 minutes long, skill gain -25% and -25% reduced effectiveness.
 - Player respawns with their worn equipment.
 - Player’s inventory (and all Coins held) is left with the corpse.
 - Player retains 10% of all skill XP gained between their last attunement and from the moment of death for all relevant skills.

If a player is dies (Executed) “Executed”

The player dies as expected.

- Player upon respawn (if unattuned) receives “**Executed by <username>**”, a debuff that lasts 60 minutes and reduces all skill gain by 50%. All skills also have a -50% reduced effectiveness (for calculation purposes, they are calculated at half).
 - If player is attuned, upon respawning at the crystal, which these effects are 30 minutes long, skill gain -25% and -25% reduced effectiveness.
 - Player respawns “naked” without any worn equipment or inventory (including Coins).
 - Player retains 5% of all skill XP gained between their last attunement and from the moment of death for all relevant skills.

Player vs Player



Tagging

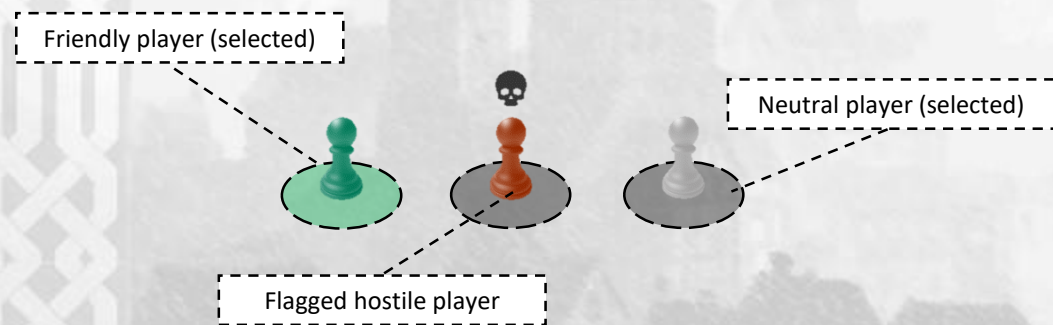
When players interact with each other, typically through trading and combat, players will be tagged with distinct icon and/or color(s) to identify if the player is friendly, allied, neutral or hostile.

Friendly tagging is not shown unless a player is explicitly targeted (tapped upon and selected), a player can then view the player's health bar and friendly status as they would an NPC. Hostile tags however are overtly shown above the head of any such flagged player so they can be identified quickly, from a safe distance and easily.

Color Coding

With a player selected, that target will have a colored aura (selection indicator) representing the tagged status.

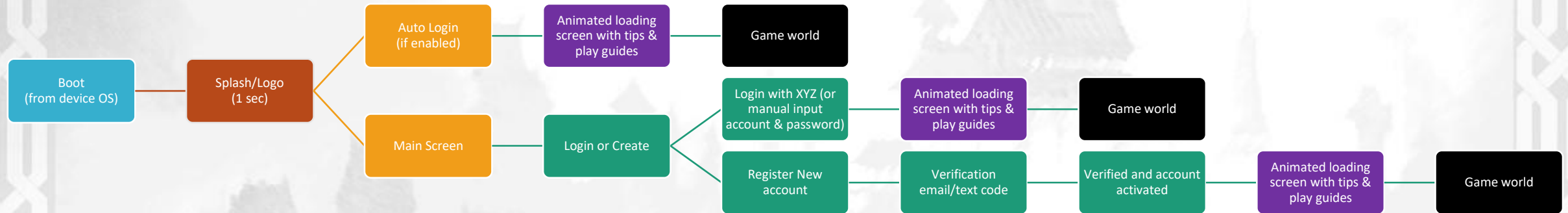
- **Purple** – Allied player (guild mate)
- **Blue** – Party/group member
- **Green** – Non-allied or grouped player that is friendly and in good standing with the town guards...etc.
- **Grey** – A player that has committed a crime and is neutral with the town guards. A player with a neutral standing can be openly attacked by any other player without flagging the attacking player as hostile.
- **Orange** – Another player that has consented to PVP dueling. This player may be hostile and can be openly attacked without being flagged as hostile.
- **Red** – A hostile player that has been flagged as a thief and/or murderer. In addition to the RED aura, this player will be tagged with a hostile indicator for all other players to see. (See reputation and social section on how to exit RED status and return to good standing).





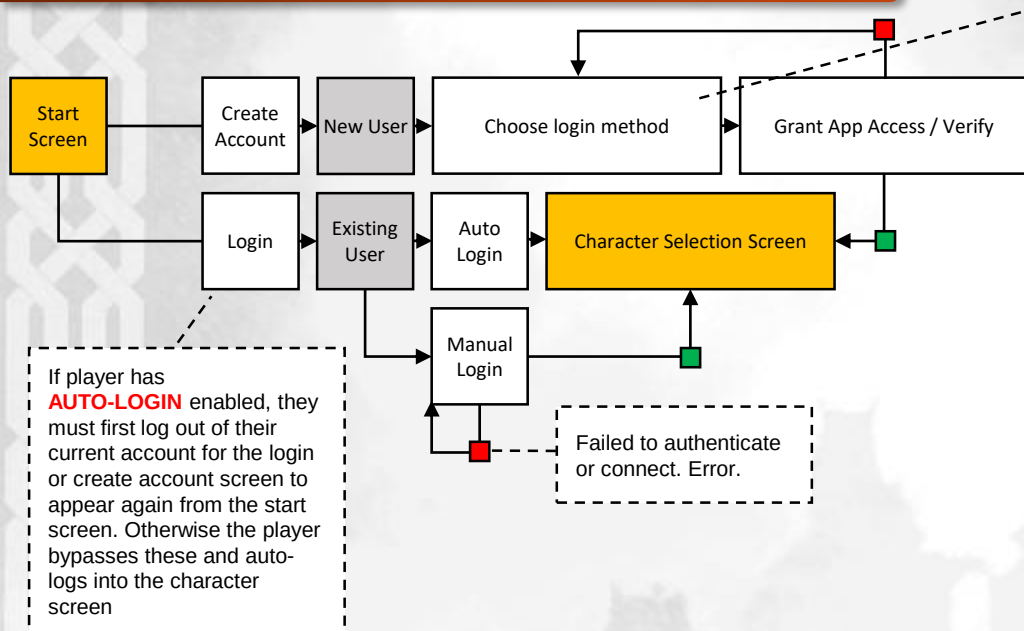
Simple UX Flow

Get into the game FAST.





Login Flow



Login Method (existing accounts with 3rd party)

Facebook / Twitter / Apple Game Center / Google
Final option is to register by EMAIL for a non-linked Dragonforge game account.

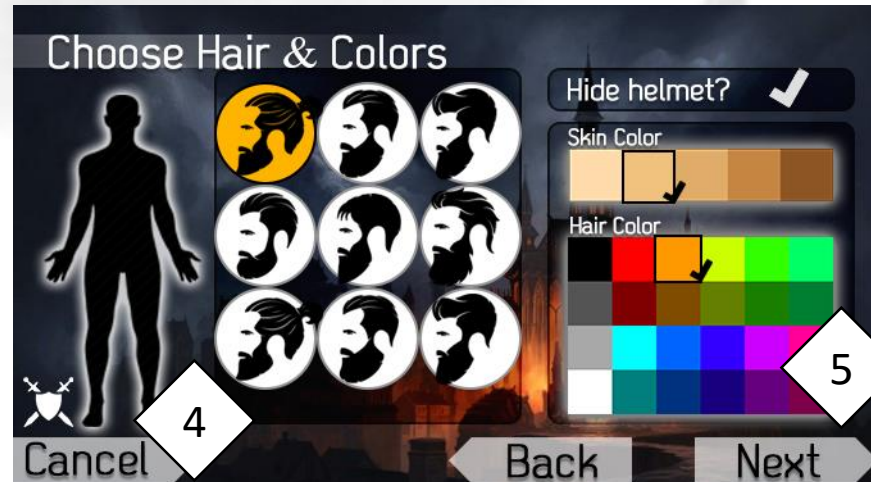
If using Facebook or Twitter – Game will ask for permission to login and link the account and game app to Facebook/Twitter.

If using Game Center / Google accounts – Game will login using that device's signed in credentials, if any. If not signed in, user will be prompted to sign-in/sign-up for Game Center and/or Google Play when choosing this login method.

Once signed in for the first time, AUTO-LOGIN is enabled for the account by default, which can be turned off and require user's account credentials for each login thereafter.



Creating a character avatar

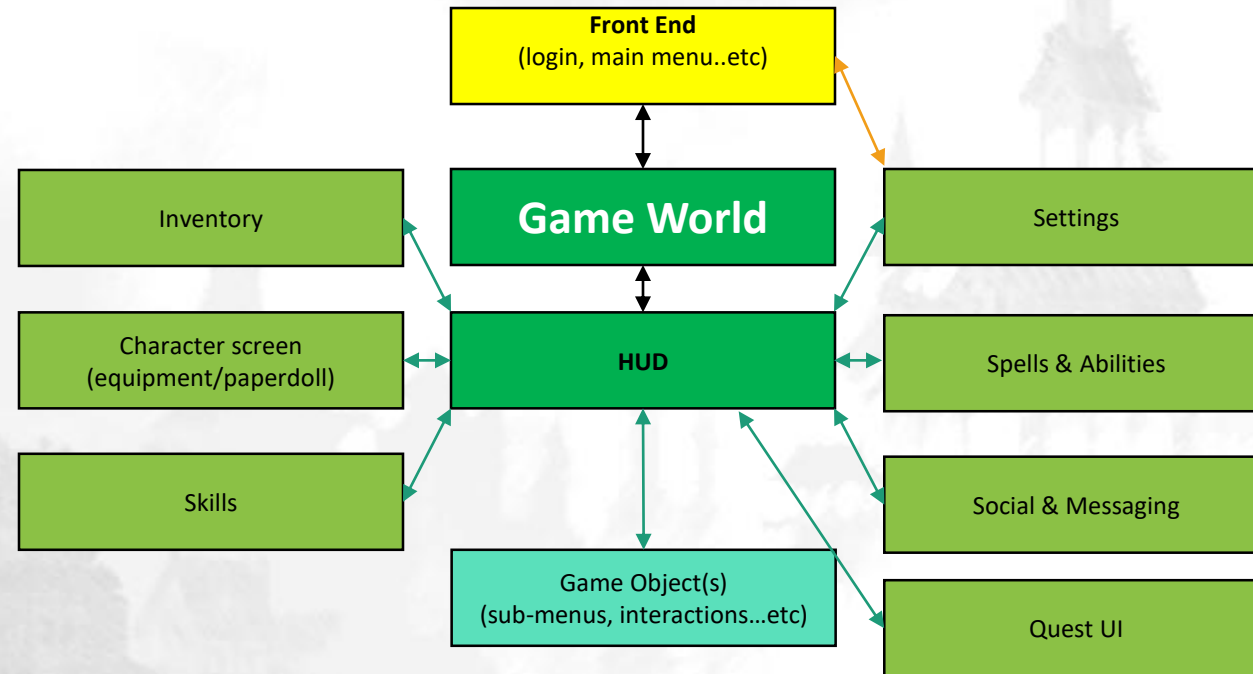


Six step process

1. Select Gender (M/F)
2. Choose body type
3. Choose hairstyle
4. Choose skin tone and hair color
5. Select affinity (Melee, Ranged or Magic)
6. Input character name



UX GUI flow chart (From HUD)

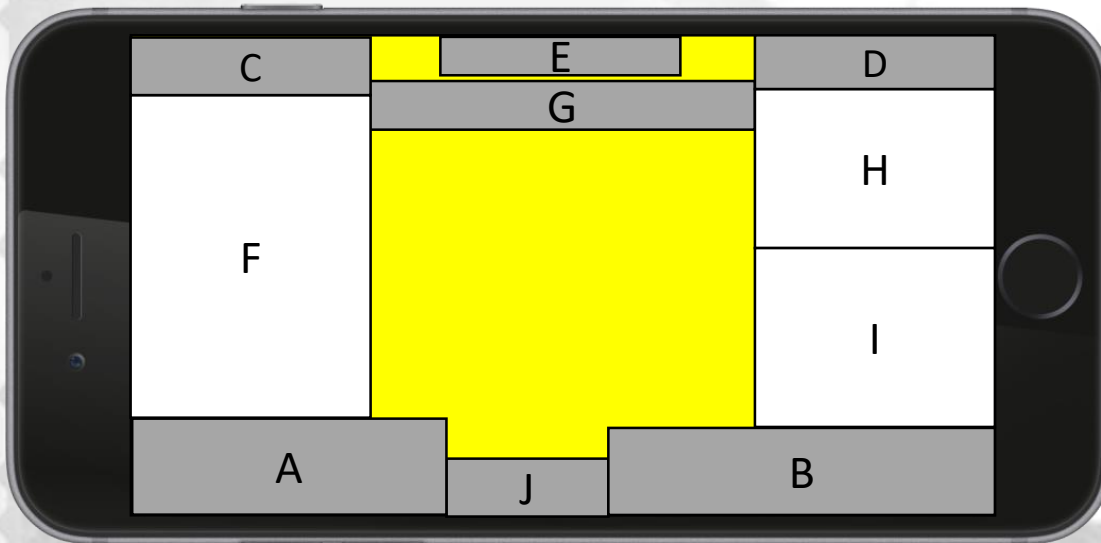


Primary Heads-up Display (HUD)

The Heads Up Display is the primary graphical interface the player will be viewing and interacting with during the game. It will consist of several groups, buttons, displays and widgets. GREY boxes represent “solid” UI elements, where as the other areas are transparent overlays onto the gameplay area.

Primary HUD Legend:

- A. Your HP/MP/ST meters (below/above) are status effect icons/timers.
- B. Spells and Abilities virtual hotkey slots
- C. Menu Cluster (Chat, Social, Actions/Emotes, Quest Journal)
- D. Menu Cluster (System Menu, Inv, Character, Tasks)
- E. Target display information (HP bar, name..etc)
- F. Social messaging / chat channel display
- G. Area announcement banner
- H. Tracked task display
- I. Action (combat) text display
- J. Party system UI and display



(HUD, with minimal panels open)



Menu Button Row (HUD)

Along the top-edge of the screen are located 6 primary buttons. Each of these calls forth either another GUI or a sub-menu of selections to choose from.

Social

- Control, create, disband and manage guilds. (UI TBD)

Actions

- Emote submenu
- Perform action submenu (such as perform an action on a target, auto action...etc.)
- Auto-action toggle
- Mute toggle

Quests

- View quest entries, quest/task details, give up quests and toggle on/off tracked quests.

Char

- Submenu contains shortcuts to access the equipment (also character info) screen, skills, spells and abilities.

Inv

- Opens the player's inventory, no sub menu.

Menu

- Submenu for all system-related controls. Settings allows the player to configure more about the game, while we also have a quick audio toggle and shortcuts to access support ticketing system. LOGOUT is also found within this submenu.

(HUD, with maximum panels open)



Open the submenu:

Tap and hold to open a submenu, hold down and drag to the desired sub-menu option, then release your finger to select it.

Scroll text (HIDE/SHOW) toggle

Tap and hold for 1s to hide or to show the combat action scrolling text log.

Chat Panel (HUD)

Along the top-edge of the screen are located 6 primary buttons. Each of these calls forth either another GUI or a sub-menu of selections to choose from.

Default Channels

- Global, Private & Party
- If in a guild, a 4th channel will be automatically created for GUILD chat.

Inputting text

1. The player must tap “Enter Message” button. This will call forth the virtual keyboard.
2. Player can simply type in their message and press SEND.



3. Optionally, the player may change the channel to send the message to as well as change the squawk mode from “Say” to “Yell”, “Whisper”, “Reply” or “Emote”.

4. A player may also input common chat commands into this text box as an override, such as

/w username OR */whisper username messagestring*

(HUD, with maximum panels open)



Open the username sub-menu:

Tap and hold upon any player’s username in the chat channel to bring up a small list of commands, such as:

Block (must be unblocked from social panel), Report, Whisper, Reply and if nearby, Trade, Party Invite, Guild Invite.

Character Panel (Paperdoll)

The equipment panel is represented by a paper doll figure with multiple equipment slots. In total, there are 8 equipment slots

Equipment slots

1. (Armor) Head
2. (Armor) Torso
3. (Armor) Arms (includes hands, arms and shoulders)
4. (Armor) Legs (includes thighs, knees, shins, ankles and feet)
5. (Accessory) Universal Slot1 (can use any accessory, rings, necklaces...etc)
6. (Accessory) Universal Slot2 (can use any accessory, rings, necklaces...etc)
7. (Weapon/Shield) Left hand
8. (Weapon/Shield) Right hand

Character Panel (Stats)

Under the general STATS section of this interface, the player can see the basic readouts of their character's attributes, including the location to which the player is attuned to.

The abbreviations in brackets are to be icons that represent that stat. For example, [HP] means a heart or blood icon to represent health.



Stats Screen

Shows player's stats and attributes



Equip Screen

Shows available items from inventory that can be placed in the selected slot.



FLICK LEFT

Fast-Navigation within the GUI

When inside the GUI menu systems, swipe left or right to change between screens. Otherwise, change screens by tapping the appropriate button displayed.



FLICK RIGHT

Character Panel (Paperdoll)



Changing Equipment

1. Tap on an equipment slot to open the side panel.
2. Scroll to locate item to equipment from the available items. This list is populated (and filtered) from all appropriate items held in current inventory for the selected equipment slot.
3. Tap on item in the list, it will then be equipped.
4. Tap on a different equipment slot/button to change selected slot or press the (X) to close this sub-panel.

Character Inventory

The inventory contains everything presently held and carried by the player's character with the exclusion of the player's worn equipment (shown on the separate Character Screen).

The player's Inventory screen shows the players "backpack" and all the item slots within. In Dragonforge, each item takes 1 slot and multiple of the same item can be stacked up to the maximum stack size.

Inventory Size and Expansion

The inventory begins with 16 slots (assuming 8 slots per row). The player can purchase additional slots and expand their inventory in town at the Pack Merchant. The maximum player inventory is capped at 128 slots, although the inventory UI has the capability of scrolling to show more slots, but be warned more bag space means more items to potentially lose upon dying!

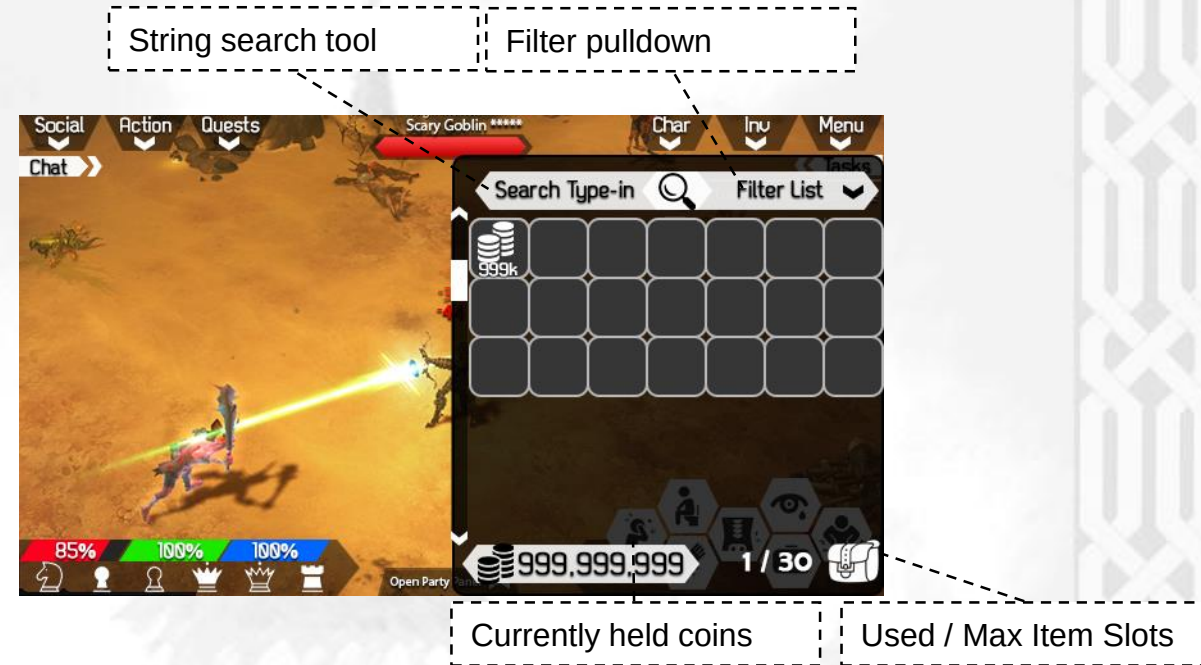
Filter Pulldown

Tap here to see a list of item Types to sort the inventory by. All items not relevant to this item type will be *HIDDEN* until the filter is reset back to "ALL ITEMS".

Wallet and currency

Dragonforge utilizes a centralized currency called "Coins". The player can carry as much Coin as they desire on their character at any time, however Coin consumes inventory space just as other items.

A stack of Coins has a maximum stack of 100k (100,000) per inventory slot. The total amount of Coin carried by the player's inventory is displayed for convenience at the top of the UI, next to the current and maximum inventory capacity.

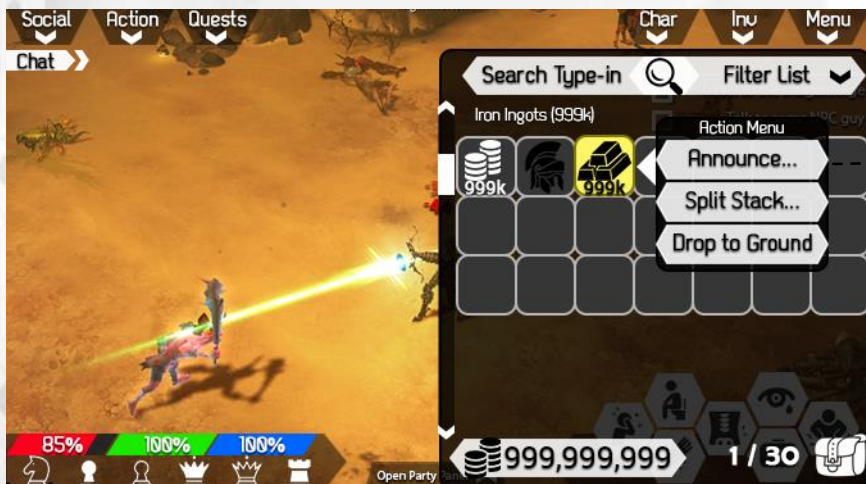


Character Inventory

Description Card

This area is reserved to display detailed information about the selected item(s). From this card the player can view stats and other information.

- **Drop button** (aka Discard or “Drop to Ground”) will allow the player to drop/dump this selected slot’s item(s) to the player’s feet in the game world, where it will be displayed as a small loot bag and accessible by other players freely.
- **Tap and Hold button** will display other sub-menu options (Action Menu)...
- **Announce...** This feature will copy and paste the item’s name and link into the current chat channel to send broadcast that item is for sale, etc.



Sub-menu appears when an item slot is tapped upon and held for 1 second.



Large description card appears on the left panel when an item is selected from the inventory slots.



Banking & Trading

All items and coin stored within a player's bank vault are safe from loss in the case that the player dies. Bank slots can be purchased within the Bank UI and each slot added increases in cost to purchase the next. The maximum limit of bank space a single player can acquire is 1024 slots.

While the player's bank vault can be only accessed from specific banker NPCs within town, a player may store additional items within containers and other storage located within their homes and on their mounts' saddle bags. However, only banked items and coin is truly secure.

Transferring between Inventory and Bank

A player may simply access the bank UI and select an item slot and press the "Transfer to Bank" button. Alternatively, a player may tap, hold and drag the item slot from the player Inventory side of the UI onto the Bank Vault side of the UI to also transfer item slots quickly.

Filters and search type-in fields allow the player to further filter and find items they are storing in either their inventory or within their vast bank vault.

Transferring between Players

There are three panels, your inventory, your offer and the other player's offer. Once both players have placed at least one item and have both tapped CONFIRM, the trade action occurs immediately. If a player does not have room in their inventory, when that player presses the CONFIRM button, it will display a warning and disable the confirm button from activating.



Trading and transfer screens are similar



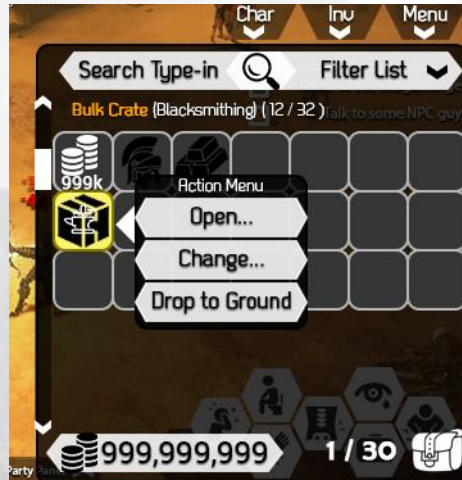
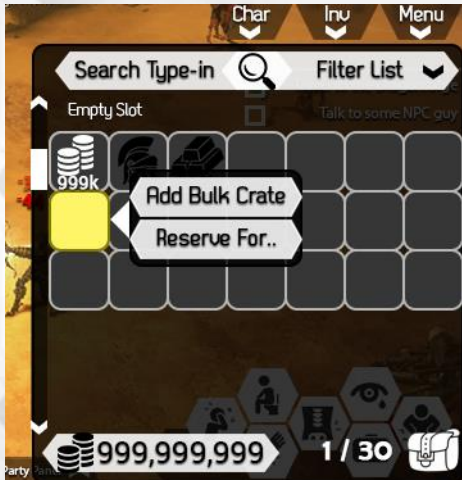
Storage

Securing Containers

In homes, a container may be tapped and held upon to choose it's sub-menu. From here the owning player can choose to "set lock" or "release lock". When setting a lock, the player can input a simple PIN or choose to generate a KEY. If the player generates a KEY, this key is then added to the player's inventory. Be warned however, keys can be lost or stolen, in which case a player may end up having to lockpick their own container!

Bulk Crafting Containers

A player may tap and hold on any empty inventory slot to create a container in which may be assigned to place items into when crafting. A player can then access and open this container to view all items within. A bulk container can contain 64 items. Only crafted items can be placed within during crafting.



Chest (1 / 16)
(unsecured)

[Open]
[Rename]
[Set Lock]
[Set Trap]
[Take All]



Example sub-menu and commands when you tap & hold on a container





Skill Panel

The skill UI is fairly simplistic as this screen is merely a display-only list of all the skills in the game and the player's current skill level and progress towards the next level shown as a simple progress bar under the skill name/label.

A player can simply scroll through this list to see their current skill levels, but also a player is able to tap on a skill to select it and read more information, such as learning what the skill does and how to increase it.

From here, the player can also navigate back to their inventory, the character screen or exit the UI entirely and return to the gameplay.



VERTICAL SCROLL

Spells & Abilities

On this screen, learned spells are sorted into a spellbook list of sorts, whereas the player's learned abilities (things such as specialized attacks, buffs and special combat actions) are located on a separate page and list.

From here, the player is able to filter and scroll through all the various abilities and spells their character has learned. All actions listed here can be hotkeyed for use from the hotbar located on the main game HUD during gameplay for ease of access.

Viewing details

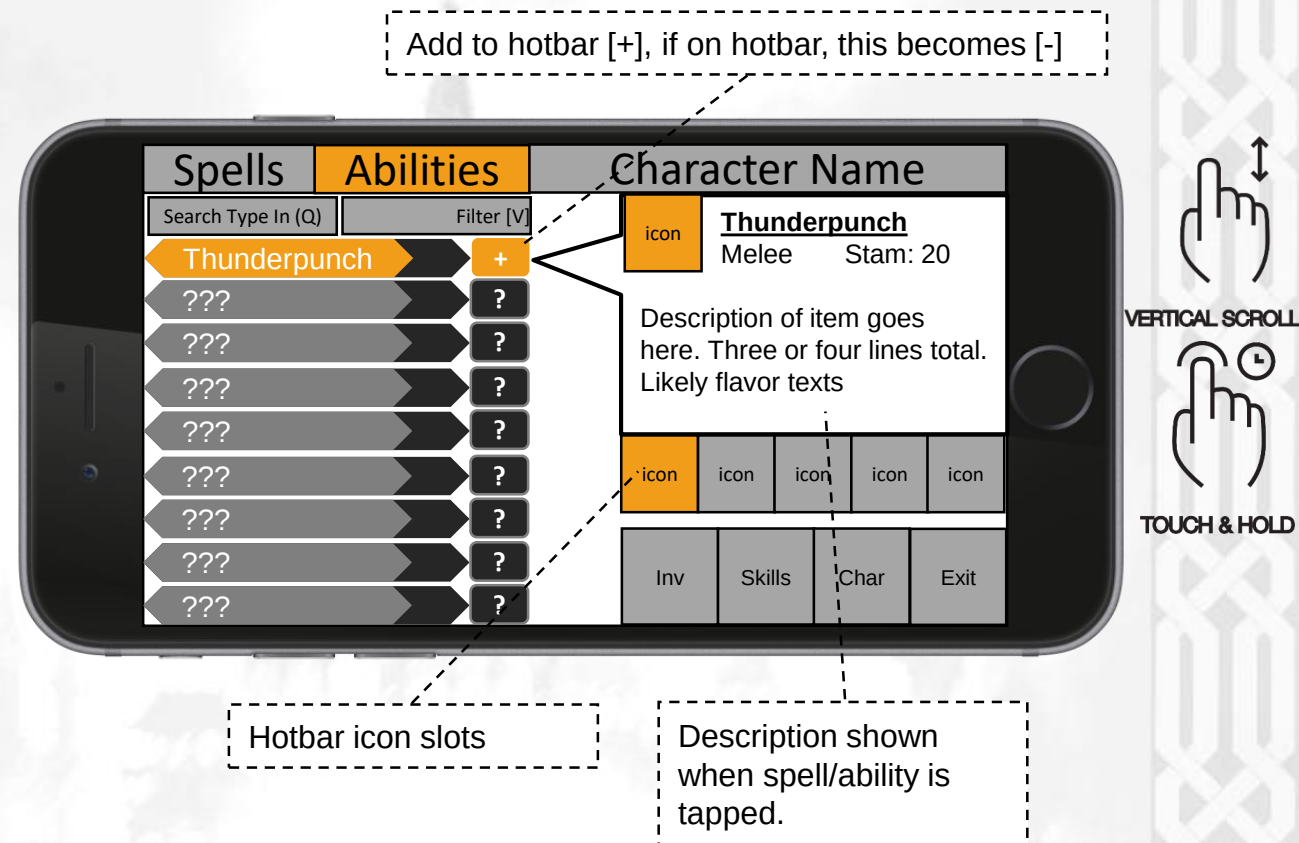
Simple TAP on a spell or ability to view it's description card. From the description card, a player can see the icon, name, cost and other parameters of the selected spell or ability.

Rearrange the Hotbar

A player can drag & drop the icons on the hot bar left and right as they desire. They can also tap and hold to remove icons and double tap a spell/ability from the list view to quickly add it.

No limits

A player may amass a large collection of spells and/or abilities as they play the game. There is no limit to the number of spells or abilities a single player may possess. New spells and abilities become available from crafting, questing, monster drops and may frequently be added as new game content at anytime.



Questing Panel

Quests are split into two categories; Quests and Dailies.

Quests are longer, multi-step tasks that are usually tied a smaller story arc or even the main primary story arc throughout the game. The Quests tab will display all completed and started, yet incomplete quest lines. Quests that are not yet discovered (or started) do not appear on this list.

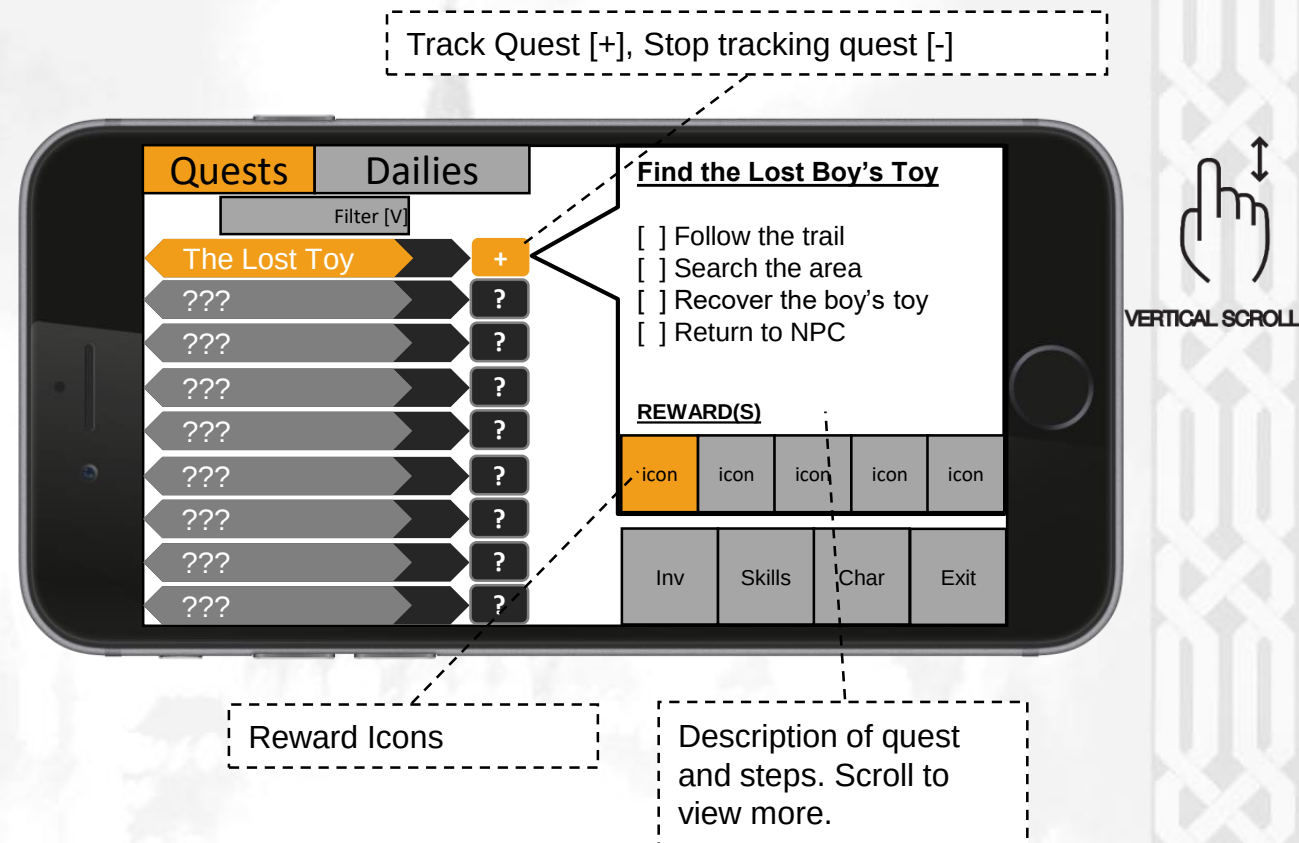
Dailies (aka Daily quests) are smaller, simple and fast tasks that a player may repeat on a daily basis. These quests may offer resources or items that are rare or may be difficult to obtain through non-quest means.

Tracking Quests

When enabled, the quest will be tracked and the status/tasks will be displayed on the main HUD in the “quest display” area. Up to 3 quests may be tracked. While tracked, a quest’s directional indicator and distance to next waypoint will appear on the HUD as well.

Rewards

The tangible rewards are shown at the bottom of the quest’s UI. Up to 5 items (stacked or single) can be awarded per quest. The items offered are defined and the player is given all items shown upon completing the quest. The player will never be given a choice between which items to take or leave. If a player does not have enough inventory space for the items, the game will drop the last slots in a player’s inventory (with the exception of quest-items or LINKED items) to the player’s feet and then take the quest rewards to ensure the rewards are given.





Social Panel

Chat channels

Party controls

Guild Leader/member controls

Social network interactions

Socializing

How to socialize

Communication

Trading

Coop

Assisting

Crafting

Guilds

Raids

Events

Mentorships/Apprenticeships (train newbs)



Settings & Options

Control options

Camera settings

About the game (version, server stats..etc)

Legal

Credits

Support

Forum/Web links



Skills Overview

Every action in Dragonforge is governed by a skill, in many cases an action is governed by more than one. When an action is performed, the skill's value is what determines not only the outcome and efficiency of the action performed, but also controls how much that skill grows depending upon the difficulty of the action as it is performed.

For example;

John has zero skill in Lumberjacking.

His chance of successfully harvesting wood from any tree will be extremely low, but still feasible. However, the act of simply performing the action will have a much larger chance of increasing the skill overall, although the action itself may be failing to produce the desired outcome (in this case, harvesting wood).

Should John keep at it and succeed to harvest wood (even if it takes some time with his low success chance), his skill gained from a successful attempt will be that much larger when it does succeed. In the meantime however, John will experience swift skill gain until eventually he consistently is able to succeed, resulting in the skill gain to slow down over time as he becomes more skilled and is required to perform Lumberjacking on trees that are harder to harvest.

As players gain higher skill, performing the same repeated action on easy tasks, (that is to mean actions that have a very high chance of success), will result in lesser skill gain than attempting to perform actions with lesser chances of success on harder tasks. Players that strive to do harder tasks will see the best skill gain results overall.





Gaining and improving skills

Skills all begin at zero (0) (for calculation purposes, we always consider zero to be 1)...This is what all skills begin at when a new character is created. As the player plays, most skills will typically develop and grow naturally, however some skills that are a little more specialized may require extra attention to increase.

For each action attempt performed, the skill's value will:

1. Calculate the chance of success or failure for the action.
2. Determine the chance of skill gained for the action (failed or successful).
3. If successful, determine the efficiency of the action performed
4. If successful, determine the outcome of the action (this will vary).

Skill Maximum limits

Any single skill has a maximum limit value of 1000. After this cap has been reached further actions utilizing that skill will cease to raise the skill value any further.

Unfortunately in a complex and competitive PvE and PvP game of this type, we simply cannot allow players to exploit uncapped skills to the point of breaking all game content and leaving the potential to grief, troll or otherwise harass lesser players. While a player can max out all skills, gain all abilities, learn all spells and be able to do everything, all maxed players will be equal in regards to the tangible skill cap limits.

Action Difficulty

Actions will be given a base difficulty value to determine how hard the action is. This value is a range between 1 to 99 and is assigned to the node, game object or other element. In the case of combat, all related skills for that attack/action are added up to determine the player's "difficulty" level check.

Example A;

Robert is mining Iron ore, an iron ore node has an assigned difficulty value of 40. Robert's mining skill is 80, which means Robert will have a very high chance of success to collect Iron Ore as well as a decent amount of it per success, but also he will experience very low skill gain from each successful harvest.

Example B;

Ken is attacking a Skeleton with a sword, the monster has an assigned combat difficulty value of 160. Ken's attack action relies on 4 skills, that are 40, 50, 20, 40. Added together, Ken's combat difficulty check is 150, which puts him on-par with the Skeleton and allows for challenging, yet equalized chance of success and decent skill gain for the skills involved in the attack action as well.

Example C;

John is a new player and has 0 skill in Fishing. John attempts to fish in a nearby pond. The act of fishing has a table of difficulty associated with the outcome, with John's low skill, the chance to catch nothing at all is very high, but each time John attempts to fish, his skill gains are considerable despite catching no fish, after about 15 minutes of fishing, John has increased his skill from 0 to 15 and is now able to catch a basic fish 15% of the time. When John catches a fish, the skill gained further advances his Fishing skill to 18, increasing his chance of success.



Success or Failure

Chance for success or failure is based on a simple check vs difficulty of the action. When a player performs the action, behind the scenes we are adding up all the player's skills related to that action.

Example A; Jim is mining raw Iron Ore. We add up the related skills for the action, which consist of Mining (80) and Smelting (40). Jim's skill score totaled is 120.

Iron Ore has a difficulty value (DV) of 40, which Jim easily matches and exceeds by +200% with his score of 120. Because we have a hard cap of 99% chance of success, Jim easily succeeds at harvesting Iron Ore from the node.

Example B; Mike is mining raw Iron Ore as well. Mike's skills are Mining (19) and Smelting (0). His total score is 20 (19+1) which is half of the Iron Ore's DV. This means that Mike has a 50% chance to succeed in harvesting Iron Ore from the Node.

Sliding scale of skill gain

For every action taken, the player has a 1% to 99% chance to gain 1 skill point in the primary related skill and half that chance to gain 1 skill point in a single secondary related skill at random (if more than one). The chance to gain a skill point is directly tied to the chance of success. The higher the chance of success, the lower the chance to earn a skill point for the action.

Example;

Jim caps out his 99% chance to mine Iron Ore as above, he only receives a 1% chance to increase his Mining skill by 1 skill point because it is too easy for him and 0.5% chance for +1 Smelting. Mike on the other hand has a 50% chance to get ore, but also a 50% chance to gain 1 skill point for Mining and 25% chance for +1 to Smelting for the same action as Jim.

Bonus chance for skill gain upon Success!

If the action attempt succeeds, the primary and secondary skills are granted a +20%/+10% bonus chance to gain + 1 skill point, respectively. The cap is still 99% however even with the bonus skill gain chance, but makes successful attempts more likely to award skill gain than failures.

Combat Success

Combat Success

A successful action during combat is how a player lands a hit on their target. In this case, it is considered the HIT CHANCE. This is determined between the related combat skills, the ACC of the weapon (or spell) and modified by the target's Avoidance skill.

Example; Kim wielding a Long Sword (ACC 40), with Tactics (20), Focus (30), Blades (40) and Anatomy (10) against a Skeleton with a DV of 160.

$160 \text{ DV} - 40 \text{ ACC} = 120 \text{ CV}$ (combat value). This is this the expected value to meet or exceed to be successful. Kim's skills add up to $20+30+40+10$ for a total of 100 CV.

$100 \text{ CV} / 120 \text{ CV} = 83.33\%$ Chance to Hit the skeleton successfully and deal damage. (once a successful hit has been determined, damage calculation then is figured out).

Player vs Player

Similar to above, PVP combat hit chances are figured out as such;

Example; Kim wielding a Long Sword (ACC 40), with Tactics (20), Focus (30), Blades (40) and Anatomy (10) against Matt with an Avoidance (200).

$200 - 40 = 160 \text{ CV}$ to meet or exceed. Kim's skills add up to $20+30+40+10$ for a total of 100 CV. $100 \text{ CV} / 160 \text{ CV} = 62.5\%$ Chance to Hit and deal damage to Matt.

On the other hand, Kim tries to attack Nathan with Avoidance (1000).

$1000 - 40 = 960 \text{ CV}$, $100 / 960 = 10.41\%$ Chance to Hit Nathan and deal damage.

** NOTE - Some worn items may further modify HIT CHANCE! Resulting in minimum 1% chance to hit the target.*



Skill Efficiency

In the case of harvesting materials, the skill's success chance also determines the rate at which the collected material is multiplied. The minimum multiplier is 0.5x (or 1, whichever is greater) maximum multiplier for efficiency is 5x.

Example A; Jim is mining raw Iron Ore. We add up the related skills for the action, which consist of Mining (80) and Smelting (40). Jim's skill score totaled is 120.

NOTE: Resource amounts harvested from nodes per action will be defined in data. In this example we assume Iron Ore has a harvest value of 1.

Iron Ore has a difficulty value (DV) of 40, which Jim easily matches and exceeds by +200% (2x) with his score of 120. This means that Jim collects 1+2 (3 total) iron ore from the node instead of the default of 1.

Example B; Mike is mining raw Iron Ore as well. Mike's skills are Mining (19) and Smelting (0). His total score is 20 (19+1) which is half of the Iron Ore's DV.

In this scenario, let's assume Mike successfully attempts a harvest. Because his skill check was 50% (0.5x) of the required DV he only gains 1 Iron Ore (the minimum) from the success.

Critical Hit

We use the difference in the HIT CHANCE to determine the critical hit chance after a successful hit connects with the target.

Example C; Kim wielding a Long Sword (ACC 40)(CRIT 20), with Tactics (20), Focus (30), Blades (40) and Anatomy (10) against a Skeleton with a DV of 160.

From this, we already know Kim has a 83.33% Chance to Hit the skeleton successfully and deal damage. Next we determine if the Hit is upgraded to a Critical Hit by using the difference plus any modifiers or buffs.

The sword's CRIT is 20(%) of the chance to hit of 83.33 that Kim succeeded with. This means she gets an additional 16.66% chance that this HIT will become a CRITICAL HIT and deal additional damage.

Example D; Bob has a 50% chance to hit the same Skeleton with the same sword, he lands the blow and is calculated (CRIT 20) out of the 50% chance to hit. Meaning he has an additional 10% chance to upgrade to a Critical Hit.

Example E; Victor has 99% chance to hit the Skeleton with a sword with CRIT 100 attribute. This would max out Victor's Critical Chance at 99% to upgrade to a critical hit.

Example F; Nathan has a 10% chance to hit the Skeleton with a CRIT 10 axe. He barely succeeds in landing a blow, but he does still have a minimum of 1% chance to upgrade to a critical hit, although how unlikely that is, it is still possible.

** NOTE - Some worn items may further modify HIT CHANCE!*



XP Gain

When a skill (or action) is performed successfully, the player is awarded a dynamic amount of a specific XP per the skill(s) being utilized. The amount of XP awarded is based on the difficulty check. The more difficult the action (and less likely the success) the more XP is awarded should it succeed.

- The primary skill gains 100% XP, where as the secondary skill earns 50% of that.
 - *Ken successfully performs a Mining action and is awarded Mining XP and half Smelting XP.* Failing a skill check (or action) awards 10% of the potential XP from a successful attempt. This way players can still try to gain XP even if they are unskilled until they are more successful and able to gain full XP rewards.

XP gain based on Difficulty Values (DV)

Example A; Jim is mining raw Titanium Ore. We add up the skills for the action, which consist of Mining (80) and Smelting (40). Jim's skill score totaled is 120.

Titanium Ore Mining Node has a difficulty value (DV) of 300 (**see Data for specific DV values!*), which Jim is far below with his score of 120. Based on this, Jim has a 40% chance of success based on his skill vs the DV of the ore node, which we use $(X / Y) * 100 = P\%$, where X is the Player's Skill total and Y is the Difficulty Value (DV).

Unable to perform the action

If the player does not meet a minimum chance of >25% to succeed, they cannot attempt the action and will be given an error such as "*Skill too low!*". This will prevent players with low skill from trying to milk much harder nodes for failure XP. *But this does not count as a successful or failed attempt!*

XP Reward Amounts

Example B; Jim is mining raw Titanium Ore. We add up the skills for the action, which consist of Mining (80) and Smelting (40). Jim's skill score totaled is 120.

- Upon Success if total skill values are **under** the DV check: Jim receives 1 BONUS XP per DV point difference (in this case, $300DV + 120SKILL = +420$ MiningXP & $+210$ SmeltingXP).

Example C; Ted is mining raw Titanium Ore. His skills are much higher, which consist of Mining (800) and Smelting (400). For a skill score totaled at 1200.

- Upon Success if total skill values are **over** the DV check: He receives -1 PENTALY XP per DV point difference with the minimum XP rewarded being 100% of the original DV. As such, in this case, $300DV - 1200SKILL = -900!$, so instead we give a minimum of $+300$ MiningXP & $+150$ SmeltingXP) with a capped success rate of 99%.

Example B; Jim is mining raw Titanium Ore. His low skill which consist of Mining (80) and Smelting (40). Jim's skill score totaled is 120.

- Upon **Failure**: Jim receives 0.1 XP per DV point difference (which is $300DV - 120SKILL = \text{ROUNDUP}(180 \times 0.1) = +18$ MiningXP & $+9$ SmeltingXP). Or in Ted's case, $+2$ MiningXP and $+1$ Smelting XP, but Ted will likely never fail even with a 1% chance due to his high skill levels.



Level Gain

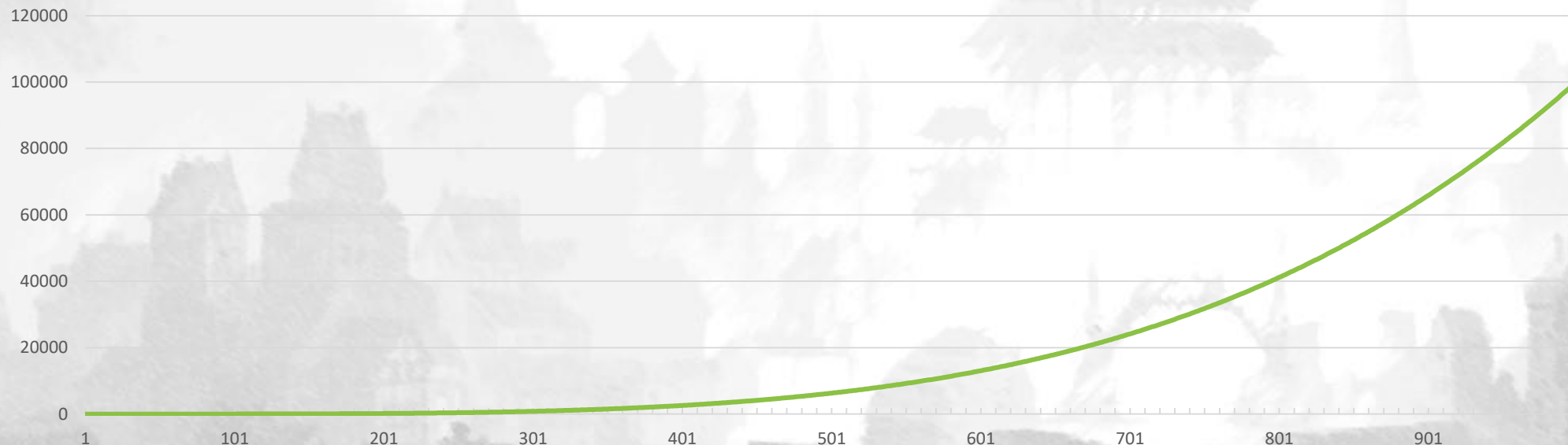
BASIC SKILL LEVELS CALC: $\text{ROUNDUP}((\text{BaseXP} * (\text{Level} ^ \text{MultVal})) / \text{DivVal}) = \text{XP for next level.}$

(Data from DragonFroge_Data_v13.xls) BaseXP=1, MultVal=4, DivVal=10000000

Skill levels increase from cumulative XP gain, that is to say that the XP acquired from the previous level up is not reset to zero. For example; skill level 120 requires a total of 543xp (in that specific skill, such as MiningXP). To reach skill level 121, requires a total of 564xp. So to go from 120 to 121 just needs 22xp (or 564xp totaled).

To reach max skill level of 1000 in any one skill, it requires a total of 20,050,551xp earned for that specific skill. Players should find the pace swift upwards towards skill level 500-600, with slowed progression beginning around 600+

Skill_Gain_Global (Curve)





Skill Master List (Combat)

Skill Name	Governing Action	Description
Force	Combat	Strength behind each blow (melee)
Tactics	Combat	Melee talent (accuracy, aim and precision)
Archery	Combat	Ranged force (punch-through power)
Aim	Combat	Ranged talent (accuracy, aim and precision)
Potency	Combat	Magical power and magnitude
Essence	Combat	Magical talent (accuracy, aim and precision)
Focus	Combat	Improves hit chance, critical hit and critical hit multiplier
Recovery	Combat	Reduces knockback, stun and interrupted actions
Unarmed	Combat	Weapon skill governs all unarmed melee (weaponless attacks)
Blade	Combat	Weapon skill governing the use of swords and blade weapons
Blunt	Combat	Hammers, Bats, clubs and the like
Crushing	Combat	Maces, mauls and warhammers
Piercing	Combat	Spears, polearms, some swords, bows & arrows, xbows
Throwing	Combat	Throwing knives, ninja stars, javelins...etc
Spellcraft	Combat	Governs learning and using spells
Anatomy	Combat/Healing	Improves attacks and healing on humanoids
Monster Lore	Combat/Healing	Improves attacks and healing on monsters
Mapping	Crafting/Searching	Governs ability to read, decipher and make maps
Healing	Healing	Governs health recovery speed and amount
Meditation	Healing	Governs magic power recovery speed and amount

Skill Name	Governing Action	Description
Unarmored	Combat, Defensive	Naked and very light clothing
Light Armor	Combat, Defensive	Leathers, heavy cloth, scale armor...etc
Medium Armor	Combat, Defensive	Chainmail, banded, ringmail, simple plate armors...etc
Heavy Armor	Combat, Defensive	Heavy chainmails, platemail, dragonscale...etc
Shields	Combat, Defensive	Governs the use of a shield (additional protections)
Defending	Combat, Defensive	General skill for taking incoming blows that connect
Avoidance	Combat, Defensive	General skill for attempting to avoid incoming blows
Magical Resistance	Combat, Defensive	Reduces/negates harmful spells.
Magical Reliance	Combat, Defensive	Increases beneficial spell effects.
Sprinting	Travel/Combat	Governs sprint bar, cooldown and regeneration
Riding	Travel	Allows player to saddle and control mounts
Traversal	Travel	Governs player's movement and very common actions
Trapping	Travel/Combat	Detect, identify, set and disarm all types of traps.
Detection	Searching/Combat	Detection of hidden things
Stealth	Travel/Combat	Ability of player to remain unnoticed, silent and invisible
Reputation	Questing	Affects quest difficulty and rewards
Commerce	Barter, NPCs	Pricing and availability of goods
Trading	Barter, Players	Lowers taxation between players
Identification	Economy	Able to reveal details and properties of an item
Writing	Crafting	Ability to read, write and craft things like books/scrolls



Skill Master List (Crafting)

Skill Name	Governing Action	Description
Alchemy	Crafting	Creation and brewing of potions and mixed substances
Medicine	Crafting	Crafting of tools, bandages and other remedies.
Cooking	Crafting	Cooking of food, brewing of ales and other consumables
Camping	Crafting	
Farming	Crafting	
Gathering	Harvesting	
Lockpicking	Access	
Thieving	Access	
Enchanting	Crafting	
Recharging	Repairing	
Armor Smithing	Crafting	
Repairing	Repairing	
Weapon Smithing	Crafting	
Tooling	Crafting	
Smelting	Crafting	
Mining	Harvesting	
Lumberjacking	Harvesting	
Forestry	Searching	
Hunting	Searching	
Fishing	Harvesting	



Questing

Primary Story Arc

Secondary Quests

Daily Quests & Activities

Event Quests



Items

Every item fits within 1 bag slot. Items can stack and stack size will vary depending on the item. In the case of GOLD, it shall always have a very large stack size.

Items can be moved by drag & drop and other actions/commands can be performed by accessing the ITEM SUB-MENU by tap and holding on the item's icon for >1 second (alternatively enabling access via a double-tap through the settings menu).

LINKED items

The term “**LINKED**” refers to items that are bound to a player character and cannot be sold, dropped, stolen, discarded, destroyed or traded away to another player. Linked items can be destroyed (deleted) by tap and holding on the item to access the sub-menu and selecting “Destroy”.

Linked items may be stored in the player character's bank or personally-owned storage container. In the case of shared storage, players may view and see the LINKED item, but cannot pick it up or take it.

UNLINKED items are those not yet bound to a specific character. Upon use or equipping of this item will then LINK it.

Obtaining items

New game objects (items) are generated in several ways:

- Purchased from NPCs (tools, basic equipment)
- Crafted by Players (usable equipment, refined materials, tools)
- Materials from corpses of slain enemies/NPCs (never alive ones!)
- Harvested/Collected from world resource game objects (nodes)
- Discovered within treasure chests/crates (game object containers)

Garbage cleanup

Items are removed from the game world regularly with priority given to items crafted by players (to help resolve inventory dumping). Any item dropped on the ground directly is given a removal timer of 30 seconds before it is deleted. This includes quest-related items, these are given a removal timer of 60 seconds (so be warned).

Items placed within static containers (such as player-owned containers placed within their homes) are not given any removal timer and shall remain forever. Items placed within dynamic containers or world game object containers (such as corpses or spawned chests) are subject to removal when the container is deleted.

Picking up an item and placing it into a player inventory or static container refreshes this timer if then it were placed back onto the ground.



Quality

Items are classified into 10 tiers of quality rating; *(not explicitly by drop rarity)*. Color coding refers to the color of the item's full name.

Tier	Quality Label	Name Label Color	Max # of Bonuses (affixes)
0	Quest & Story Items	Light Grey, #8c8c8c	n/a
1	Basic	Tan-Brown, #aa9570	0
2	Ordinary	White, #ffffff	1
3	Average	Green, #00c000	2
4	Superior	Aqua/Teal, #00c0c0	3
5	Exceptional	Blue, #0060c0	4
6	Marvelous	Purple-Blue, #8000c0	5
7	Mythical	Purple-Red, #c00080	6
8	Legendary	Yellow, #c00080	7
9	Godly	Gold-Yellow, #ffb400	8

The tier is determined by the item's overall properties. Higher tier items have greater potential.

This rating includes the item's overall purpose, items such as quest-related items are precious and receive their own classification in the case of very specific items.

Improvements & Upgrading

Under certain circumstances, most every item can be upgraded to the next tier (and/or downgraded). Items can be “re-rolled” as well as re-configured for optimal stats, properties and/or attributes through crafting methods.

Methods of improving an item include:

- Tiering – Refers to altering the item's general quality rating tier.
- Rerolling – Refers to regenerating the item's random properties.
- Retooling – Changing the base item's standard properties.
 - Reinforcing – Improving durability
 - Armoring – Improving armor value
 - Sharpening – Improving damage value
- Affixing – Manipulating the item's individual affix properties.
- Enchant/Bless – Granting bonuses to the item (or remove curses).
- Stripping – Returning the item to it's very basic version.
- Smelting/Reclaiming – Refers to destroying the item in exchange for raw materials.

Example: Robert up-tiers his Long sword from T1 to T2 with his high Blacksmithing skill. Allowing the sword to have improved properties and more affixes.



Dragonforge | Item Generation

Affixes and Tiers

Like with items that allow more affixes per tier, affixes themselves have the capability to also be increased in tiers, allowing for the affix to become more powerful.

Example: Kevin has received a +Cold Damage affix on his newly crafted broadsword. This affix begins as Tier 1. The affix's starting value is randomly generated based on it's Tier 1 min and max range (see data tables).

As Kevin up-tiers his Broadsword from T1 to T2, the affix remains unaffected and at Tier 1.

To increase the affix to the now max item Tier of level 2, Kevin must use his Enchanting skill to raise the +Cold Damage affix from T1 to T2. Upon reaching T2, the affix's value is re-rolled for T2.

In the end, a T10 item can have up to 8 affixes total, each of these affixes can reach T10 for an "item level" of 90 ($\text{itemLVL} + (\text{affixLVL} * 8)$)

Improvements & Upgrading

- Players may only add affixes if the item allows additional affixes. Up to the item's maximum affix limit (based upon item tier).
- Players may not remove affixes once placed on an item.
- Affixes always begin at Tier 1 and have a maximum of Tier 10.
- Enchanting skill is utilized to manipulate affixes.
- Affix values are re-rolled/re-generated upon tier increase.

Example: Robert up-tiers his Long sword from T1 to T2 with his high Blacksmithing skill. Allowing the sword to have improved properties and more affixes.



Vendor purchased items

Basic weapons, armor, materials, tools and other entry-level items are available for purchase from NPC vendors in every city and town. These include everything a new player with low skill would need to get started in the world of Dragonforge. Items purchased from an NPC vendor are always Tier 1 equivalent and do not randomly generate affixes, but rather are standard “basic” items.

For example; buying a sword from an NPC results in the same basic, non-magical Tier 1 weapon without randomized stats.

Found items (chests, nodes & corpses)

Only raw, un-refined materials are found on corpses (*for example; coin, scales, hide, bone...etc*). Monsters/Enemies never drop item-generated ready-to-use equipment, this includes humanoid NPC enemies wearing equipment. Instead, humanoids drop purses containing randomized amounts of coin rather than a raw body part.

- Dynamically spawned treasure chests (cache nodes) on the other hand, have a chance to contain a randomized, item-generated piece of equipment depending on the “level” of the chest as well as other tangible goods such as refined & raw resources, coin and other goodies.

The purpose of preventing item-generation on corpses is to reduce waste and garbage collection, instead of generating thousands of “worthless” items found on enemies (thus requiring more filtering), instead we find usable materials to feed and fuel the crafting system so players can improve skills, establish a player-run economy as well as create specific items and equipment they need. Hunting enemies, farming, mining and harvesting resources is then more structured and intentional.



Crafting with Materials

Each recipe/blueprint consumes a number of raw materials (including upgrading). These materials can also be translated into finished goods that are also used within the crafting system.

For example; Milling raw logs into usable lumber or smelting raw ore with flux to create ingots of pure metal. Another example is cooking raw meat to create roasted and edible meals.

Material substitutions

When crafting, if a recipe or blueprint requires specific material type (such as Ingots), the player may place any valid “*Ingot*” item type into this recipe slot. Not every recipe will allow substitutions however and may require explicit materials.

Example A; Bill is blacksmithing a long sword, he can place Iron ingots into the ingot slot or use his Adamantium ingots to craft the sword. The end results will vary depending on the quality of materials Bill uses.

Example B; Ken wants to mill lumber to craft bows, he can choose to mill his Maple logs, but decides he wants to craft powerful Etherpine bows instead, so he begins milling his Etherpine logs with the same “lumber” recipe as he would use with Maple.

Using Tools

Some recipes/blueprints require the use of tools to process or refine materials. Tools have a “Uses” durability to them before they are consumed and break. The quality level of the tool itself increases the chance of successful crafting (does not affect quality of the end product).

Some materials require explicit tools for crafting and making use of specialized workbenches, forges or other workstations. These requirements will be listed within the recipe/blueprint itself.

Finished Goods used

Many recipes require and consume finished goods to produce the desired item. An example of this is “ingots” used in nearly all metalworking recipes. These are fully consumed immediately once the item crafting ends (either successfully or failed).



Material Effects

Crafting items with different materials will result in changes in the finished item. We know that creating a sword with iron ingots will result in a basic iron sword and repeating this recipe with adamantium ingots results in the same type of base sword made out of adamantium.

Plans: Swords - Tier 1 – Cutlass

Ex: 2 Iron Ingots = Iron Cutlass, (T1)

Ex: 2 Adamantium Ingots = Adamantium Cutlass, (T1)

Additionally, the material used applies bonus properties to the base item, these bonus properties also scale with the item's tier rank.

(These values not final, see data tables)

Ex: Iron grants +1% additional damage and -1% accuracy, per tier for weapons (+10%/-10% @ T10)

Ex: Adamantium grants +2% additional damage and +3% additional durability, per tier for weapons. (+20%/+30% @ T10)

Ex: Etherium enhances affix values by +2%, per tier and has a -2% damage per tier. (+20%/-20% @ T10),

* Enhancements - if an affix granted +10% damage, for example, it would be additive (10%+20%=30%) @ T10).

Each material shall have benefits and penalties, the player can choose the material to craft their item with the effects and bonuses they desire and that best fit their character build.



Equipment Generation, THE BASICS

When an item is crafted, it shall generate and randomize it's properties to match the assigned tier.

**See data tables for values, these are not included in the GDD due to space and complexity.*

Material

This determines the item's base attribute properties, this includes damage/defense values, durability and the item type. Materials can be used to craft a Tier 1 weapon or Tier 10. *(for example, a player can use Iron to craft a T1 Iron sword or a T10 Iron sword)*. Every material has both strengths and weaknesses inherent to that material's properties. Some materials are better for specific items, recipes or situations. Material used is independent of Tier (ie; Iron is not restricted to making T1 swords) it is usable for everything that needs ingots/metal as a resource). This ensures that even "simple" materials are still tangible in the beginning as well as the end-game or meta-game.

Tier

By default, all items are crafted and generated as **Tier 1** unless the recipe explicitly states the output Tier will be higher. *(ie; the recipe is a T9 helmet explicitly).*

- Item tier controls the number of affixes allowed. *(Tier 1 disallows any affixes, where as Tier 10 can have a maximum of 8 affixes).*
- Tier also determines the minimum and maximum property range values on an item when the item is generated. (ie; range of each individual attribute, such as damage/defense).
- Higher tier items also have a required skill value to equip and use. *(ie; a T8 sword may require Blade skill of 80+).* This is to prevent new characters from wielding extremely powerful items given to them from veteran players.

Base properties

Refers to the standard item values; Item Style, it's Damage/Crit for weapons, Protection value for armors as well as Durability.



Equipment Generation, AFFIXES

Affixes

Affixes refers to both **Prefix** and **Suffix** attachments to the item (specifically the item's name).

**See data tables for values, these are not included in the GDD due to space and complexity.*

*Example 1: **Flaming** Mithril Sword **of the Heavens***

An item can have up to two Prefixes and Suffixes (between 0-2 affixes), obviously this requires the item's Tier level to be adequate. For example, there shall never be a Tier 2 sword with both affixes as it can only support one prefix or one suffix, never both.

*Example 2 (no prefix): **Mithril Sword of the Heavens***

*Example 3 (single prefix, double suffix): **Flaming Mithril Sword of the Celestial Heavens***

*Example 4 (double affixes): **Extreme Flaming Mithril Sword of the Celestial Heavens***

Affixes add bonuses (up to 4, one for each affix given). These can alter properties and are always generated before any further bonuses. Affixes are always more powerful than additional Bonuses.

Creating affixes

Affixes are generated and assigned at random. These cannot be defined by the player during crafting.

- The number of affixes (0-2 prefixes and 0-2 suffixes) are random, if the item tier supports them.
 - Starting at Tier 5, the item has the potential to have the maximum number of both prefixes (2) and suffixes (2).
- Prefix/suffix assignment is random (50% chance for prefix, 50% chance for suffix, per affix roll up to the Tier maximum).
- Higher Tier items grant improved affixes and values to the affix bonuses. **see data tables.*
- Rerolling an item removes all affixes (and bonuses) and replaces them entirely.



Equipment Generation, BONUSES

Bonuses

Bonuses are additional attributes added/granted to the item after affixes are generated. An item can have a maximum number of bonuses as the item Tier allows for (see Tier table). For example a T10 item can have up to 8 total bonuses. The number of available bonuses is deducted from the number affixes generated.

- Ex; [T10] **Flaming** Mithril Sword **of the Celestial** adds 2 bonuses in total for “*Flaming*” and “*of the Celestial*”, which then only allows for a maximum of 6 additional possible bonuses for a total of 8.
- Ex; [T10] **Extreme Flaming** Mithril Sword **of the Celestial Heavens** adds 4 bonuses in total. Leaving room for only 4 additional bonuses.

Adding additional bonuses

- Bonuses from affixes are automatically generated to the item when it is crafted, these can only be changed by rerolling/removing.
- Affix bonuses consumes a “bonus slot” on the item. This will take away from the total bonuses allowed on an item.
- The remaining bonus slots on the item can be manipulated by the player by:
 - Enchanting, Smithing, Alchemy
 - *Some other appropriate skill, recipe or method.

The available bonuses (and values of the applied bonus a player can apply shall be determined by the recipe itself, tools used, skill level as well as materials utilized during the upgrade process.

- Players may stack one additional bonus of the same type as long as the affix itself is unique. (ex; “**+damage%**” and “**+damage(val)**” are two *unique* affixes, but both apply the same kind of bonus damage increase)
 - Players may never stack two of the same unique affixes (ex; disallow more than a single “**+damage%**” affix).
- This is to make affixes appealing for bonuses of the same type rather than stacking-up the same unique, single affix.



Dragonforge | Item Generation

Materials - Raw

For the sake of sanity, let's begin with 8 materials per material category.

Raw								
Metals	Copper (ore)	Iron (ore)	Platinum (ore)	Titanium (ore)	Chromium (ore)	Etherium (ore)	Mithril (ore)	Adamantium (ore)
Stones/ Gemstones	Stone (chunk)	Granite (chunk)	Obsidian (chunk)	Marble (chunk)	Ruby (uncut)	Emerald (uncut)	Sapphire (uncut)	Diamond (uncut)
Wood	Pine (log)	Maple (log)	Ironwood (log)	Birch (log)	Ashen (log)	Titanwood (log)	Eitherpine (log)	Castle Fir (log)
Flesh	Skin/Hide	Bones	Scales	Raw Meat	Claws/Horns	Heart	Brain	Fur/Hair
Veg/Fruit	Husk (corn)	Root (carrot)	Melon (gourd)	Berry (mixed)	Grain (wheat)	Leaf (cabbage)	Pods (peas)	Fruit (orange)
Mineral	Crystal (shard)	Sand (dirty)	Clay (dirty)	Salt (rock)	Coal (damp)	Sulfur (rock)	Flux (powder)	Pixie Dust (powder)
Reagent	Spider Silk	Lizard Eye	Venomous Fang	Batwing	Darksoil	Bile	Mushroom	Nightshade
Gathered	Worms	Fish	Seeds (various)	Feathers	Eggs (various)	Wool	Cotton	Aether
Liquids	Water (jug)	Blood (vial)	Acid (flask)	Oil (flask)	Wine (bottle)	Holy Water		



Materials - Refined

For the sake of sanity, let's begin with 8 materials per material category.

Refined								
Metals	Copper (ingot)	Iron (ingot)	Platinum (ingot)	Titanium (ingot)	Chromium (ingot)	Etherium (ingot)	Mithril (ingot)	Adamantium (ingot)
Stones/ Gemstones	Stone (block)	Granite (block)	Obsidian (block)	Marble (block)	Ruby (polished)	Emerald (polished)	Sapphire (polished)	Diamond (polished)
Wood	Pine (board)	Maple (board)	Ironwood (board)	Birch (board)	Ashen (board)	Titanwood (board)	Eitherpine (board)	Castle Fir (board)
Flesh	Leather (various)	Bone (powder)	Scales (cleaned)	Cooked Meat	Claws/Horns (sharpened)	--	--	Pelt (skinned)
Veg/Fruit	--	--	--	--	--	--	--	--
Mineral	Crystal (refined)	Sand (refined)	Clay (molded)	Salt (ground)	Coal (dried)	Sulfur (powder)	Ash	Pixie Dust (refined)
Other	Cloth	Thread	--	--	--	--	--	--
Gathered	--	--	--	--	--	--	--	--



Quality & Materials

Repairing Durability

ddd

Enchanting, imbuing and socketing



Quality & Materials

Repairing Durability

ddd

Enchanting, imbuing and socketing



Provisions

Player provisions such as food, drink, medicine, ammo and spell-related reagents.

Supplies, fuel & feed

Consumable goods and materials to fuel or feed mounts, vessels and power things like gateways.



Consumables

Potions, bandages, scrolls, foods, mana gems...etc

Non-Consumables

Tools, parts & goods, stationary objects, furniture, decorations



Raw Materials

Things gathered from the world, not dropped by men or monsters.
Ore, raw lumber, earth, water, plant products, animal parts/meat...etc.

Finished Goods

Processed and usable materials.
Leather, boards, ingots, vegetables, clean water, cloth...etc



Quest Items

Special non-transferrable, non-droppable, soulbound & unique.
Stored and sorted into it's own storage/inventory (tbd)

Other Items & Objects

Static objects, buildings, signs and world-objects.
Likely interactive like doors/levers, but mostly just for visual looks.

Creature & Monster loot

Common wealth and parts found on a monster's corpse. Keep it simple!

Elite & Boss loot

Rare items, exceptional wealth and unique things found on these corpses.

Human NPC loot

Things of value aside from clothing/armor/weapons. Such as coin, gems, jewelry and precious items.

Animal loot

Meat, bone, skin, parts...etc

Pre-generated items

Default things NPCs sell, such as basic materials, tools and consumables.

Player crafted items

Some items can only be obtained from crafting and/or other players.

Crafting GUI

When a player selects the type of crafting from the ACTION menu from the main HUD, such as [action→crafting→alchemy], the appropriate crafting UI will be displayed.

Crafting steps:

1. Select a recipe/plan/blueprint from the list on the left.
2. Tap each “component/ingredient” slot in the middle panel. The appropriate items will be displayed on the right side that are compatible with that selected ingredient slot.
3. Tap and choose the Material item(s) from the right panel to place into the selected recipe slot. A check mark will appear showing it was accepted.
4. Choose Craft ONE or Bulk Craft.
5. (optional) Toggle on to place the items into a bulk container (if one is created in the player’s inventory). If no container was created or the current container is filled/full, a new container will be created.



Example Crafting GUIs

Example layout of various crafting skills.





Dynamic Dropped loot

Do we want generated loot from enemies or just valuables/parts?

Item Keyword Definitions

Enhancement + **Prefix** + **Material** + Base Type + **Suffix**

Example: Enchanted Swift Iron Broadsword of Flame

Cities & Towns

World map diagram perhaps?

Level designer will need to flush out the street maps of each city/town.

The Wildlands (wilderness)

The generic outlying wilderness between and around civilization

Likely have farmsteads and whatnot.

Dungeons, Ruins & Crypts

Places to adventure and explore, pinpoint where the entrances and areas are located on the world map...

Interior Dungeon Layouts

Room-by-room flow, final map/layout to be done by level artist.

Spirit Realm

Another plane of existence we could expand to/add more content to.

Other Realms, Dimensions and Realities

A way we can bridge to other worlds/timelines if needed (expansion?)



Player Housing

Instanced player “space” accessible via NPC or teleport stone would be easiest...

If player housing takes up “public land” we could run short on housing space... so its first come first serve...

Arenas & Event Zones

Special zones/event areas that are used for special occasions. Maybe separated instances for holiday events?

Too many players in any one zone can lead to heavy server load and poor latency...

NPC Overview

Types of NPCs, monsters, creatures, animals..etc

Animal NPCs

Domesticated animals, farm/livestock, pets, rideable mounts...etc.

Human NPCs

Citizens, shopkeepers, guards, government..etc

Misc NPCs

Signposts, quest narration, spirits/ghosts, gods...etc

NPC GUI

When interacting with an NPC, a contextual GUI will be displayed to the player containing various information and selections.

Quest Giver GUI

Example below of when an NPC gives you a quest/task.



NPC Vendor GUI

Example below of when you buy/sell from an NPC.



Standard Monsters & Creatures

Every NPC is categorized into several class archetypes in addition to being sorted into one of the Melee, Ranged or Magic combat affinities. In total, there are 12 class archetypes:

Archetype	Description
Humanoid	Includes any NPCs that are human, elf, dwarf and so on. Nearly all townsfolk are classified as Humanoids. Enemies include human-like NPCs such as bandits, pirates, mages..etc.
Undead	Encompasses all NPCs that are now zombies, skeletons, ghouls, mummies and so on. This includes other NPCs such as Undead beasts, serpents and ex-living creatures.
Spirits	Any apparitions, ghosts, spirits, phantoms, angels or other extra-planar being are included in this classification.
Demons	Includes all creatures of the pit, such as imps, lesser & greater demons, hellhounds and demon spawns.
Beasts	A wide category to include animals both wild and domesticated (horses, farm animals, cows...etc) as well as wild animals such as bears, tigers, lions...etc.
Serpents	Includes all creatures with scaly skin and lizard-like features. This includes all snakes, wurms and dragons.
Insects	Every NPC that features insect-like qualities. Includes ants, spiders, bees, centipedes...etc.
Mythical	This category applies to fabled creatures, such as Manticores, Unicorns, Cyclops, Minotaurs, Phoenix...etc.
Soft	Specialized category for any misc NPC with a soft-body, such as slimes, oozes and jellies.
Hard	Specialized category for any misc NPC with a hard-body, such as rock golems, metal monsters and gemstone-laden critters.
Elite	This category overrides all other categories if the NPC is of Elite-status, for example an “Elite Spider”.
Boss	Unique category only assigned to boss NPCs.

Story Encounter Monsters & Creatures

Boss monsters that are involved in a primary/secondary quest story arc

World, Dungeon & Realm Bosses

Boss monsters that are located within dungeons, challenges and generally do not require a quest to fight.



NPC Artificial Intelligence

All NPCs are governed by several basic AI scripts and behaviors are called “brains”, these are to be defined as the NPC’s default behavior and can be overridden with more advanced scripts and patterns as desired to create more complex behaviors.

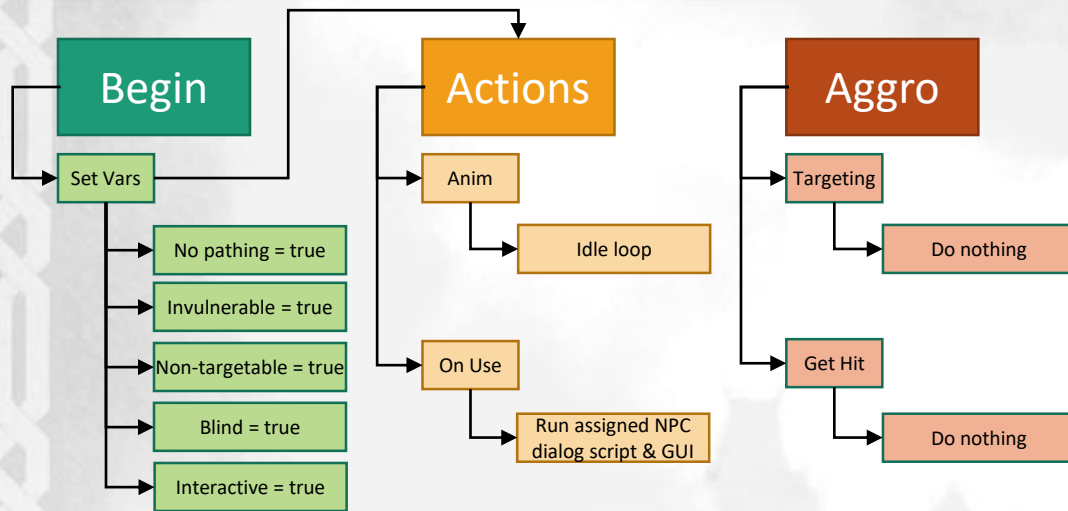
Non-Combat Brains

- **Static** – This script dictates the NPC shall remain stationary. Primarily used to control town NPCs, shop keepers and quest givers. While stationary, this script shall not seek targets or react when attacked. For the most part, any NPC governed by the “Static” brain is invulnerable (and may not be targeted by players or other NPCs) unless switched to another brain type.
 - In the case if we need “Static” NPCs to path find/move, then they will be set to pathing enabled, else remain invulnerable.
- **Idle** – NPCs assigned to this behavior shall path-find randomly and target seek within line of sight. This is the most basic default behavior of every enemy NPC. Within the combat brains, shall be a behavior branch that returns the NPC to “Idle” under a set of conditions.

Combat Brains

- **Melee** (Combat) – The script that directs the NPC to target nearest (and strongest) and rush to approach. Once within close distance, begins initiating melee combat. Melee brains are very aggressive, focused and are unlikely to retreat.
- **Ranged** (Combat) – This behavior is one where the NPC is to target the nearest threat and maintain distance from that target (and switching targets frequently), which means they will often cease attacking in favor of moving away. After a set distance, the NPC will no longer flee and will hold it’s ground to prevent long and annoying chase situations.
- **Magic** (Combat) – Similar to the “Ranged” brain, this script will aim to keep the caster at a distance while casting a variety of spells both offensively and defensively. This brain likes to target the weakest victim first.

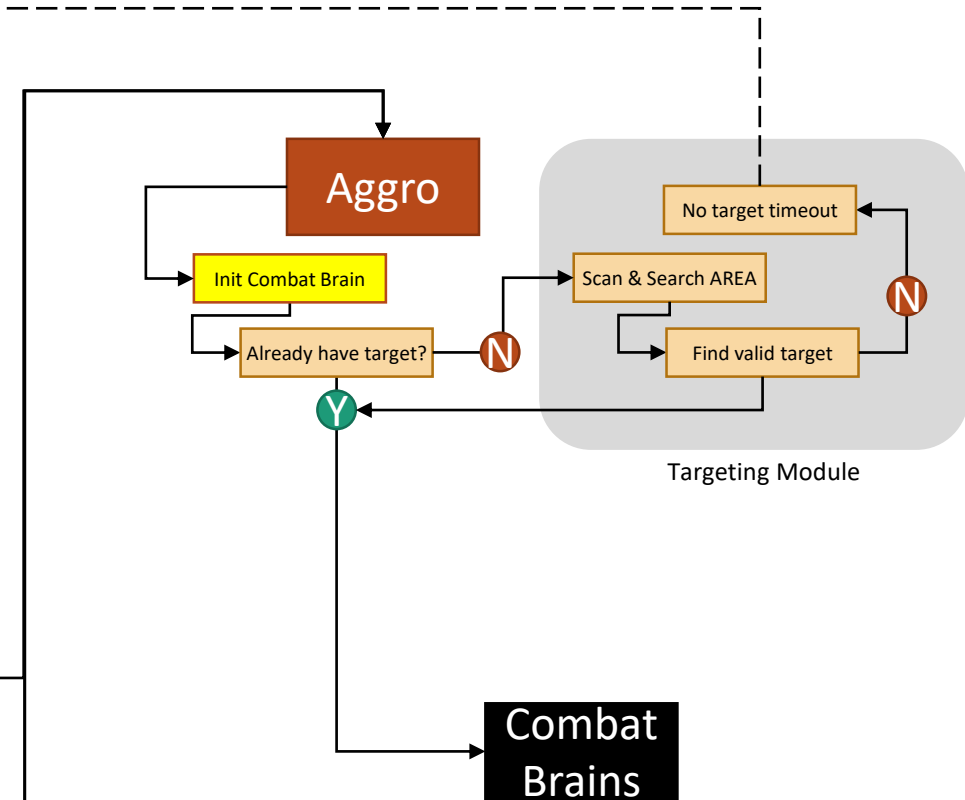
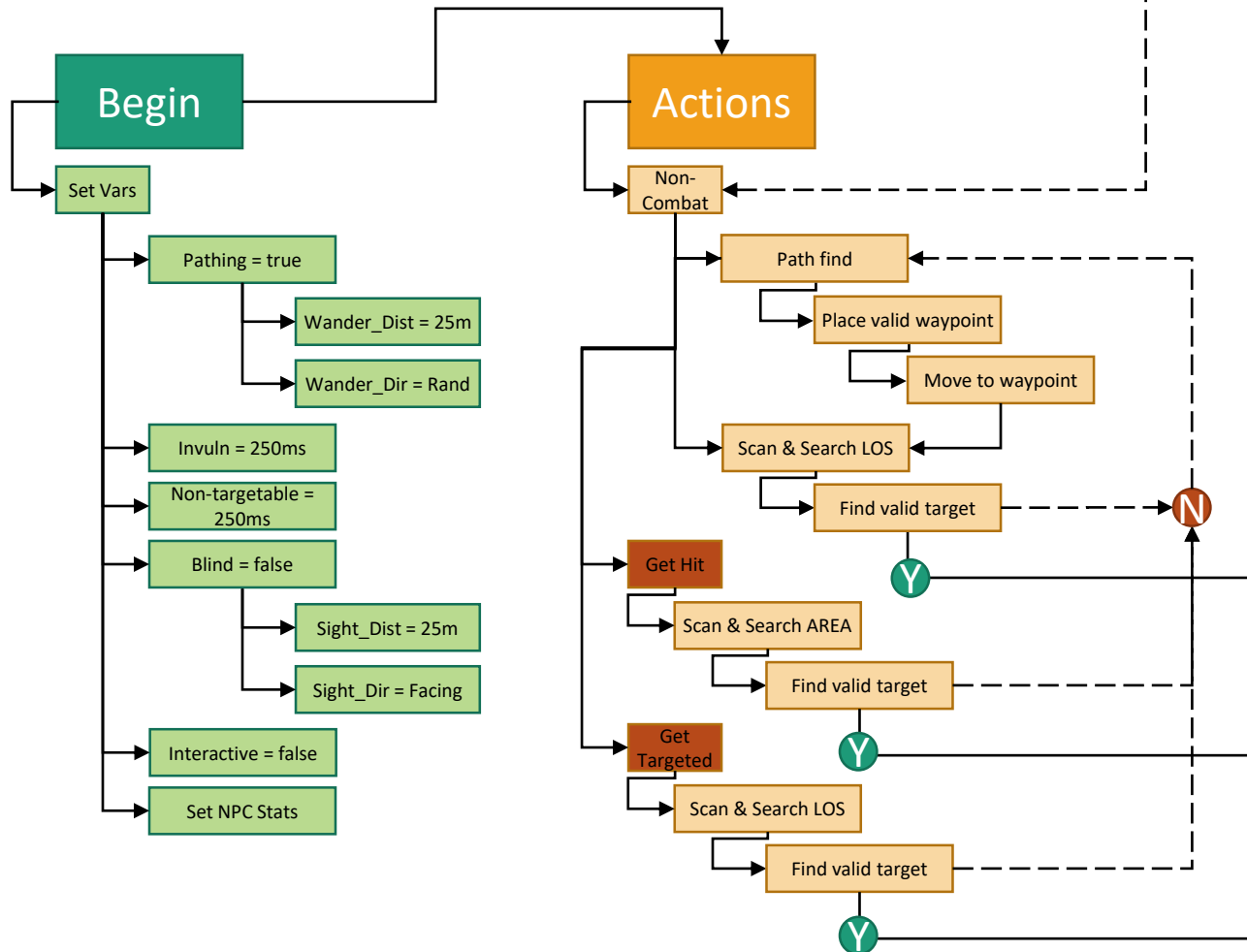
Static Brain (Town & Quest NPCs)



Static brain NPCs can be switched to Idle brains (and then to combat) if desired, however this is the most basic version of the brain to instruct the NPC to take no actions other than to be still, animate and be interactive if activated by the player.



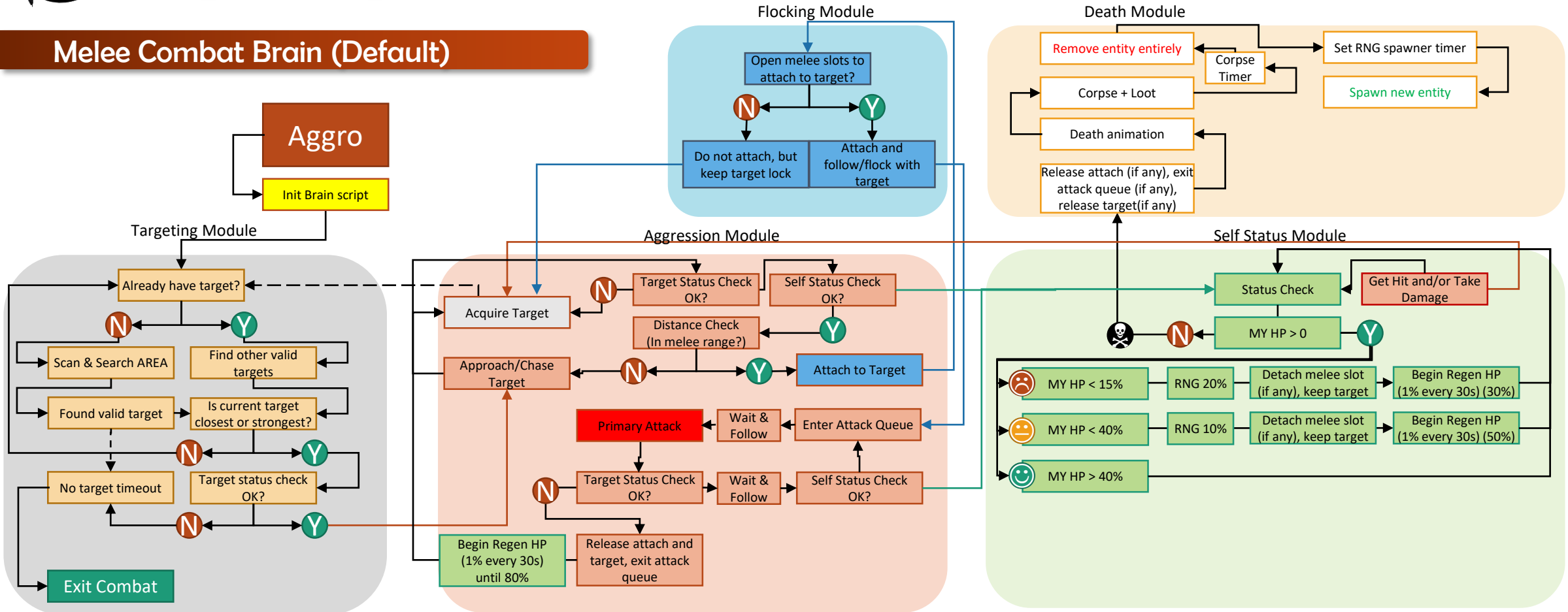
Idle Brain



All NPCs begin with idle brains with the exception of those with default Static brains. When an NPC becomes aggressive (aggro'd) towards a player, it switches from its idle brain routine into a combat brain depending on its combat affinity.



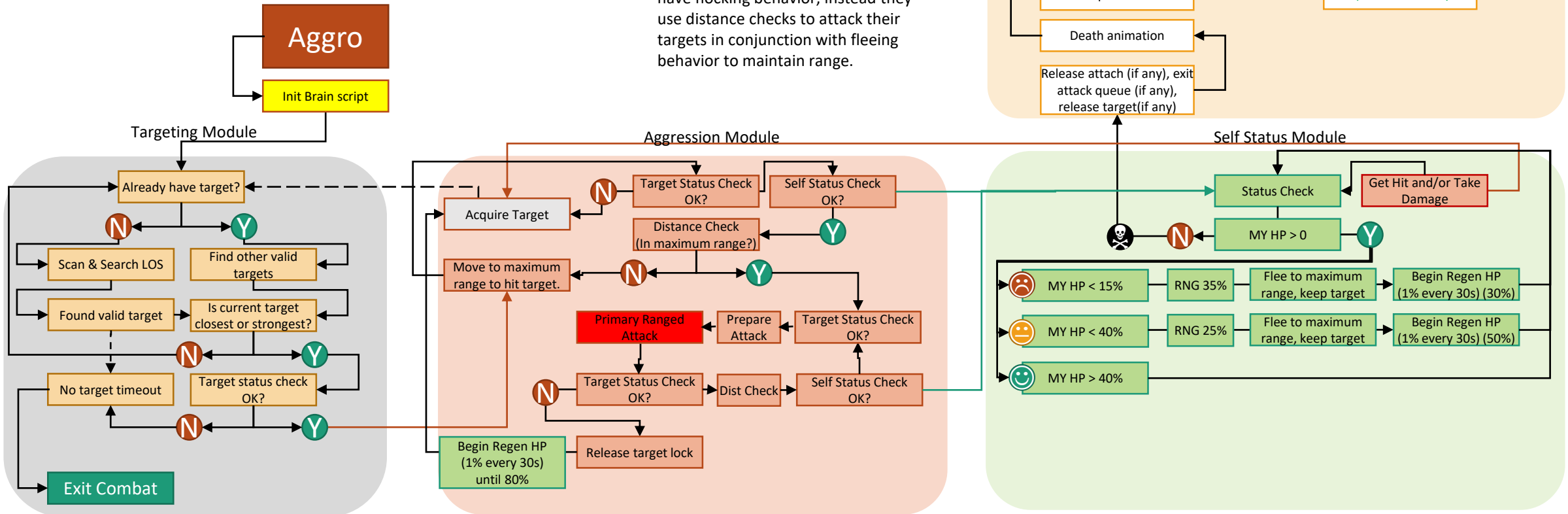
Melee Combat Brain (Default)



- **Melee (Combat)** – The script that directs the NPC to target nearest (and strongest) and rush to approach. Once within close distance, begins initiating melee combat. Melee brains are very aggressive, focused and are unlikely to retreat.



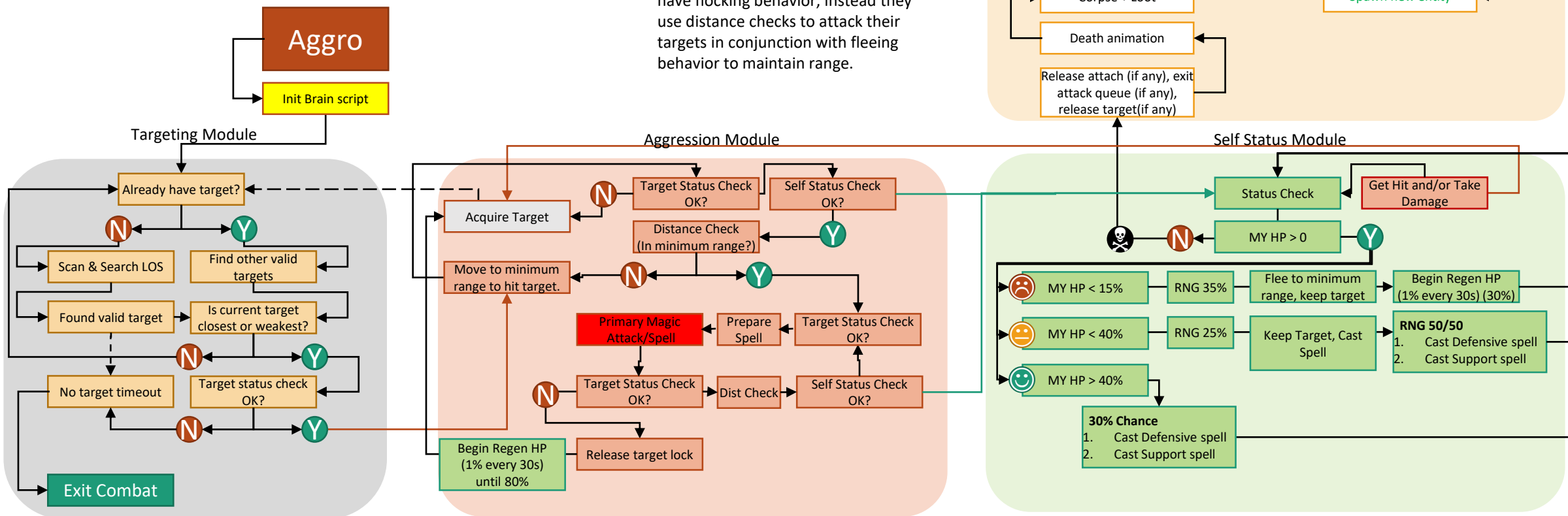
Ranged Combat Brain (Default)



- **Ranged (Combat)** – This behavior is one where the NPC is to target the nearest threat and maintain distance from that target (and switching targets frequently), which means they will often cease attacking in favor of moving away. After a set distance, the NPC will no longer flee and will hold it's ground to prevent long and annoying chase situations.



Magic Combat Brain (Default)

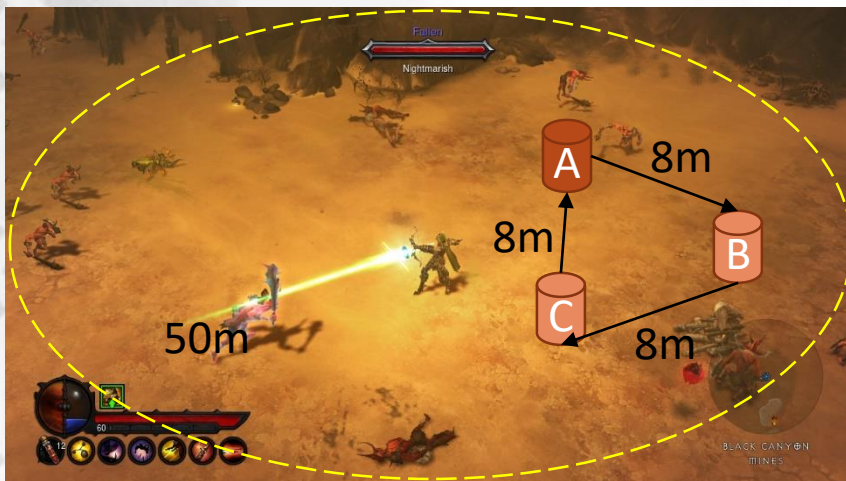


- **Magic (Combat)** – Similar to the “Ranged” brain, this script will aim to keep the caster at a distance while casting a variety of spells both offensively and defensively. This brain likes to target the weakest victim first.

NPC AI, Pathfinding

NPC pathfinding relies upon the entity setting a series of valid waypoints once it spawns and as it patrols. These can be both pre-arranged “patterns” as well as randomized “scattered” paths.

Defined pattern from point of spawn



In this example, the enemy entity spawns in a random location at point “A” and assumes all waypoints are valid/clear and reachable with no targets in sight...

1. Upon spawning, the AI checks to see if it has a clear path 8 meters to the SE. If it does, it sets this as it's next destination in it's pathing/patrol queue if it finds no targets to chase. It stores point A as it's “starting point” for later use.
2. Upon reaching point B, it does another target scan and if nothing is found, it then sets another waypoint at C, which is another 8m from it's present position to the SW.
3. After reaching point C, and the target scan returns false, the NPC then will return to waypoint A (which it remembers/records upon spawning) to complete the patrol loop.
4. The NPC can also “rotate” or “shift” the defined pattern to find a new waypoint should it be blocked or unreachable. This can also allow the NPC to have a pattern, but also “wander”.

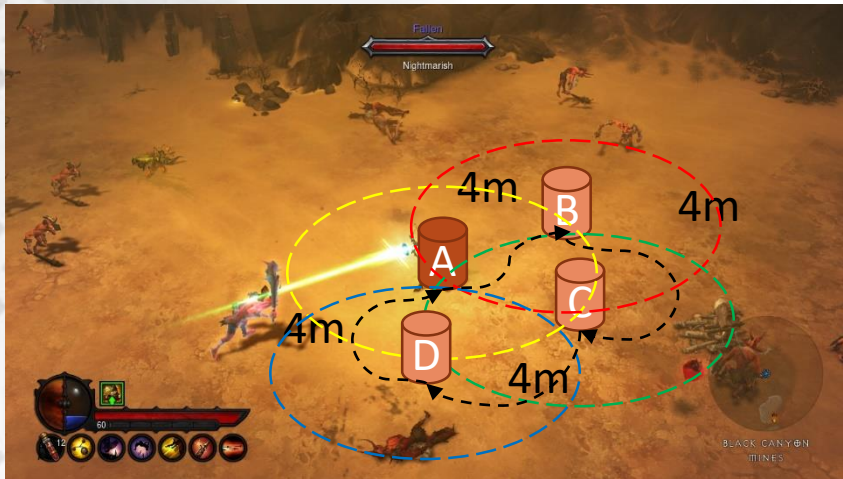
NPCs should have a 50/50 chance of taking the “shortest distance” to a waypoint or the longest distance if a route is blocked, if the point is still reachable. Otherwise, the pathing should time out after 2 seconds and the NPC should recalc it's defined pattern from the point of time out as if it spawned there.

* An NPC can have more than merely 3 total defined waypoints, however we should hard cap this at 8 for performance reasons.

NPC AI, Pathfinding

Scattered pattern from point of spawn

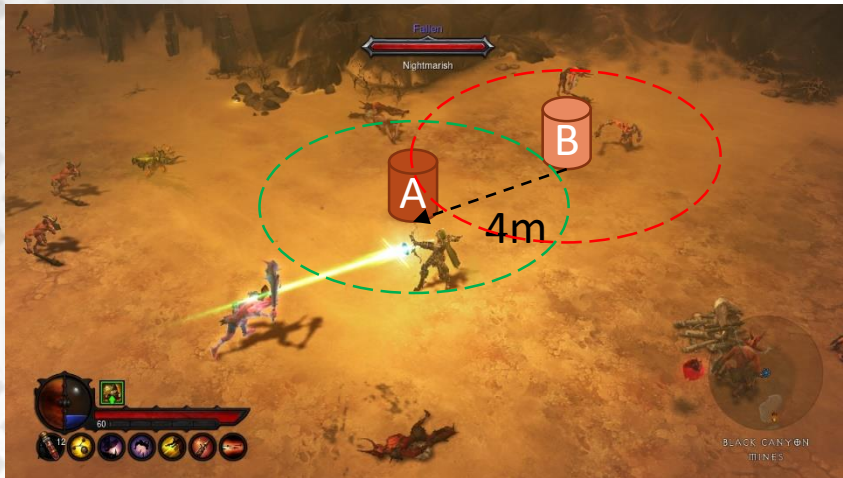
In this example, the enemy entity spawns in a random location at point “A” and assumes all waypoints are valid/clear and reachable with no targets in sight...



1. Upon spawning, the NPC places a waypoint at the outside range of it's “pathing” radius (shown in the image as 4 distinct colors) and chooses a random direction from 360 degrees. In this example, point B is 4m and towards the NE. It stores point A as it's “starting point” for later use.
2. When the NPC paths towards point B, it follows a non-linear curve path (Bezier spline), however if this is too taxing on performance/server load, a linear path can be used to walk between way points.
3. Upon reaching B, the NPC then generates a new RNG waypoint from this location and repeats it's pathing process for points C and D.
4. When reaching the final waypoint (D), the NPC will attempt to return to the starting waypoint “A” to close the loop. If the starting waypoint is unreachable or invalid, upon pathfinding timeout, the NPC will recalc the pathfinding and using the current location/waypoint as it's new “point A”.
5. Upon returning to the starting waypoint, all other waypoints are cleared and this cycle starts over, with all waypoints being re-generated at random, except for the “starting point”.

NPC AI, Following

Following is something that both Ranged and Magic combat brains will attempt to do when a player moves out of their range. By using distance checks to maintain and close the gap, these NPCs will try to path the shortest distance and most direct path to keep the player in range. That being said, if the distance check range exceeds the NPC's line of sight and the NPC targeting times out, the NPC will stop following and return to its pathfinding and target acquisition routines.



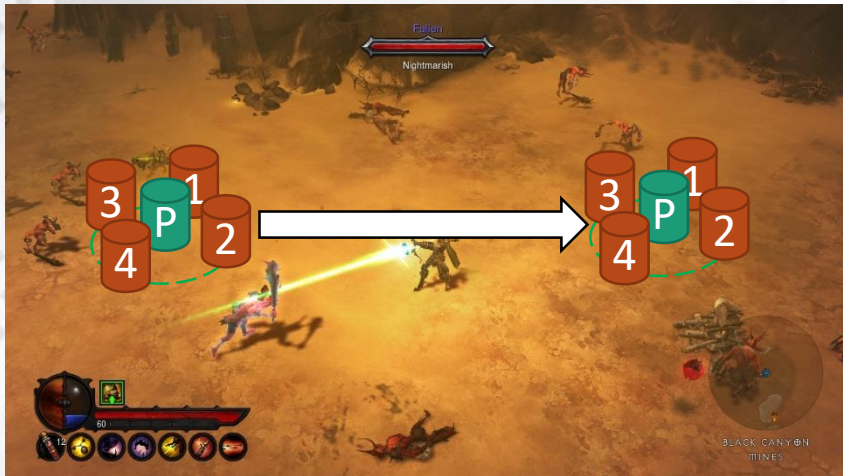
In this example, the NPC (B) has locked onto and is maintaining a follow distance to the target (A). When locked onto a target in this manner, the combat brain is in “aggro” mode. NPCs do not follow targets when idle and not aggro’d.

NPC AI, Flocking

Flocking is a behavior that is only applicable to Melee combat brains. The purpose of flocking is to allow enemies to “attach” to the player and move along with the player to prevent the player from becoming “stuck” or blocked by NPC collision, yet not pass-through NPCs.

A player has 8 total “attachment” points for melee combatants (one in each cardinal direction, N/W/E/S, etc). Once these 8 slots are filled, other NPCs that wish to engage in melee will path find and look for other targets until this target has an open slot.

NPCs will attach to the closest open and available slot to them. This may result in 4 NPCs in the “front” of the player usually, but some NPCs will then attach and move around to fill in slots in the rear as necessary.



1. The player (P) has been surrounded by 4 melee NPCs.
2. If the player were to path/move to the right the NPCs would flock and follow along with them. Maintaining their slot assignment and relevant position.
(NPCs facing the player would walk backwards for example).
3. When a player is being “flocked”, they move at -10% speed per NPC flocked to them. This means that when a player is being crowded by 8 total NPCs they move at -80% speed, but are still able to reposition and move around.
4. While in motion during flocking, attached NPCs still perform attacks and run their behavior scripts as normal.
5. Flocking slow down effect and “attachment” occurs when the player is targeted by a melee brain NPC, it does not assume the NPC is adjacent to the player. This helps prevent players from running past NPCs before they become adjacent.

Enemy Data



Overview

this section is full of temporary and incomplete variables!

Enemy data is defined within the DragonForge data tables (see XLS), which will go onto be a live database on the game server. This data contains defined variables and other referencing that will be applied to NPCs (especially enemies) upon the spawning of an NPC entity instance in the game world.

Spawning

When an NPC is scheduled to be generated in the game world (through a local spawner object), it then pulls and assigns the relevant enemy data from the database/server. The NPC then appears in the game world with all stats, abilities and behaviors ready.

Grace Period

Upon entering the game world, all NPCs will have a grace period to allow the clients (and server) to ensure everything is in sync. As such, newly spawned NPCs will be given a 500ms window of invulnerability before they initiate their brain and behavior routines or be able to be targeted and/or killed.

Example:

Graveyard spawning object determines it needs to spawn an NPC. It makes server job and queues to the server to spawn a Skeleton that is appropriate to that area (which is setup in the spawner object). The spawner collects and caches all the data needed for the Skeleton then creates the entity within it's spawn radius and assigns that data to the new NPC.

Keep Alive

Spawner objects also are responsible for tracking, storing and keeping it's owned NPCs active when players are nearby. When no player is present, the spawner object will "put to sleep" all active NPCs to save on server load and awake all owned NPCs to resume their routines when a player is detected in the same zone.

Enemy Data



Data assignments

this section is full of temporary and incomplete variables!

Enemy data is assigned as follows from the **Enemy** tab within the data.

1. Spawner object's configuration determines in the spawner's area of influence
 - Total number of live enemies allowed Total number of corpses allowed
 - Type of enemies to be spawned (either by classification or explicit definition)
 - Difficulty value range of enemies (which can be used to either override or filter the type to spawn)
 - *Example, setting the type to "Undead" and difficulty value filter to "Low" will result in the spawner only spawning low-difficulty undead NPCs in the given area.*
 - Area in which to spawn (radius around spawner in valid non-collision locations)
 - Spawn count (either single, one-at-a-time, or spawn by groups)
 - Spawn timer(s) (for singles or groups)
2. The spawner then requests from the server to spawn X number of enemies. Starting with the first enemy, the spawner then pulls that enemy's primary data set (from the **Enemy** tab within the data table) to assign to the blank NPC entity. Secondly, the spawner then determines which (if any) additional affix and modifiers are applied to the NPC.
3. Once the enemy primary data is assigned, it then will enable all referenced data for that enemy (such as it's pathing pattern, if any, its abilities/attacks and it's brain scripts).
4. Upon death, the enemy entity will then reference that enemy's loot template data to populate and then spawn a corpse. Upon becoming a corpse, the spawner will deduct the enemy from the total count and begin a spawn timer countdown to replenish the missing NPC back to the spawner's maximum number of live enemy count.

Enemy Affixes



Enhancing an enemy

this section is full of temporary and incomplete variables!

When an enemy is spawned, it has an equal chance to be granted a single additional affix, appended to it's name. Each affix modifies the enemy's base stats, attributes and values and is defined within the data table [EnemyAffixes](#).

Example;

A skeleton is to be spawned in the nearby graveyard, the spawner assigns the "Skeleton" stats to the NPC, then randomly rolls on the affix table to also modify the skeleton with the "Fast" affix, which results in a "Fast Skeleton".

The affix list can have weighting (TBD), and up to 5 modifiers to the enemy for the defined affix effects.

Example;

The "Fast" affix, grants additional movement speed and lowers attack cooldown timers.

Enemy Ability Templates



Abilities

this section is full of temporary and incomplete variables!

Enemies have defined attacks, abilities and spells defined within the data table under [EnemyAbilities](#). This is to differ these from player usable abilities for both exploit segregation and for easy patching/balancing later on. Many of these abilities may be simply “copied” from player abilities (for example, spells), but have their own entry so we can control these more tightly.

- **AbilCount** should define how many of the abilities are to be assigned (and used) by the enemy.
- **Abil#** references the ability definition on the [EnemyAbilities](#) table.
- **Abil#Weight** is used to determine which ability the enemy uses more, the larger the number value the more frequent the enemy will use that ability when it acts (such as cast a spell instead of do a melee attack). The weighting is independent of the AI brain behavior, however we should consider linking these systems for additional behaviors (such as casting healing when low on HP, instead of “at random” as it is designed now).

Each ability entry contains defined data for that ability’s intent. In most cases, attack abilities are very straightforward (does XYZ damage, cooldown and repeat). In some instances, an ability needs to reference additional effects, such as “Regen over time” or “AddHeat damage”. We’ll need a way to setup if an effect is a one-time or over-time occurrence.

Enemy Loot Templates



Defined loot tables

this section is full of temporary and incomplete variables!

Enemies are assigned a loot template to be referenced when that enemy is slain within **Enemy** data tab. A loot template as shown within **EnemyLoot** data table is a list of item(s) that have a chance to appear within the enemy's corpse.

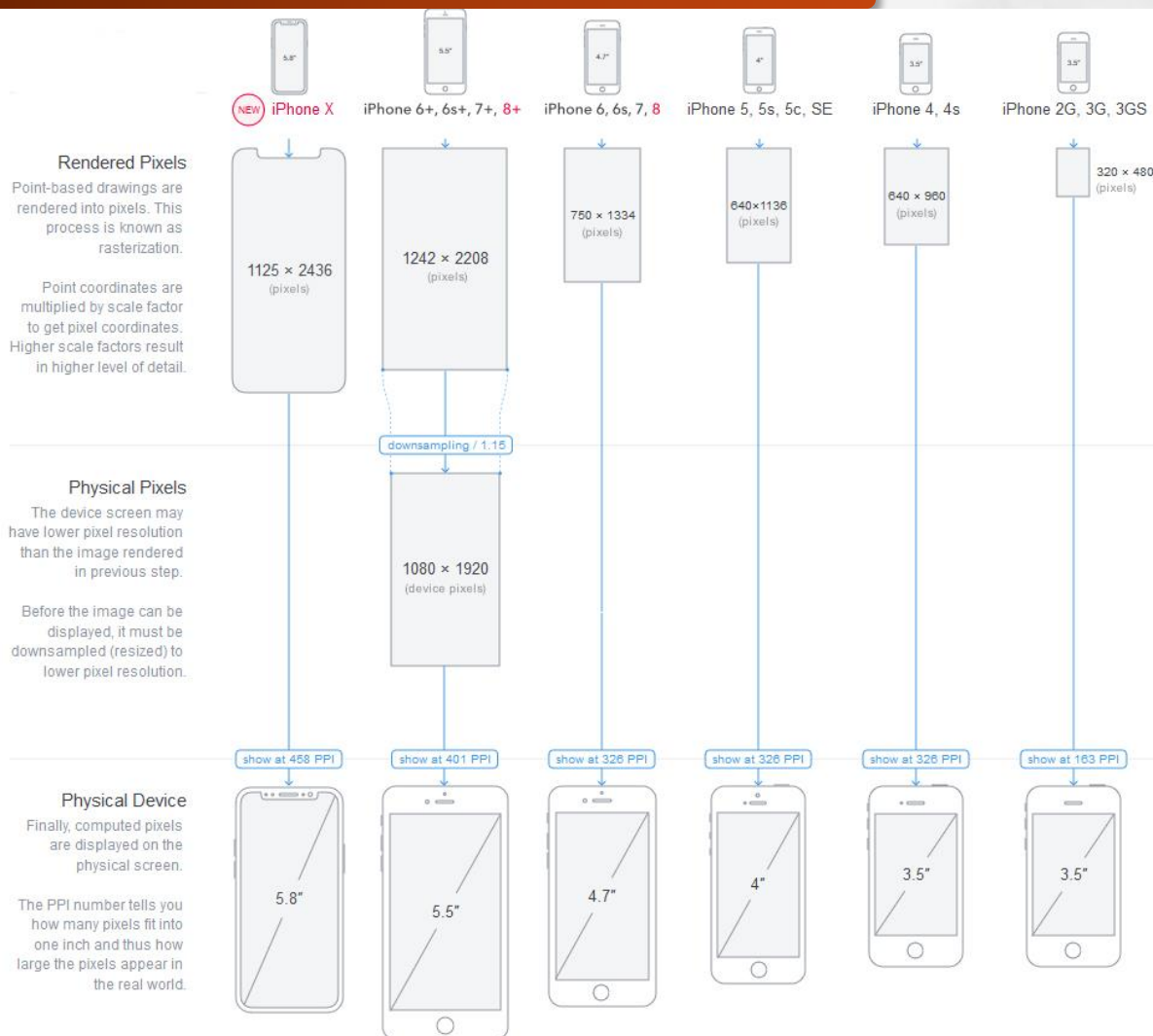
- Each item "slot" can be explicitly defined, such as entering "Coin" or "Iron Ore".
- Wildcard categories should be allowed in the item slot, such as "rand_Consumable", which would give a chance at a randomized item for each instance of itemCount from the Consumables table with equal weighting (we may need to narrow this down further and allow for a sub-list or further filter, such as "rand_Consumable_Food")
 - We may need to define a list of sub-categories, such as "Foods" or "Materials"...etc for the randomization tables.
- There should be no limit to the number of items allowed in a single template, however for development purposes the data is default to 5 total item slots in the template data (Item1 ~Item5)...
- Loot templates should be allowed to reference OTHER loot templates to be included (recursive loot). Such as within "BossLoot2" template I could define it to use template "BossLoot1" and "BossLoot3". Naturally we have to be cautious about recursive loops and self-callbacks (ex; BossLoot2 should disallowed by default from being able to read BossLoot2). In any instance where a loot template references itself, it should ignore itself and skip to the next.
- itemCount is to ensure that item (or reference) only appears X times. In most cases this will be set to "1". However if we wanted to allow a random roll to roll more than once, we can set itemCount to 3 and it will reroll and add 3 items before moving on to parse the next item spawn.
- ItemChance is the chance that the item# has to appear in the corpse inventory, 0-100%, and is per item# definition and repeats for each itemCount

Scripted AI and Patterns

Boss monsters should follow a routine when fighting



iPhone



iPad

iPads compared



